TEAM INVICTUS

Systems Thinking & Design Phase 2 Submission

The Breakfast Paradox

IIIT Hyderabad Breakfast Mess System

Activity	Document	Starts on page	Pages
Activity 6	System Map	2	19
Activity 6	Causal Loop and Feedback Analysis	21	16
Activity 7	Systemic Problem Analysis	37	17
Activity 8	Reframed Systemic Problem Statement	54	13
Activity 8	Design Opportunity Statement	67	12
All five	Corrections Since the First Submission	79	1

System Map

IIIT Hyderabad Breakfast Mess System

Contents

Overview 1

Actors, Process, Resources and Constraints 2

- The Operator Layer 3
- Registration, Procurement and Service 4

Three Kitchens, Not One 5

- Two Operator Subsystems Side by Side 6
- The Yuktāhār Book System 7

Waste, Environment, and the Dogs 9

The Service Timeline 10

Conflicts and Dependencies 11

- Food-Safety Checkpoints 12

Formal and Informal Structure 12

The Class Schedule, and Who Else Needs Breakfast 15

The System Already Solved This Once 16

Bottlenecks, Tensions, and What Actually Closes the Loop 16

Culture, Health, and What Isn't a Problem 18

What Stands Out Most 19

What's Still Missing 19

Overview

This map shows who is in the breakfast mess system, how a registration becomes a plate of food, what shapes that process, and where it strains. It draws on an interview with the Campus Facilities Services Chair, a walkthrough of the Kadamba kitchen, four dining-service posters, six student interviews, interviews with the three caterers, Yuktāhār's registers photographed page by page, a team member's first-hand account of the Kadamba and Yuktāhār visits, and the institution's scan record of every Kadamba breakfast in April 2026.

Claims are labelled confirmed (read off a record, stated clearly by a named source, or seen on site), single-sourced (one account), candidate (a reasonable mechanism nobody has verified) or assumed (a judgement offered as such). Kadamba's figures were heard first-hand, so the numbers are not in doubt, but they are the operator's description of its own practice. Bakul's come from a team member's synthesis of its recordings, and no Bakul records exist.

Two findings frame the rest. The institution has already run a version of the system in which billing follows who shows up, and it worked. And the gap between registration and turnout is measured twice, in the scan record and in the operators' own registers, and acted on inside the kitchens every day. What is missing is any route that carries that measurement to whoever sets the registration and billing rules. Yuktāhār is the most data-rich actor and one of the least powerful. Kadamba keeps gram standards and a live feel for the counter but no record that carries one meal into the next. Bakul's manager adjusts his batch weekly from past attendance, and whether any of it is written down is unknown, so there the measurement may live in one head (candidate).

Actors, Process, Resources and Constraints

Students are the registrant, the billed party and the eater. Campus Facilities Services (CFS) sits above Campus Dining Services (CDS), which has a faculty Committee with final say, a Student Council that drafts and first reviews proposals, and one admin per hall. Its head is open: the CFS Chair named one person as CDS Chair and the team's field account names another as CDS head, and these may be two roles or one wrong name. Whether CDS and its bodies are the Mess Office, Mess Committee and Student Parliament of Phase 1, and whether the Warden's vendor role belongs to CFS, is also unresolved. Operators call the institution's side ”the CDS team.” Catering is outsourced by open tender, one operator per hall or pair of halls. Academic administration sets the class schedule and has no role in mess governance. Students run an informal resale market in a WhatsApp group, Mess Cell.

Several actors sit on the boundary: suppliers of groceries, vegetables, milk and paneer; the garbage contractor who collects Kadamba's waste; and the institution's site team, which relays complaints into each operator's WhatsApp group. A second demand stream eats the same food. Kadamba cooks about 30 extra portions for 30 to 50 people from outside the student registration, such as night-shift security, and about half its own staff also eat at the mess (both single-sourced). They are planned demand and an uncounted sink for surplus (candidate). At Yuktāhār faculty and staff diners number zero to two a breakfast (confirmed). Whether the campus dogs' feeders take some scraps is assumed, with no evidence.

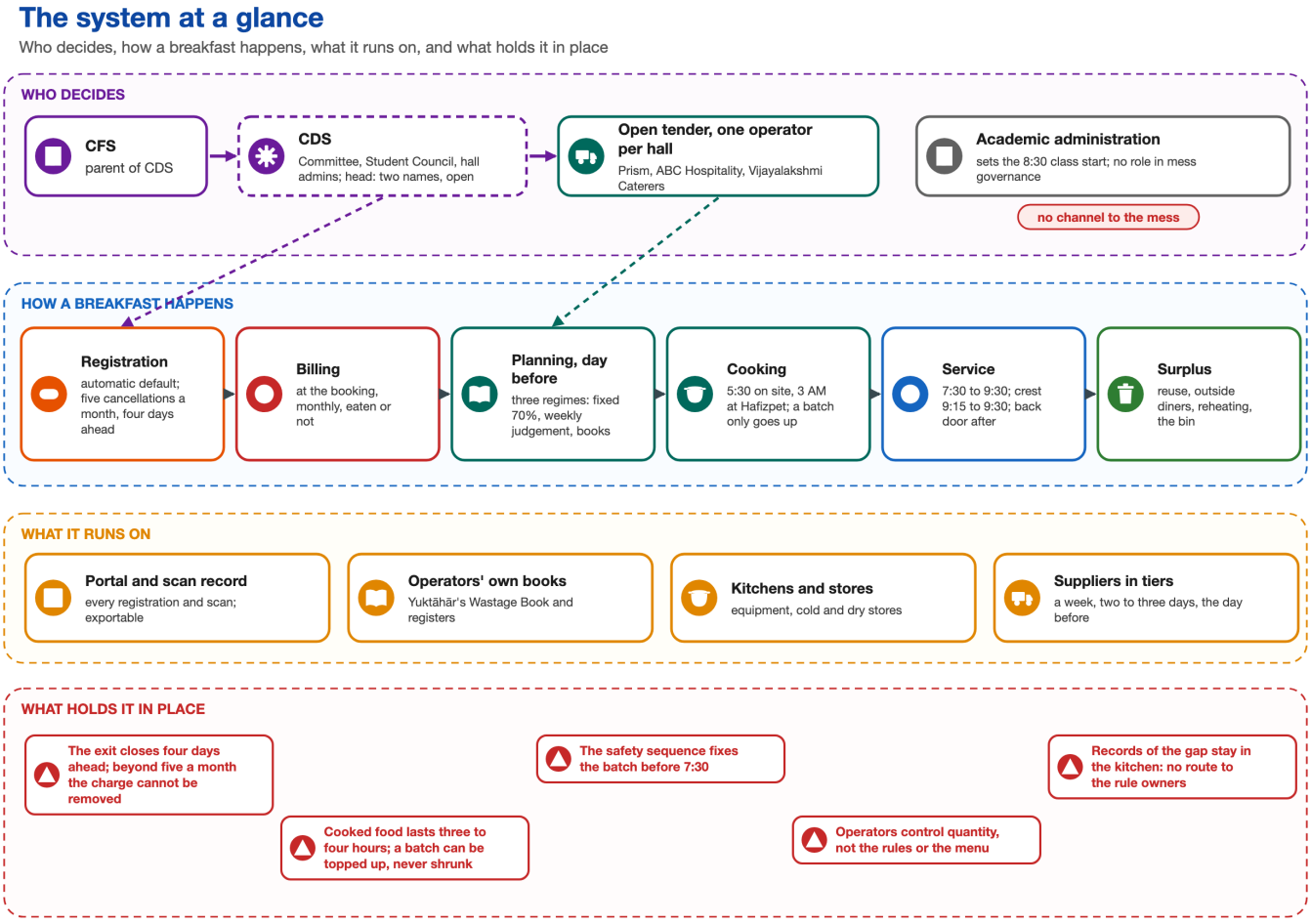


Figure 1. The system at a glance: who decides, how a breakfast happens, what it runs on, and what constrains it.

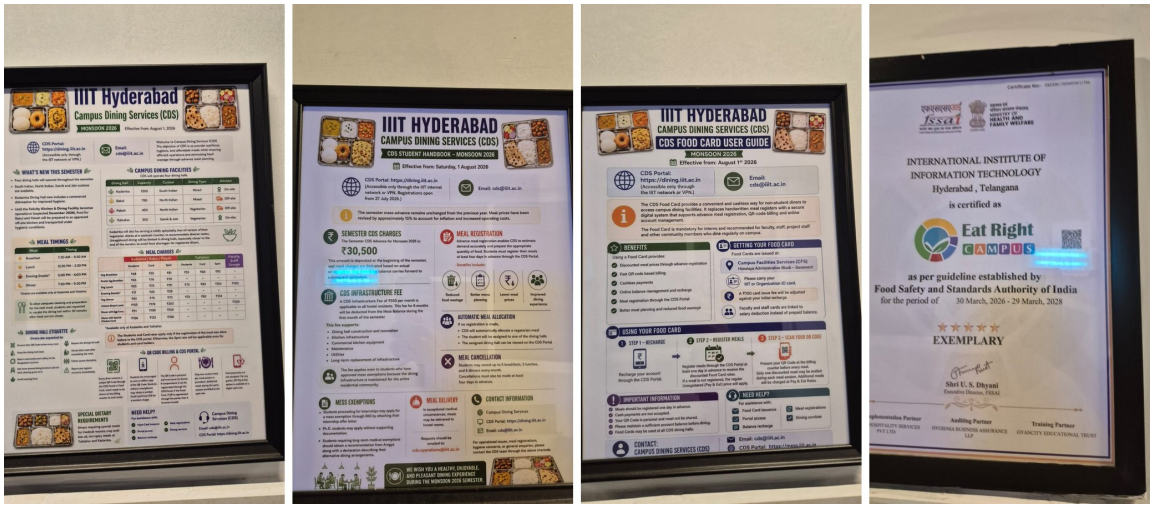


Figure 2. The institution's own statement of the system: overview poster, student handbook, food-card guide and food-safety certificate.

The Operator Layer

Three operators turn registrations into cooked food in three ways. Each reads the closed count from the portal ahead of service and applies its own discount.

Yuktāhār, run by ABC Hospitality Services, serves vegetarian and Jain food. A planning meeting twice a day, with the storekeeper, coordinator and kitchen staff, fixes tomorrow's breakfast and lunch, then snacks and dinner. Its inputs are portal registrations (regular, Jain and cash meals), the Wastage Book entry for the matching day of the two-week menu cycle, a weekday attendance register, the month's pattern, and holidays or events, but only if someone raises them: ”We don't come to know... until we sit in the meeting.” During service the first half hour is a sample for projecting the rest of the meal (single-sourced). Two project managers run it, with about three site staff sharing the work and 16 kitchen staff without a hierarchy. The stated goal is best quality, less waste, lower cost and higher profit, so waste is a cost lever. It tried a commercial ERP and dropped it, because paying lakhs for prediction made no sense with the data already in its books (”I want to remain close to the data”) and its staff are not digitally inclined. Paper is a design choice. Weighing waste began last August, staff first read it as surveillance (”they are catching us”), the managers made it blame-free, and by December staff reported waste falling.

Kadamba is run by Prism, a vendor that won the contract by tender and runs everything on site (single-sourced). It cooks a first batch of about 70% of registrations, veg and non-veg planned separately, and keeps the rest of the raw material ready. Each dish has a gram standard (four idlis from about 120 g of batter, 100 g of sambar), but some items ”float” above it: bonda, puri, vada and idli. A supervisor watches what is left of each item against the time and tells the chef how much more to cook, ready in 10 to 15 minutes. The manager says there is no forecasting method: ”We don't have that idea... it was wasted many times.” The operator cooks on registrations because ”it is a responsibility and they are getting paid for it.” Biryani lunch is the contrast, cooked at 100% (a lunch, single-sourced).

Bakul is run by Vijayalakshmi Caterers, which cooks nothing on site (single-sourced). Its manager sets the first batch at 70 to 80% of registrations by item, from past attendance reviewed weekly, and keeps food for about 70% even on very low weekends. His rule is ”there should not be a shortage of food, even if some wastage occurs.” He checks stock about every 30 minutes and phones the Hafizpet kitchen for more, on a lead time nobody has stated. He estimates that at most about 60% of registered students come, with no record to test it. Whether Palash shares the operator and kitchen is a candidate.

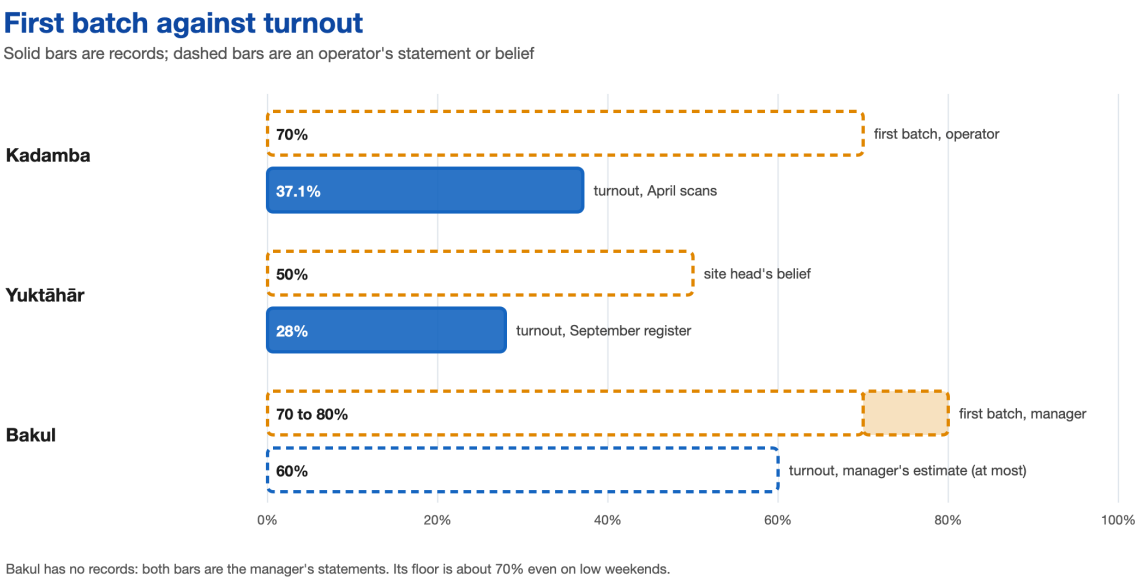


Figure 3. *First batch against turnout at each hall. Only Kadamba's turnout and Yuktāhār's register are records.*

A 70% batch at Kadamba is about 1.9 times the 37.1% who eat, before any top-up, and about 2.6 times on a Sunday (candidate arithmetic: a stated rule applied to a month whose batches were not recorded). The operator puts waste at 5 to 10% a day without saying of what (single-sourced). That fits a batch near twice turnout only if outside diners, staff and reheating absorb most of the difference, or top-ups run smaller than the batch implies, or the base differs. The evidence does not say which. On Bakul's own numbers its batch and weekend floor both exceed the highest turnout its manager expects, and if Bakul resembles the two recorded halls, 60% is an upper bound (candidate). Yuktāhār's site head believes breakfast runs at about 50% where his register averages about 28%, so even the operator with the fullest records misjudges its own level.

Registration, Procurement and Service

Registration is an automatic default with a narrow exit. Students are registered for every meal whether or not they are on campus, holidays included, and the only way out is five cancellations per meal type per month, each at least four days ahead. Beyond that, in a student's words, ”we don't have any opportunity to nullify it.” A student who never registers is still assigned a vegetarian meal (confirmed, from the poster and student experience). Kadamba's operator describes counts reaching it in weekly blocks, so whether that and the students' automatic monthly booking are one mechanism is unconfirmed. Billing is monthly and follows registrations. A walk-in pays 70 to 90% more (Rs. 81 against Rs. 48 for a veg breakfast), which nudges students to register.

Registration therefore behaves as a standing booking. Kadamba's breakfast registrations held near 1,070 a day on every day of April 2026, Sunday included (range 928 to 1,123), while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays. Of 32,162 registered breakfasts, 11,941 were eaten: 37.1% turnout and 20,221 billed breakfasts nobody ate (confirmed, scan records). Yuktāhār's September breakfast averaged about 28% (17.7 to 40.2%, Sundays among the lowest) while lunch on the same days ran at 56 to 89% (confirmed, register). The gap belongs to breakfast: about 63% unused at Kadamba and about 72% at Yuktāhār, where an earlier working figure of 30% fits only lunch. The CFS Chair's 35 to 40% is now confirmed for Kadamba.

The four-day lock applies to registration, not to buying. Yuktāhār's vegetables arrive two to three times a week and milk daily (confirmed). Its stock register shows groceries bought fortnightly while the operator describes buying the week before use, and the two are not reconciled. Kadamba buys in three tiers: dry goods a week ahead, some vegetables two to three days ahead, and perishables the day before (single-sourced). Its team challenged the four-day cut-off directly, and only dry goods, which keep, are committed before it closes (candidate). What binds waste is the cooked first batch, sized as a ratio of registrations. Raw stock is recoverable, cooked food lasts three to four hours, and a batch can be topped up but never shrunk. At Bakul it leaves Hafizpet before any sign of the morning's

turnout and a top-up only adds food, slowly. At Kadamba the team judged that the rest of a batch is held, reheated and moved between vessels, so some of the cost of surplus arrives as poorer food (candidate).

The institution's account, backed by a second source, is that Skip Meal declarations reach the kitchen and change how much is cooked. The toggle does not remove the charge. No operator names it as a planning input, and Kadamba's says flatly that it cooks on registrations (single-sourced). The conflict stays open. One reading that fits both is that a declaration lowers the count before the operator reads it, and the operator treats whatever count it gets as the obligation (candidate).

Beyond that: exemptions exist for internships, PhD students and long-term medical need, though the monthly infrastructure fee still applies. When the scanner fails, Kadamba's staff write the ID and name by hand (single-sourced). The portal is named dining.iiit.ac.in on three posters and mess.iiit.ac.in on the fourth. Kadamba's toaster and grill are broken (single-sourced). Yuktāhār has a store function but no storekeeper-only role, and its daily gas, electricity and water logs show for example that Wednesday's electricity peak comes from grinding idli batter (confirmed).

Hall	Capacity	Cuisine	Kitchen	Operator
Kadamba	1,200	South Indian, mixed	On site, automated	Prism
Bakul	750	North Indian, mixed	Off site, Hafizpet, about 8 km	Vijayalakshmi Caterers (single-sourced)
Palash	450	North Indian, veg	Off site; hall under construction	Same kitchen as Bakul (candidate)
Yuktāhār	350	Satvik and Jain, veg	On site, manual	ABC Hospitality Services

Bakul and Palash share one approved off-site kitchen until a new shared kitchen opens, expected around December 2026, so this is a system mid-transition. Kadamba's two capacity numbers, 1,200 on the poster and ”420 seats against 1,200 turnout” in an interview, read best as registration capacity and physical seats: April registrations ran near 1,070 a day while about 398 breakfasts were eaten.

Three Kitchens, Not One

Kadamba cooks batch items such as sambar and upma from 5:30 and live items such as dosa in front of diners from 7:00 (single-sourced). Yuktāhār cooks by hand from 5:30 for a 7:30 finish, mostly in one batch with poha in two, with handmade oil-free rotis and a second roti counter at peak (confirmed). By the team's judgement Kadamba's main quality problem is food cooked ahead, reheated and moved between vessels (candidate, the team's assessment and not the operator's).

Bakul runs on a supply chain the other two do not have (all single-sourced). Labour-heavy items such as aloo paratha start at Hafizpet from about 3 AM to reach Bakul around 7. Food travels about 8 km, 20 minutes early and 40 later, in insulated boxes holding about 60 to 72 °C, then stands in a hot-water setup at the hall. The chain protects heat and not texture: hot parathas steam in the closed box and soften. The food kept back for late arrivals is held longest, so late diners probably get the softest paratha (candidate). Other items are untested. A finding about turnout, waste or quality at one hall should not be assumed for the others.

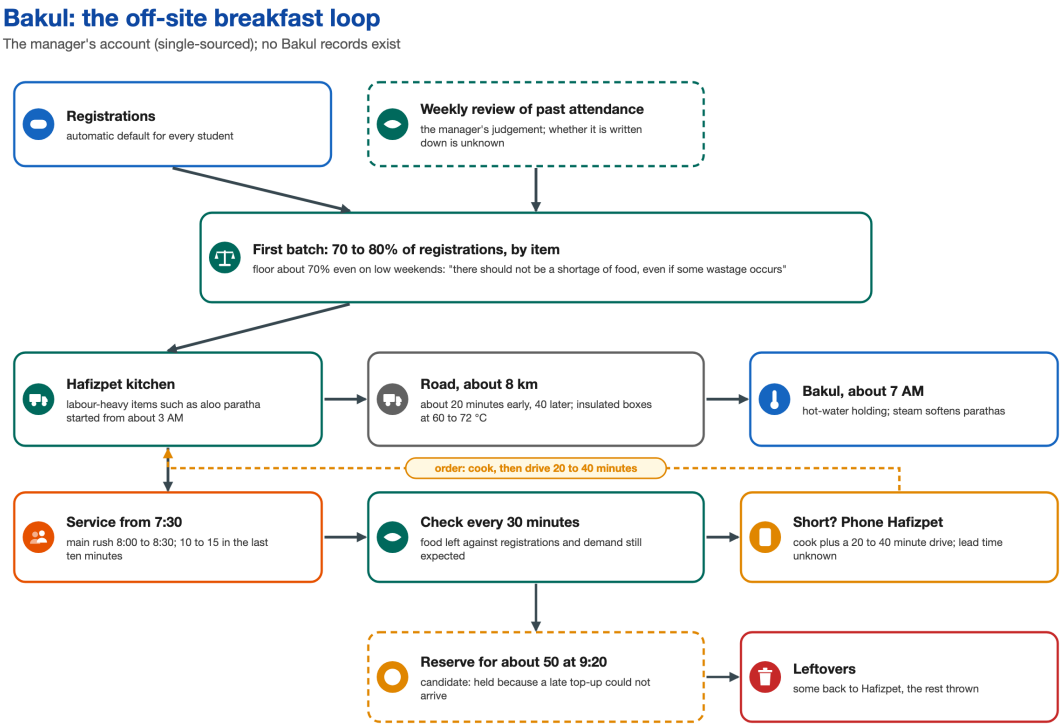


Figure 4. Bakul's breakfast loop, from the weekly review to the reserve held at 9:20 (manager's account; no records).

Kitchens adapt to supply problems in documented ways. Yuktāhār cleans and sorts grain, weighs the stones removed and reports them to the supplier (confirmed). An expected grams-of-waste norm per item triggers a complaint or a change of supplier, and the operator visits the mills that make its oil (single-sourced). Bakul switched paneer brands after complaints. During the cooking-gas shortage Kadamba cut quantities and bought food outside. Yuktāhār reuses leftovers across meals: idli into idli upma or poha for lunch, batter back to the fridge, surplus milk set as curd, paneer made in house. Bakul returns some leftovers to its kitchen and throws the rest.

Each operator won its hall by tender and is paid monthly as a lump sum on plates, with deductions (single-sourced). Whether ”plates” means registered or served decides who bears a no-show. Bakul's operator says served, in one word of a normalised account, and Kadamba says it is ”getting paid for it” on registrations. The contracts may differ, ”served” may mean billed, or Kadamba's remark may describe a duty and not a formula. The basis is unconfirmed for all three and nothing is carried across. If Bakul is paid on served plates it bears the cost of cooking above turnout and accepts it, which would mean it judges shortage costlier than waste (candidate). If plates follow registrations, billing that ignores attendance protects revenue and students partly pay for a batch above turnout (assumed). Kadamba's automation probably means lower labour and higher equipment cost than Yuktāhār's manual kitchen, which would favour steady volume (assumed).

Two Operator Subsystems Side by Side

Kadamba and Yuktāhār share a campus, registration and billing rules and a portal count, yet turn it into food differently, which makes them the cleanest comparison in the project. Bakul is a third column.

	Kadamba (Prism)	Yuktāhār (ABC)	Bakul (Vijayalakshmi)
Structure	Layered: operations manager, manager, supervisor, staff	Flat; peers cross-check	Manager at the mess, kitchen off site, a managing director above
First batch	70% of registrations, veg and non-veg apart	Matching-week lookup, weekday register, month pattern, events	70 to 80% by item, adjusted weekly by judgement, floor about 70%

	Kadamba (Prism)	Yuktāhār (ABC)	Bakul (Vijayalakshmi)
In-service	Supervisor reads what is left	First-half-hour projection; checks every 30 minutes	Check every 30 minutes; phoned top-up, 20 to 40 minutes plus cooking; reserve of about 50 at 9:20
Record forward	Gram standards only	Wastage Book, read again two weeks later	Weekly review; written down unknown
Waste	5 to 10% a day, base unstated	Weighed per dish and meal, blame-free	Not stated; no record
Stance	Cooks to registrations as a paid obligation	Waste is a cost and profit lever	Shortage is the failure to avoid; waste the accepted cost
Surplus to	Outside diners and staff; reheated (candidate); the bin	Lunch dishes, batter, curd	Some back to the kitchen, the rest thrown

The difference is in the learning, not the data. Both on-site kitchens know on the day how many are eating. Only Yuktāhār writes the outcome down in a form that returns to the next decision about the same meal, and only Yuktāhār treats a smaller leftover as money saved. Bakul corrects against surplus too, by one person's judgement and only down to a floor held against shortage. Yuktāhār's projection has its own limit, since a first-half-hour sample under-reads a crowd that crests at the close (candidate for Yuktāhār, which keeps no arrival times).

The Yuktāhār Book System

Yuktāhār runs on paper books that feed each other. Every book was photographed (confirmed), and how they connect is the operator's account (single-sourced). The day's registrations go into the header of the Grocery Book. At the planning meeting the matching-week Wastage Book entry, the weekday register and the month's pattern become a per-dish list in kilograms from the store's ingredient book. That list is the issue in the Grocery Book, which the stock register balances against receipts and which drives purchasing. The vegetable book and indent track stock and spoilage (”3 spoiled”), a further register covers milk, fruit, gas, electricity and water, and each fridge has a temperature log and cleaning checklist. After each meal the outcome goes into the Wastage Book: raw issued, cooked, left, percentage, run-out time, re-cook, and plate waste plus leftover as the day's wastage. Its figures go on a wastage board, where the coordinator's target is to keep pushing the waste share down, and the page is read again when the menu day returns two weeks later. It works as a rule: an item that runs out at 1:30 is re-cooked, one that runs out at 2:20 is not.

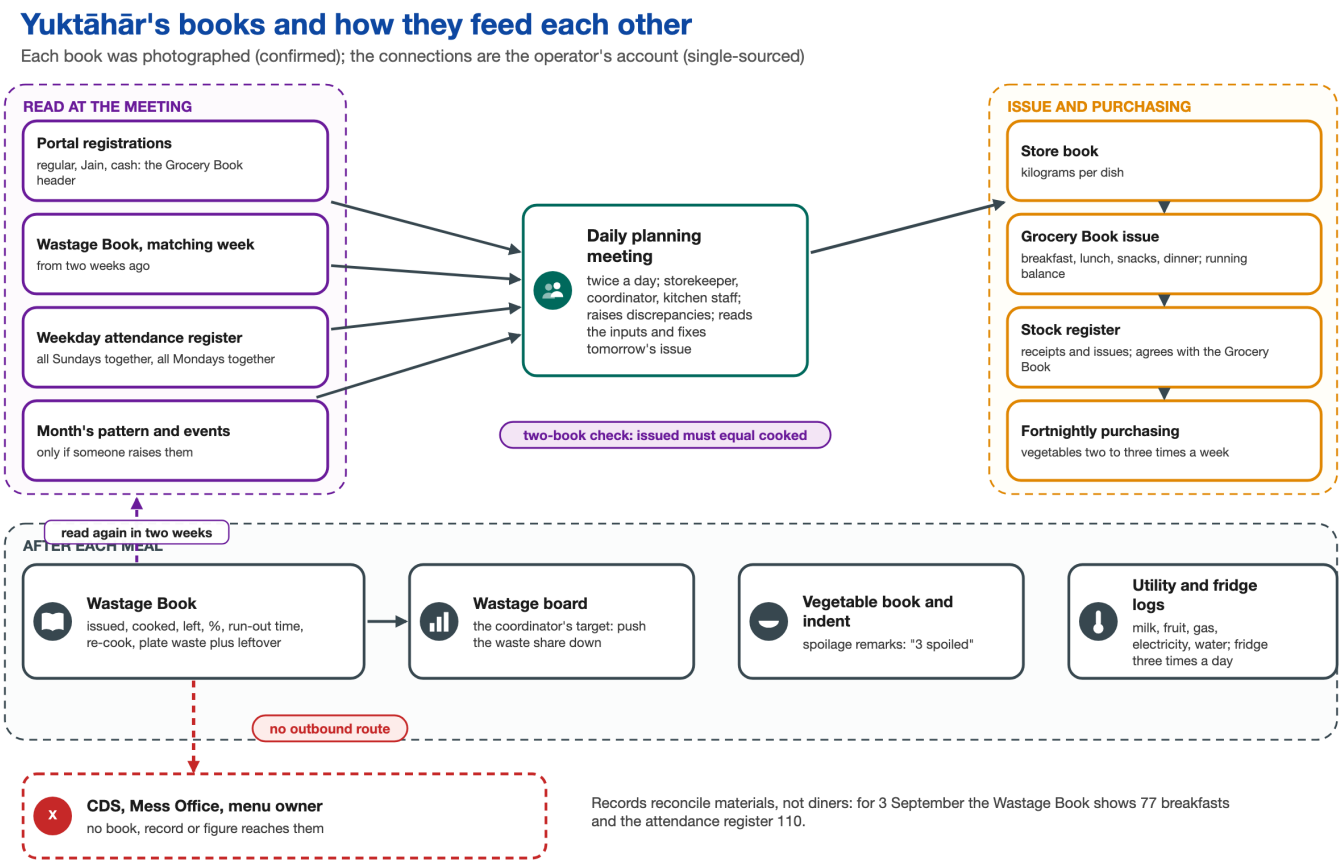


Figure 5. *How Yuktāhār's books feed each other, and where the chain stops.*

Two rules hold the books together. What the store book says was issued must match what the physical book says was cooked, and a mismatch goes to the daily meeting (they agreed on the day checked: 4.5 kg of toor dal issued, 49.45 kg balance). And every book is kept by one person and cross-checked by another, "not to find fault." The books reconcile materials and not diners. The Wastage Book and the attendance register have different keepers, and for 3 September they record 77 and 110 breakfasts (confirmed).

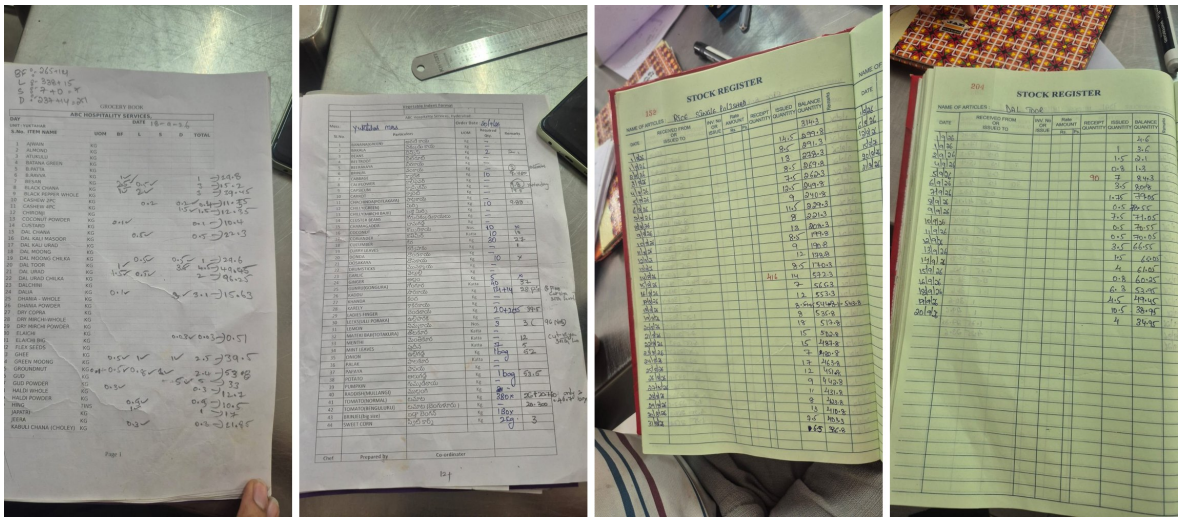


Figure 6. *The books on the table: Grocery Book, vegetable indent and stock register pages.*

Waste, Environment, and the Dogs

Kadamba's waste goes into a dustbin emptied by a garbage contractor, with no composting (single-sourced, but plain and repeated). The CFS Chair had described waste going to an outside compost facility. The accounts are unreconciled, the operator's is first-hand and more recent, and the contractor's onward route is unverified. Yuktāhār's route was not stated. Suspected contamination still goes to a laboratory, and Kadamba holds an external five-star Eat Right Campus rating.

Waste is measured, at least by Yuktāhār, which weighs it per item and meal and uses the record to set the next matching day's issue (confirmed). Kadamba's figure is a quoted percentage with no record seen. What does not exist is any route from those measurements to the registration rules, the billing rules or the menu owner, and nothing suggests the institution knows operators hold the data. Waste feeds the kitchen's next decision and stops there.

Surplus is absorbed before anyone who sets rules sees it: reuse at Yuktāhār; at Kadamba the outside diners and half the staff, and reheating (candidate); the walk-in buffer; latecomers served after the nominal close; and finally the bin. Each does real good for real people, which is why the problem persists without anyone ignoring it, and why no fix should work by removing one: cutting Kadamba's first batch would take food from the outside diners and staff unless it planned for them. A contractor's bin is even less visible than a compost stream, and it has the shape of the resale market on the demand side, a sensible local mechanism that quietly removes the evidence of the gap.



Figure 7. Yuktāhār's kitchen on the day of the visit: the registers laid out, the cold room, and staff sorting vegetables.

Two smaller notes. Bakul has been described as untidy, with insects near food, which has nothing to do with registration and might lead someone to avoid that mess whatever the menu. At Kadamba plates are washed on a staff-run conveyor and the same plates serve all three meals, so some students distrust them despite a strong safety rating, a trust problem and not a hygiene one. Food sent to general garbage also carries a larger landfill and methane cost than a compost stream, and nobody has measured the cost of the 20,221 Kadamba breakfasts registered and never eaten in April. The campus dogs' feeders are a student body; nothing in the interviews says any scraps reach them.

The Service Timeline

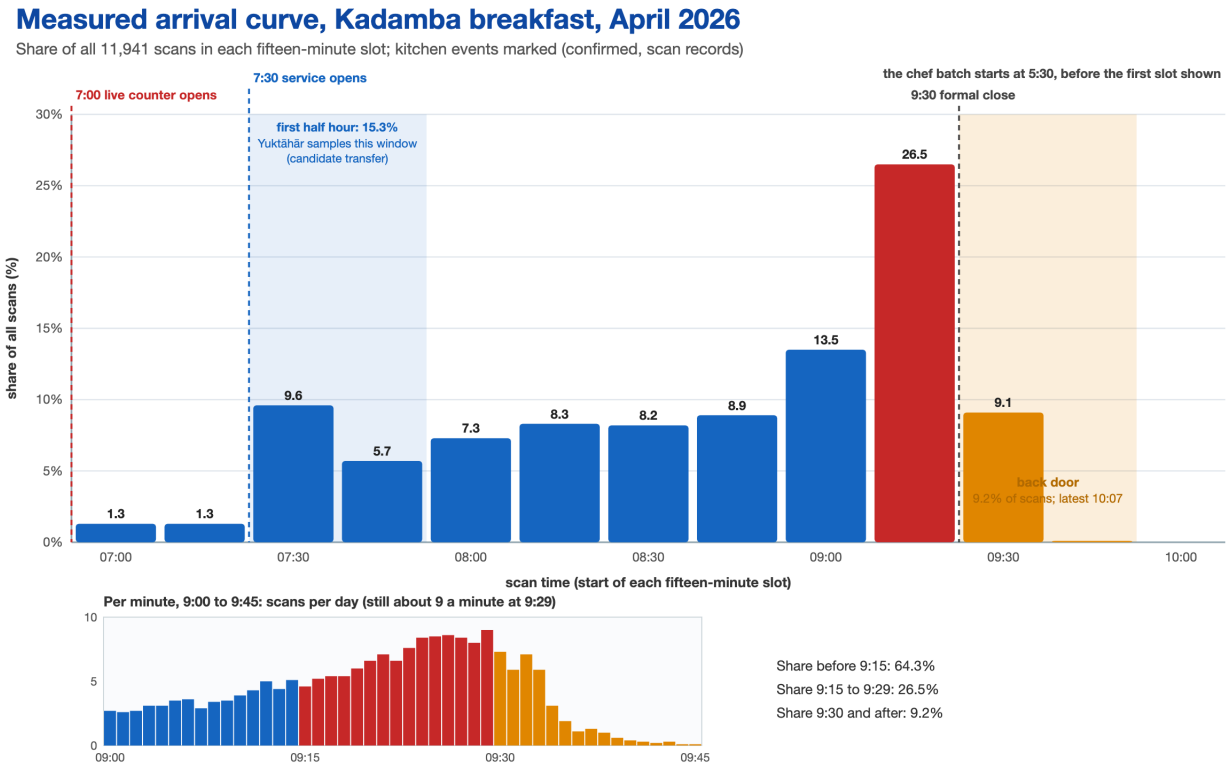


Figure 8. *At Kadamba 26.5% of April scans fall between 9:15 and 9:30, and the rate is still rising at 9:29.*

The Kadamba crowd is measured. Scanning starts at about 7:28, half of the breakfasts are scanned by about 9:00, 13.5% of scans fall between 9:00 and 9:15 and 26.5% between 9:15 and 9:30, climbing to about nine a minute at 9:29. About 9% come after 9:30, service closes between 9:35 and 9:45, and the latest scan was at 10:07 (confirmed). It is one continuous build to a crest just before close. The operator's rule of thumb has the same shape, about 100 early, 100 through the middle and 150 in the last ten minutes. The records show 19.8% of scans, about 79 a day, in the last ten minutes against about 43% in his example, so the loudest minutes loom largest in the kitchen's memory (candidate). Yuktāhār's site head describes the same closing rush. Bakul's manager places his main rush at about 8:00 to 8:30, with 10 to 15 people in the last ten minutes, and keeps food for about 50 at 9:20 (single-sourced). Arrival curves may differ by hall (candidate).

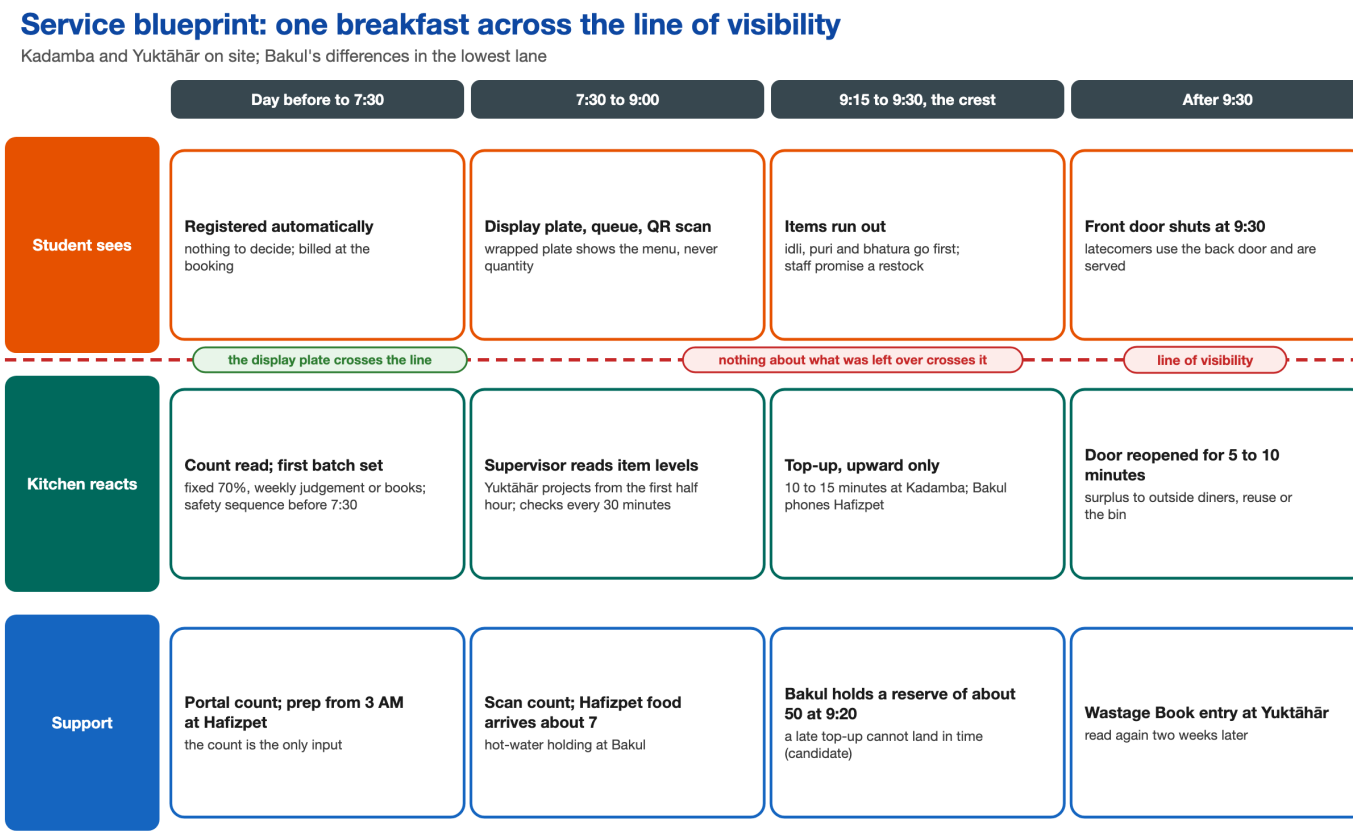


Figure 9. *Service blueprint: what the student sees above the line, what the kitchen does below it, and the one signal that crosses it.*

Behind the scenes the kitchen reacts to the counter continuously within a meal. At Kadamba the supervisor reads what is left of each item beside the scan count, tells the chef how much more to cook, and the counter asks diners to wait five minutes. Yuktāhār projects from its first half hour, checks every 30 minutes from about 45 minutes before close and opens a second counter in surges (checks and counter confirmed, projection single-sourced). Bakul's manager checks and phones Hafizpet. Kadamba reopens its front door for five to ten minutes for latecomers (25 students came after close on one biryani day), and after 9:30 latecomers come in by the back door and the mess cannot refuse them (confirmed, from a team member's experience as a diner and the operator's account).

That response runs in one direction. It adds food, counters and minutes and cannot take back a cooked batch. What does not cross the line of visibility is the signal across meals: nothing tells the student, or whoever sets the rules, what was left over. One artefact does cross it every meal, a wrapped display plate at the entrance showing the full menu (confirmed), and it carries the menu and never quantity. Otherwise the student crosses only by entering a cancellation or Skip Meal on the portal or leaving a rating. Every decision that matters to them happens on the visible side.

Conflicts and Dependencies

Students depend on the operator for food, and no operator depends on any one student. Operators receive no forecast. They receive a count into which students are placed by default, which they know to be inflated, and each converts it privately into a first batch. CDS depends on students registering accurately though nothing in billing requires it, so the side with less to lose controls an input the other cannot replace. Operators depend on the institution for the menu, the rules and the exam and event calendar, and get the calendar only informally. Bakul also depends on the road from Hafizpet. The institution depends on operators for the only item-level record of what is eaten and

wasted, and has no channel to receive it.

The tensions are these. Billing certainty runs against attendance uncertainty, a normal part of student life. The flexibility students want runs against the Mess Office's workload, the stated reason for its one policy tightening (candidate, from secondhand reports). The class start and the breakfast window are two systems each solving its own problem. Registration closes four days out while perishables are ordered one day out. The operator controls quantity and not the rules or the menu. The three operators work under one billing basis with three stances: Kadamba cooks to registrations as an obligation, Yuktāhār treats waste as lost profit, and Bakul puts no-shortage first. Bakul's reserve of about 50 at 9:20 sits against the 10 to 15 late arrivals it describes and reads as insurance against a shortage a late top-up could not fix (candidate). Menu formation pulls three ways, as described under Formal and Informal Structure. Off-site holding trades temperature against texture, and holding and reheating surplus avoids discarding it at a cost in quality (candidate). The safety sequence commits the first batch before service, against any response to that morning's turnout (candidate). And cutting Kadamba's surplus runs against feeding the people who eat it.

Food-Safety Checkpoints

Assurance is routine and runs before any laboratory step. At Kadamba every meal passes the same sequence. The kitchen checks the cooking temperature and keeps a sample for 72 hours, so the food can be tested if anyone falls ill, including to tell whether the illness came from food eaten elsewhere. It sets a wrapped display plate at the entrance, measures the serving temperature of each item, and the facility team tastes the food with two Prism staff before corrections and service. The plate and tasting are confirmed by the team's observation and the rest is single-sourced. A quality-control inspection comes weekly, and cooking and drinking water is read daily for dissolved solids, a failure triggering a full check. Whether faculty visitors to the store are the weekly inspection is unknown, as is who the facility team is. Yuktāhār has a tasting team and logs fridge temperatures three times a day, all within 1.4 to 3.5 °C in September (confirmed).

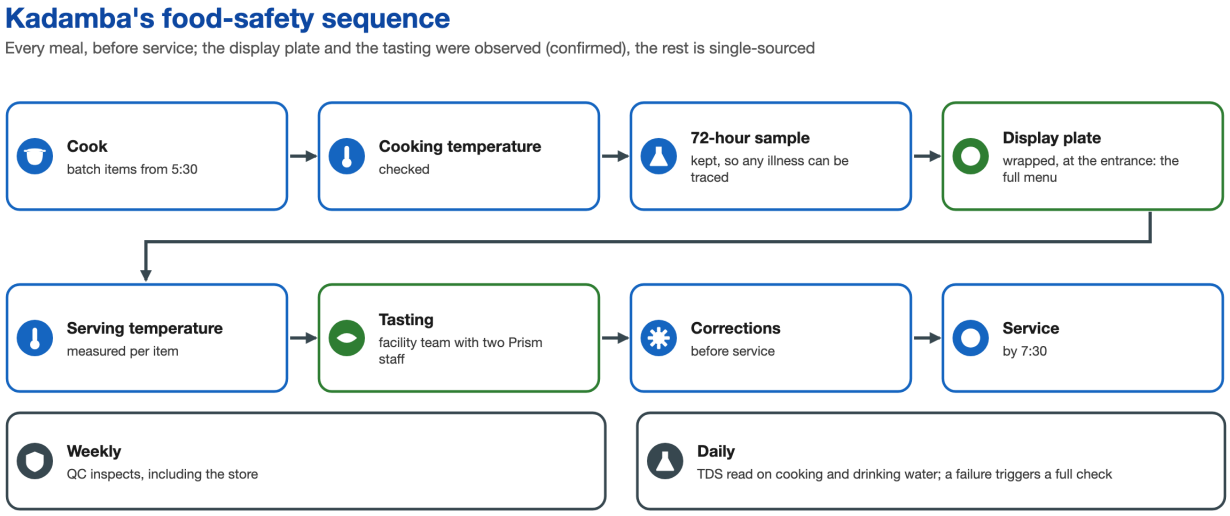


Figure 10. The Kadamba food-safety sequence that fixes the first batch before 7:30.

All of this covers safety and taste, never quantity. The batch has to be cooked, sampled, displayed and tasted before service opens at 7:30, from cooking that starts at 5:30, so its size is committed before a diner arrives (candidate). Nothing seen during service can shrink it.

Formal and Informal Structure

CDS sits under CFS. Low-intensity feedback goes straight to the CFS Head, and high-intensity feedback escalates through the Student Council to the Committee. Below CDS is a contract relation that governance diagrams usually

leave out: CDS tenders each hall to an outsourced vendor, which runs the kitchen.

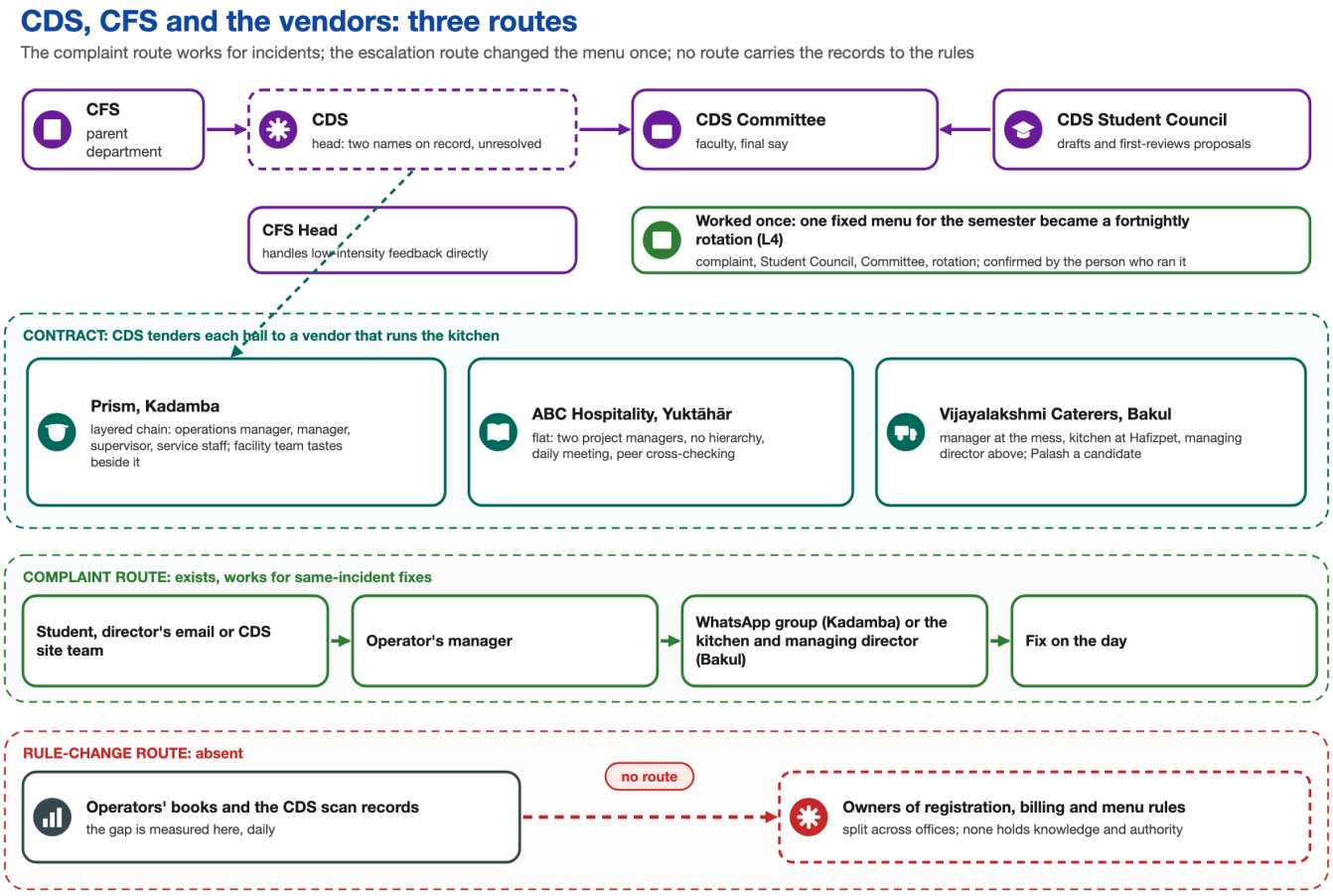


Figure 11. *CDS, CFS and the vendors: the complaint route that works for incidents, the escalation route that produced the menu rotation, and the rule-change route that does not exist.*

Each operator has a structure of its own. Kadamba's is a layered chain from CFS through CDS and Prism to an operations manager, a manager, a supervisor and the service staff (single-sourced, heard first-hand), with a facility team tasting beside it. Complaints reach operators by their own routes. At Kadamba an email to the director passes to the manager and into a WhatsApp group the site team also reports into. At Bakul the route runs from the CDS team to the manager, the kitchen and the managing director.

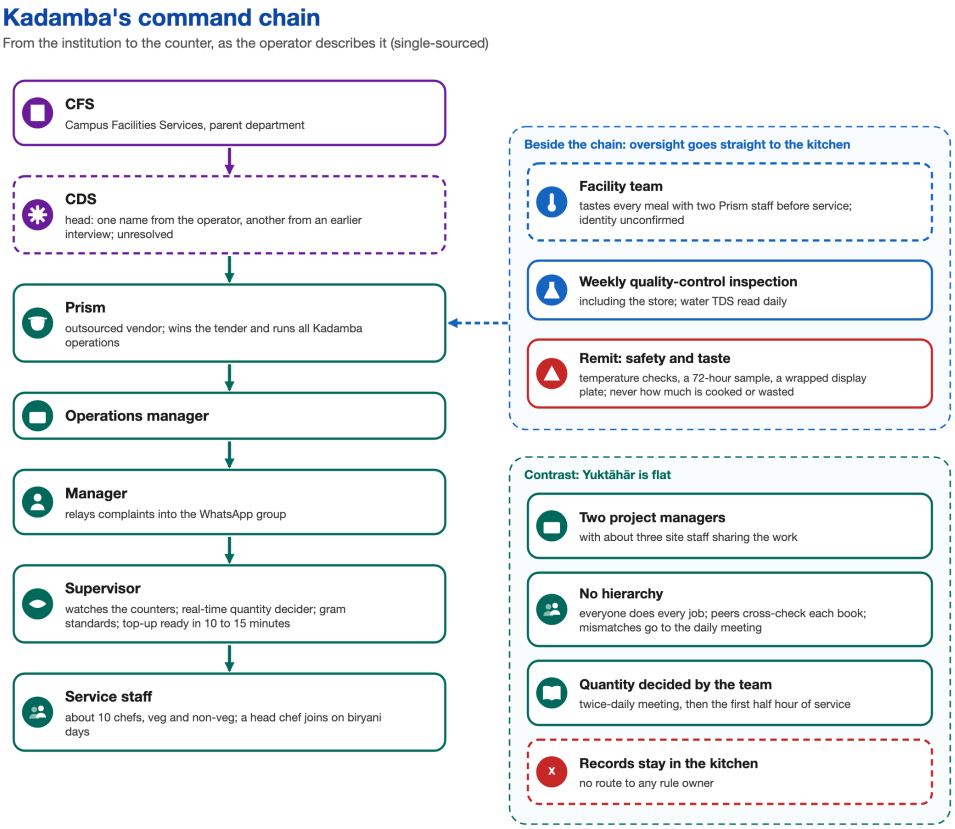


Figure 12. *Kadamba's chain, with the facility team's side channel, and Yuktāhār's flat contrast.*

Yuktāhār has no hierarchy, and everyone does every job (confirmed). Its informal structure makes the books work. There is no leave policy: anyone who needs leave raises it at the daily meeting and colleagues absorb the load. One project manager took the staff to his farmhouse for training on what good quality looks like, and staff describe a family-like culture (single-sourced). The mess in-charge named in the team account and the ”Adit Amma” in the recording are probably the same person, since *amma* is a Telugu honorific, but that is unconfirmed.

The escalation route has produced one real result. A long-standing complaint about a single fixed menu for the semester went through the Student Council to the Committee and came out as a menu that rotates every two weeks, and the operators confirm the cycle from their side. It is the clearest working feedback loop in the project, and it makes Yuktāhār's matching-day lookup possible, so a governance loop enables a kitchen loop. The menu is chosen by students and the committee (single-sourced). Operators see which items waste every time they appear (uttapam, set dosa and upma at Kadamba, while puri, bonda, vada and idli are eaten) and have no route to the menu owner. Kadamba dropped dinner sambar and a student complained it was removed while charges rose, and Bakul's manager asked the interviewers for menu input.

Bakul's manager makes this a question of how a menu is formed (his view only, with the students' side unheard). In his account students choose the combinations and his objection is to pairings, not dishes: kachori belongs with chole, and Raj Kachori is listed with chana where he would serve aloo curry. He describes students reasoning from balanced diet, protein and preference, and himself from traditional and regional combinations, taste and what is eaten, with about 20 years in North Indian food. He mentioned a protein target of 30 to 40 g a meal, set by someone unknown. He also sees consumption the menu owner does not: pav bhaji brings more students and muesli and Chocos ”go easily.” Whether residents share his view is untested. The menu owner lacks two kinds of knowledge the operator holds, what is eaten and what goes together, and neither has a route to it (candidate).

Outside the formal structure, Mess Cell is a resale market, Ping is the student publication with no confirmed power to act, and at least one unofficial tool manages registration outside the portal. Yuktāhār's books are formal inside

the kitchen and invisible to the formal governance of the mess.

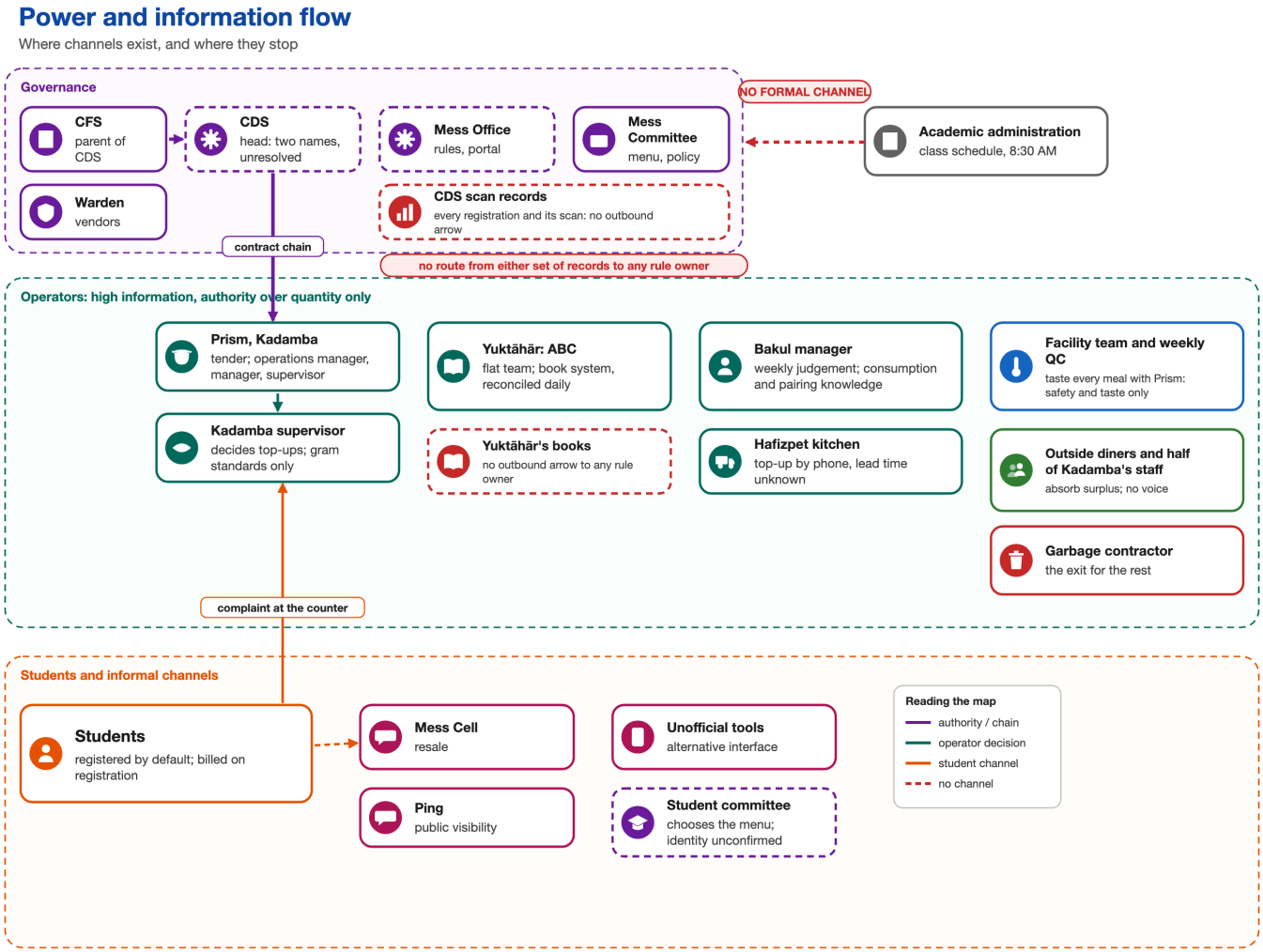


Figure 13. Where information and power sit: operators hold the most information and authority only over quantity; the records have no outbound route.

Academic administration has no formal or informal relationship to any part of mess governance, a confirmed absence and not a research gap.

The Class Schedule, and Who Else Needs Breakfast

The project has treated 8:30 classes as an obstacle, which is probably unfair to the academic side. The timing may put harder lectures when students are freshest, let a faculty member teach once and keep the day free, or fit every programme's hours into five days on shared halls. This map does not recommend moving class time. It would treat the academic system as if its only job were to ease breakfast. The 8:30 start is a fixed condition the mess works around, not a lever. The exam and event calendar is different. It reaches a kitchen only if someone raises it in the planning meeting, which is an absent information flow and not a scheduling clash.

Faculty and staff demand is small and counted at Yuktāhār and larger at Kadamba, where the outside diners and half the staff eat the same food and so are part of how surplus is absorbed. Whether that is a written entitlement or a norm, and whether anyone counts or bills it, is unknown, as is whether grounds and housekeeping staff who start before seven eat at any mess in volume.

Mess Cell shows that students can organise around a shared problem, and there is a past case of an organised email

effort producing a change. Today that energy goes into reselling registrations, not into pushing for a fix to registration or billing. Kadamba's manager independently describes students who register ”even if you are in the village” while a friend eats in their place (single-sourced). Whether Mess Cell ever raised billing with the Committee or CDS is unconfirmed.

The System Already Solved This Once

During a cooking-gas shortage the mess cut to one item at the same price and the cancellation limit rose from five to ten a month. During holidays and long weekends such as Felicity, registration stopped being required, students could walk in and scan or register a day or two ahead at any of the four messes, and someone who neither registered nor came was not charged (single-sourced; the holiday part is disputed, since the team's own account is that students stay registered through holidays).

If it holds, the institution has already run a different way of matching food to demand, aimed at this project's exact gap, more than once, and has not kept it. That turns ”nobody has worked this out” into ”this was worked out and switched off,” and it sharpens the point that responsibility is scattered across offices.

The usual reason for switching it off, protecting the vendor's four-day lead time, is weakened. The kitchens already absorb daily unpredictability by cooking a fraction of registrations and topping up, and Kadamba orders perishables one day ahead. The likelier reason is commercial. If operators are paid on registered plates, a walk-in model moves the cost of a no-show from student to operator (candidate, pending the plate basis). The exception also worked partly because it was rare and short, and partly because the live top-up was already doing most of the work (candidate).

Bottlenecks, Tensions, and What Actually Closes the Loop

The bottleneck is the cooked first batch, sized as a ratio of a count that is itself a standing booking, which can be topped up but never shrunk, against a crowd that at Kadamba crests in the last fifteen minutes. At Bakul the arm is longer: the batch leaves Hafizpet before any turnout signal exists, and late in the window only food already held can meet late arrivals. At Kadamba the safety sequence fixes the batch before 7:30, so the constraint costs quality as well as food. The 9:30 close is a precisely timed cut-off against a crowd that peaks right before it, and in practice runs to 9:35 or 9:45. No formal channel joins Academic administration to mess governance or carries the calendar to operators. Waste is measured in detail by one operator and is a signal for nothing outside that kitchen, not because anyone ignores it but because no route exists. Bakul's manager adds that a shortage is visible and routes a complaint to the managing director while a surplus is silent, so his floor reads as the kitchen-level form of a system that reacts only to what is loud (candidate).

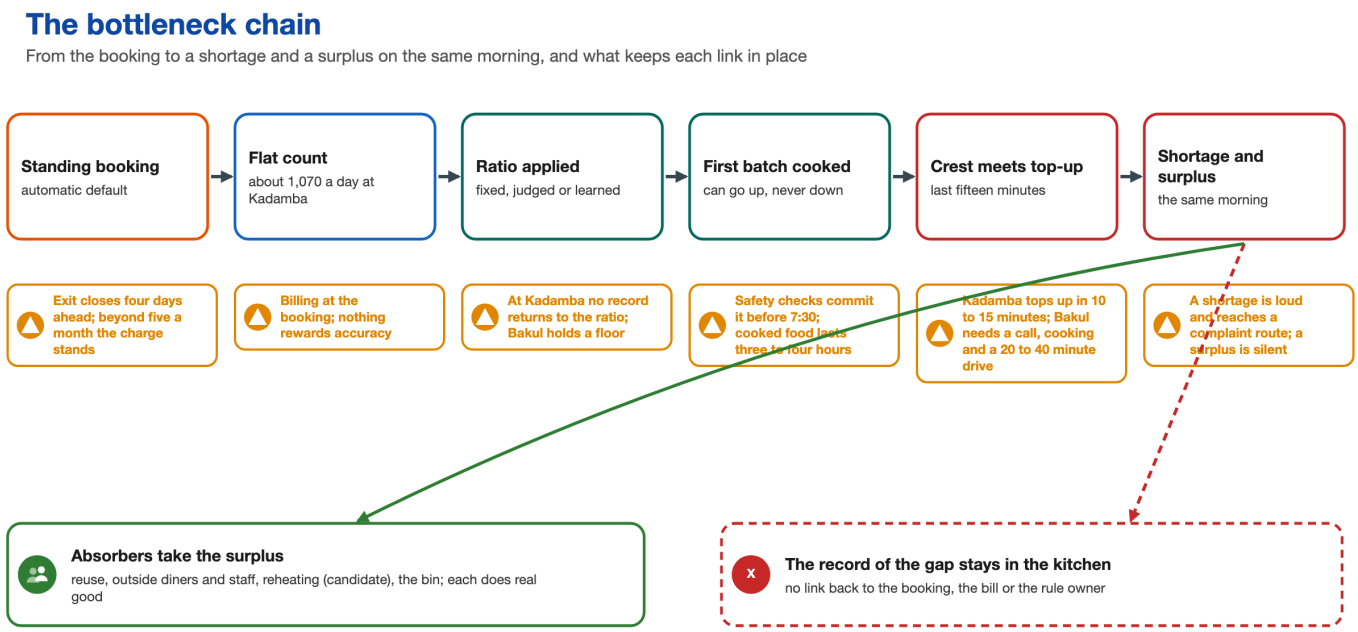


Figure 14. *The bottleneck chain from the booking to the same-morning shortage and surplus, with what keeps each link in place.*

Eighteen mechanisms were tested for whether they close into loops, numbered L1 to L17 plus L3b and tested in full in the Causal Loop and Feedback Analysis. Three close with every link evidenced, all balancing: L4, the menu-rotation governance loop; L11, Yuktāhār's Wastage Book learning, with a delay of one to two weeks, confirmed at one hall, Bakul's weekly review a candidate second instance bounded by its floor, and absent at Kadamba; and L12, counter top-up within a meal, which corrects only upward, with delays of 10 to 15 minutes at Kadamba and 20 to 40 plus cooking at Bakul. L12 protects against shortage and cannot reduce surplus, which is why the 9:20 reserve exists. Five are candidate loops with an assumed closing link: resale (L1), Kadamba demand lock-in (L3b), waste absorption (L9), the soft close (L15) and kitchen competence shifting the burden (L17). Six are open chains. The core one is L13: an automatic booking exited only four days ahead keeps the count flat, all three kitchens start from that count, turnout falls on some days and not others, so over-preparation is worst where turnout is lowest, and nothing runs from the surplus back to the booking or the rule. The others are Skip Meal (L2), menu transparency (L5), shock relaxation (L6), late nights to skipping (L7) and the closing rush to a just-in-case batch (L14). Capacity redistribution (L3) is contraindicated, since scarcity at Kadamba locks people in harder. Vendor discipline (L8) is unresolved until the plate basis is known. Operator waste reaching the menu (L16) is absent. The measured-but-unrouted gap (L10) is a stable stuck state, with a second missing inbound link from the institute calendar to the planning meeting.

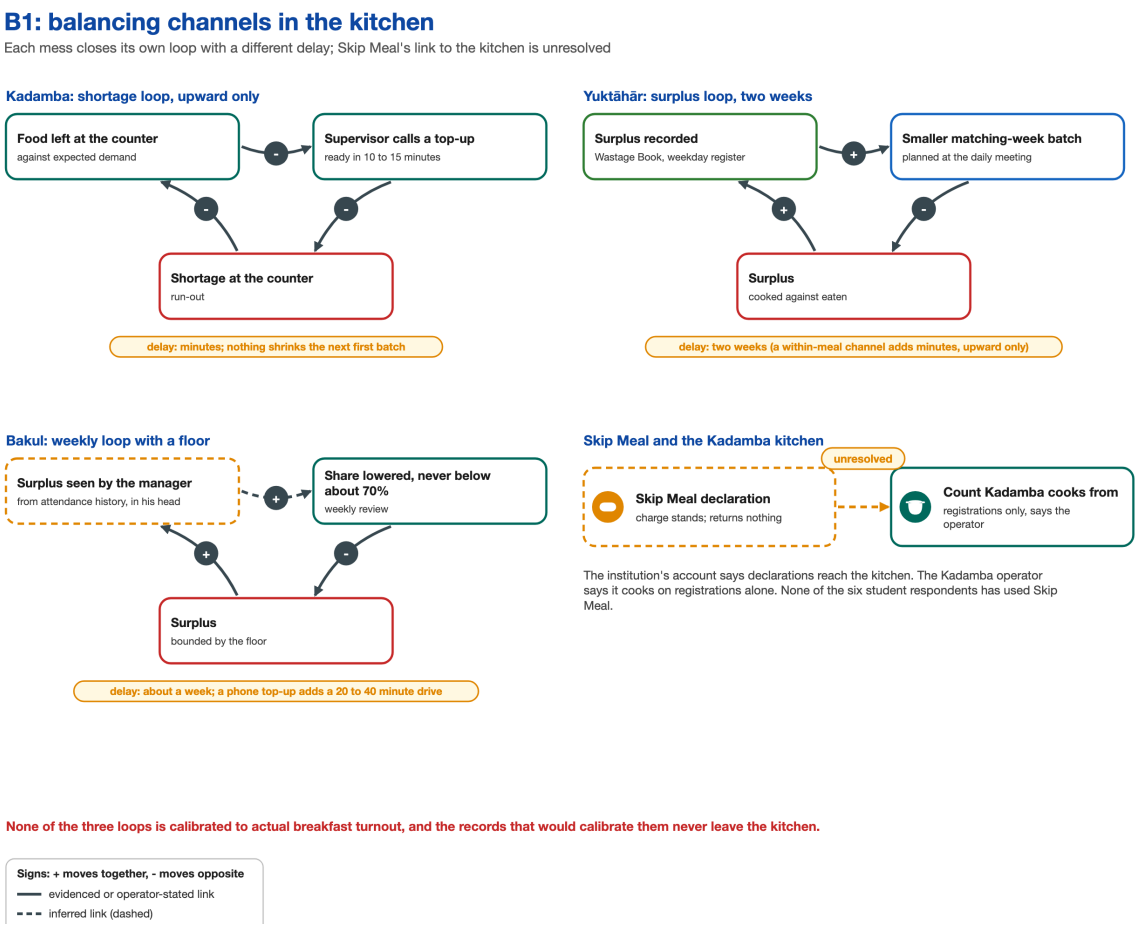


Figure 15. *The kitchen loops that do close, each with its own delay (B1), and Skip Meal's unresolved link to Kadamba's kitchen.*

Culture, Health, and What Isn't a Problem

Breakfast menus lean toward nutrient-focused items such as dal, sprouts and plain rotis, and not the food people crave, a plausible separate reason turnout is lower than at lunch, alongside the billing incentive. Yuktāhār now confirms that breakfast runs far below lunch on the same days under the same rules, so both explanations can hold. Bakul's manager offers an operator's version: students may dislike the taste or menu, or may not wake up or want to come (his view). The second agrees with what interviewed students say most often, late sleep. The first is his hypothesis, since the six student interviews do not support food quality as a primary reason and none of those students is known to live at Bakul.

A recent outbreak of hand, foot and mouth disease reportedly spread across both hostels, cause unconfirmed. Skipped breakfast did not cause it, since the disease spreads by contact. Irregular eating plus shared spaces such as mess halls and washrooms would make an outbreak worse once started.

Some students have reasons for skipping that have nothing to do with the system: intermittent fasting, a belief that they are healthy enough, family or regional habits with no formal breakfast, dietary or religious practice, or discomfort eating certain foods such as egg somewhere visible and shared. None of that is dysfunction, and a meaningful share of low turnout is how a diverse group eats. It does not weaken the billing and governance findings, but it means no fix should be judged by bringing turnout near 100%. The measure that matters is whether what is cooked tracks what is eaten.

What Stands Out Most

The institution has already built and run a working fix for the core problem, and it is not the standing default. The reason now looks less like the kitchens needing four days' notice, which they do not, and more like who pays for a no-show. The gap between registration and turnout is measured and acted on inside the kitchen every day, and no route carries it to the people who set the rules. Yuktāhār is the most data-rich actor and one of the least powerful. The three halls run three planning rules, so a finding from one should not be assumed for the others, and the two on-site kitchens under identical rules are the clearest comparison available. Breakfast is the problem meal. At Kadamba and, by description, Yuktāhār, arrivals crest in the last fifteen minutes, which is why shortage and waste share a morning, while Bakul's manager describes an earlier rush. Reuse, the outside diners and staff, the bin and the resale market all absorb a cost quietly enough that nobody is forced to fix its cause, and each does real good while doing it. The class schedule probably has good reasons of its own. And a real share of the paradox is legitimate variation in how people eat.

What's Still Missing

The operators have now been interviewed, and the kitchens' planning, the crowd curve and the scope of the CFS Chair's breakfast figure are answered by their accounts and by records. Still open:

- The contract's plate basis for each operator. It decides who bears a no-show, whether L8 could close, and whether any lasting walk-in model is commercially possible.
- Composting against the bin, since the institution and Kadamba's operator disagree and Yuktāhār's route was not stated. Disposal has not been observed directly.
- Whether Palash shares Bakul's operator and kitchen.
- At Bakul: whether the weekly adjustment is written down, the true top-up lead time, whether the reserve of about 50 rests on data and how much is left at close, why its own residents skip, and which menu decisions students control.
- Whether Skip Meal lowers the count before Kadamba's kitchen reads it, and whether any operator has ever changed a standing ratio, which tests L14.
- Whether the 9% of Kadamba scans after 9:30 are door reopenings, back-door entrants or a later real close, and whether back-door entrants are scanned at all.
- Scan records for lunch and dinner and for Bakul, Palash and Yuktāhār, and Yuktāhār arrival times to test its projection against a closing crest.
- The base of Kadamba's 5 to 10% waste figure, and whether holding and reheating really happens, which needs direct observation.
- Whether the 30 extra portions for outside diners and the 30 to 40 portion walk-in buffer are one allocation, and whether their meals are an entitlement, counted or billed.
- Which Yuktāhār record drives planning when two disagree, whether its groceries are bought fortnightly or the week before use, and whether registration is in weekly blocks, monthly, or both.
- Who the facility team is, whether faculty visits are the weekly inspection, whether one person runs Kadamba's inventory, and which portal address is correct.
- Who heads CDS, with a conversation with that person and with the mess committee still pending.

Causal Loop and Feedback Analysis

IIIT Hyderabad Breakfast Mess System

Contents

Overview 1

The Loop Status Dashboard 2

Reinforcing and Balancing Feedback Loops 3

L11: Operator Waste-Learning (Yuktāhār) 3
L12: Counter Monitoring and Top-Up 4
L4: Menu Rotation Governance Response 5

Candidate Loops 6

L1: Speculative Registration / Resale Safety Net 6
L9: Waste Absorption 6
L17: Kitchen Competence Shifts the Burden 7
L3b: Kadamba Demand Lock-in 7
L15: Soft Close 8

Causal Chains That Do Not Close 8

L13: Standing Booking to Fixed-Ratio Over-Preparation 8
L14: Closing Rush to Just-in-Case Over-Preparation 9
L2: Skip Meal: a Designed Mechanism That Does Not Close 11
L5: Menu Transparency Effect 11
L6: Shock-Adaptive Registration Relaxation 11
L7: Late-Night Workload to Skip to Crash 12

Absent and Disconfirmed Mechanisms 12

Unintended Consequences 13

Structural Causes and Systemic Tensions 13

What This Means for the Reframe 15

What Remains Untested 15

Overview

A causal loop exists only if the last variable in a chain causes the first. This document tests every candidate feedback mechanism in the Breakfast Paradox against that rule. A loop with an even number of negative links reinforces whatever starts it. One with an odd number pushes back toward a steady state. A chain that never closes is neither, so it is drawn as a chain with a dashed closing link.

The evidence is the student interviews, the CFS Chair's account, the CDS poster data, interviews with the operators of the three messes, the team's first-hand account of the Kadamba and Yuktāhār visits, Yuktāhār's photographed registers, and the CDS scan export of every Kadamba breakfast registration in April 2026. Kadamba is run by Prism, Yuktāhār by ABC Hospitality Services, and Bakul by Vijayalakshmi Caterers, which cooks at its Hafizpet kitchen about 8 km away. Whether Palash shares that kitchen is a candidate. Anything read off a register or the scan export, or seen on site, is confirmed. The Yuktāhār operator's statements are confirmed as his account of his own practice. Kadamba and Bakul operator statements are single-sourced: the figures are not in doubt but they are the operators' own claims, and Bakul has no records of any kind.

Eighteen mechanisms are tested, numbered L1 to L17 plus L3b, a reinforcing variant of L3. An earlier analysis had no operator-side evidence and treated the kitchen as a black box. Opening it shows that the best-evidenced feedback in the system lives there.

The Loop Status Dashboard

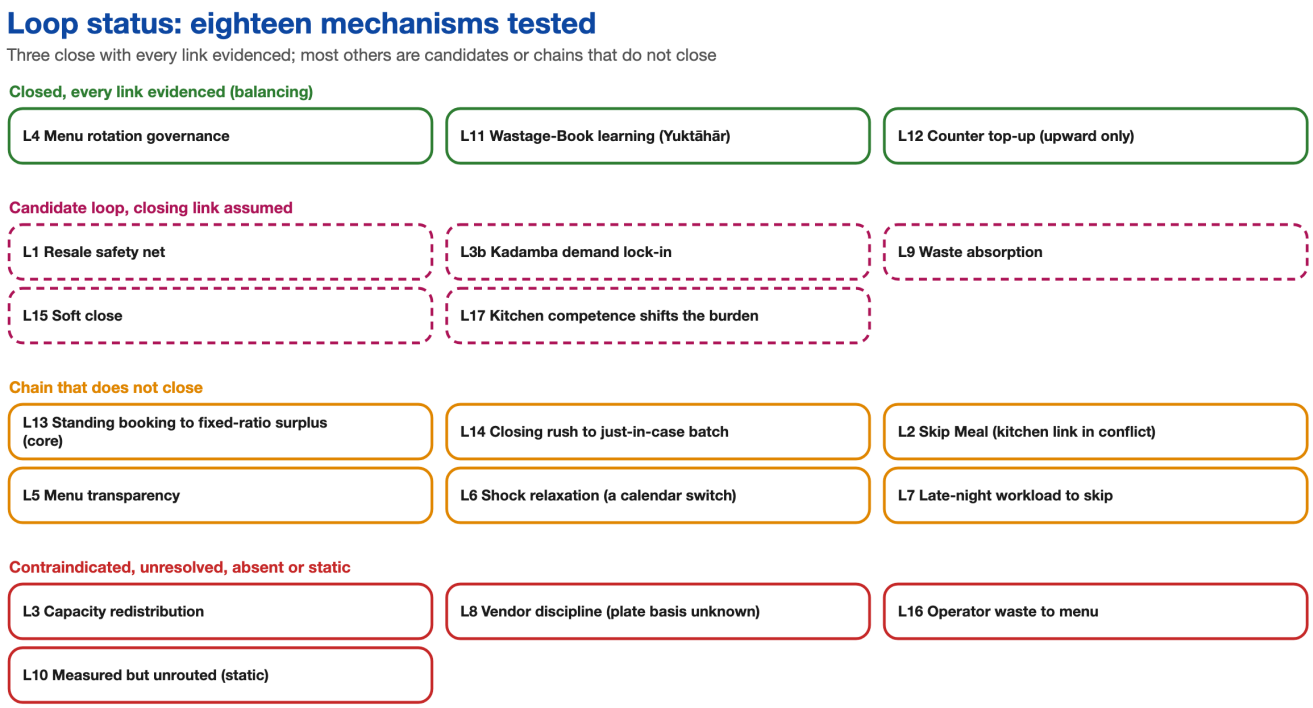


Figure 1. *The eighteen mechanisms by status.*

ID	Mechanism	Status	Delay
L1	Speculative registration and resale	Reinforcing, candidate	Cancel 4 days ahead
L2	Skip Meal	Chain; kitchen link in conflict	Plan the day before
L3	Capacity redistribution	Strongly contraindicated	None found
L3b	Kadamba demand lock-in	Reinforcing, candidate	Registration cycle
L4	Menu rotation governance	Balancing, closes, confirmed	Semester to policy
L5	Menu transparency	Chain	Menu posted 2 weeks ahead
L6	Shock-adaptive relaxation	Chain; a calendar switch	Length of the shock
L7	Late-night workload to skip	Chain; first link confirmed	Overnight
L8	Vendor quality discipline	Unresolved; depends on plate basis	Monthly payment
L9	Waste absorption	Reinforcing, candidate	Cooked food lasts 3 to 4 hours
L10	Measured-but-unrouted gap	Static condition	No route
L11	Operator waste-learning	Balancing, closes, confirmed at Yuktāhār; absent at Kadamba; candidate at Bakul	Next-day plan; 1 to 2 week lookup
L12	Counter top-up	Balancing, closes, confirmed; upward only	10 to 15 minutes on site
L13	Standing booking to fixed-ratio surplus	Chain, the core; open quality branch	4 days, 1 day, 5:30
L14	Closing rush to just-in-case batch	Chain; candidate ratchet	Last 10 to 15 minutes
L15	Soft close	Reinforcing, candidate	Unknown
L16	Operator waste to menu choice	Absent	No route
L17	Kitchen competence shifts the burden	Candidate	Daily

Three close with every link evidenced: L4, L11 and L12. Five are candidate loops (L1, L3b, L9, L15, L17), six are evidenced chains that stop short (L13, L14, L2, L5, L6, L7), three are disconfirmed, absent or unresolved (L3, L8, L16), and one, L10, is a static condition. The split is the finding. The gap is measured and acted on every day inside the kitchen by loops that work, but no route carries those measurements to whoever sets the registration rule. Every working loop corrects the kitchen's conversion of bookings into food, and none touches the booking.

Reinforcing and Balancing Feedback Loops

L11: Operator Waste-Learning (Yuktāhār)

At Yuktāhār, cooked quantity above what is eaten raises the leftover and run-out time written in the Wastage Book. A planning meeting held twice a day reads that record against the matching past day (the menu repeats with weeks one and three identical and weeks two and four identical, so a week-three Thursday is planned from the week-one Thursday), the month's turnout pattern, any events, and registrations counted as regular plus Jain plus cash meals. Its output is the next day's issue list. A larger leftover lowers the next issue and a lower issue lowers the surplus. One negative link makes the loop balancing, and every link is evidenced. The photographed pages record per item the raw quantity issued, cooked, left, percentage wasted, run-out time and re-cook, and the site head confirms their use in planning. The rule has a stated form: ”If it runs out at 1:30 we re-cook; if at 2:20 we don't.” Shown the 18.7% attendance of Sunday 6 September, he said the quantity would be cut ”not even next month, next week.”

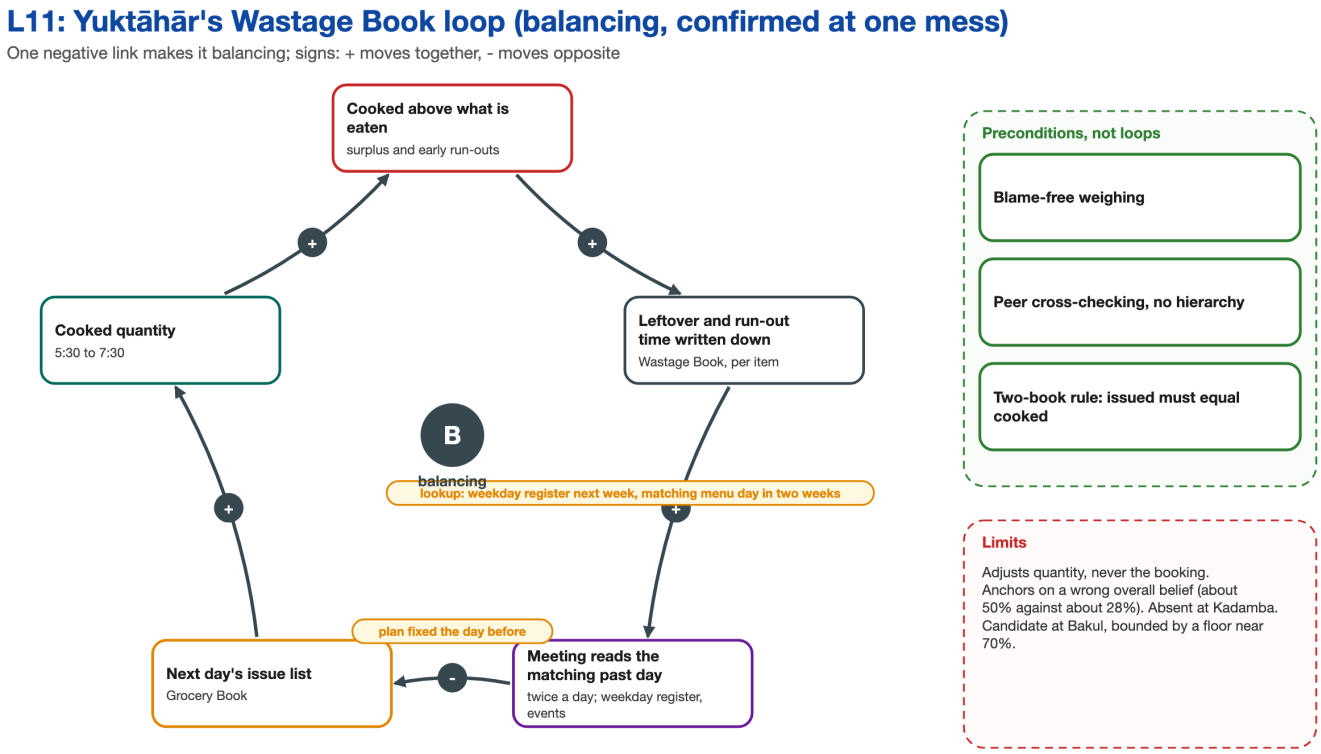


Figure 2. *L11: Yuktāhār's Wastage Book loop, its two delays, and the preconditions that keep the record honest.*

The loop has a goal. The coordinator keeps a wastage board and works to push the waste share down, and the operator names lower cost among its aims, so waste is a cost lever for this vendor. Two delays apply, the plan fixed the day before and a learning lag of one to two weeks. A slower one sits behind the loop's existence. Weighing began in August, staff first feared blame (”they are catching us”), the managers made it blame-free and discussed it daily, and by December staff themselves reported lower waste (self-report; no before-and-after figures seen). Three conditions keep the record honest, and they are preconditions, not loops: blame-free weighing, peer cross-checking by staff who rotate across every job, and a two-book rule that what the store book says was issued must match what the physical book says was cooked. The rule held on 18 September, when 4.5 kg of toor dal and a balance of 49.45 kg agree in the Grocery Book and the stock register (confirmed). It all runs on paper, since the operator tried a commercial ERP system and dropped it as too costly for data already in its books (single-sourced).

Four limits apply. The site head believes breakfast turnout is about 50% while his own register shows 17.7% to 40.2%, about 28% on the legible rows, so the loop corrects item by item while anchoring on a wrong overall figure. It adjusts quantity only, never registration. It rests on one operator's account from one visit, and it is confirmed at

one mess of three. Kadamba keeps per-dish gram standards (four idlis from about 120 g of batter, 100 g of sambar) but no record that learns from turnout: ”We don't have that idea… it was wasted many times.” The three operators work under the same registration rule and hold three stances. Yuktāhār treats waste as lost cost to be learned away, Kadamba cooks on registrations ”because it is a responsibility and they are getting paid for it”, and Bakul's rule is ”There should not be a shortage of food, even if some wastage occurs.” No contract term behind any stance has been seen, so operator culture is a candidate explanation for why the loop is present at one hall, absent at another and bounded at the third.

Bakul is a weaker, bounded candidate second instance. Its manager sets the first batch at 70 to 80% of registrations by item from weekly attendance history (if Mondays have run near 70%, he cooks for about 80%), so a surplus can lower next week's share. The loop closes against surplus only down to a floor: even on weekends, when he calls breakfast attendance very low, he keeps food for about 70% of registrations. Whether the review is written down is unknown, so the measurement may live in one person's head. His estimate of at most about 60% turnout has no record behind it, and on his own numbers the first batch and the floor already sit above it.

L12: Counter Monitoring and Top-Up

Within a meal, food at the counter falls, a periodic check registers the risk of running out, the counter asks for more, and food rises again. One negative link makes the loop balancing, and all three operators describe it.

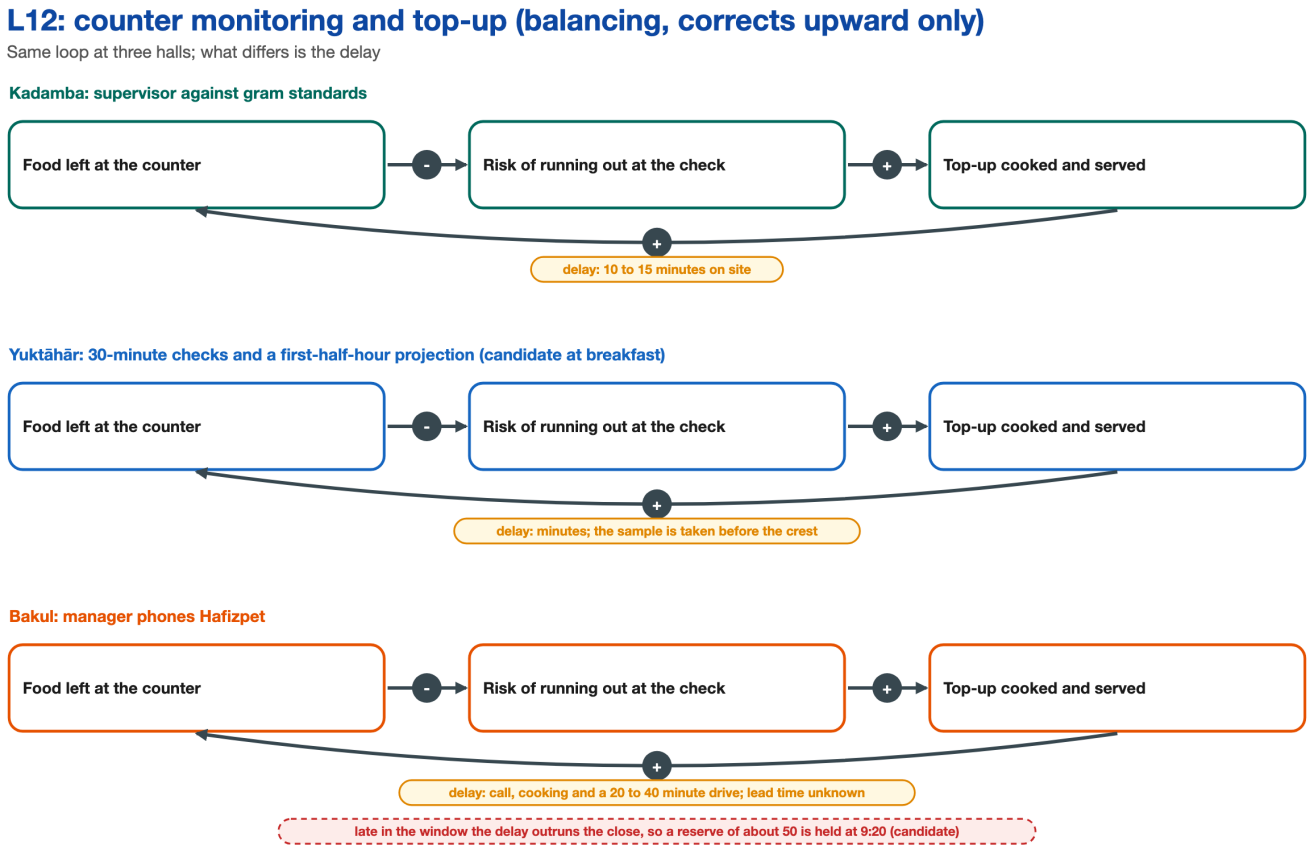


Figure 3. *L12: top-up at three halls, and the delay that decides how well each works.*

At Kadamba a supervisor watches each item's level against its gram standard and asks the chef for a number of portions, combining registrations with turnout so far; the raw remainder is held ready, and top-ups are ready in 10 to 15 minutes (single-sourced). At Yuktāhār staff check every 30 minutes from about 45 minutes before close, open a second counter at surges and re-cook by the run-out rule, and the 9:00 re-cook on 3 September is on record (confirmed). Yuktāhār also runs a forward-looking second form: the first half hour of service is a sample that projects

the rest, which drives top-up cooking (its use at breakfast is a candidate). The weakness is the sample, taken from the part of the morning least like the close. On Kadamba's April curve 15.3% of scans fall between 7:30 and 8:00, so a flat projection of that rate gives about 240 of the roughly 398 daily scans, about 40% short (candidate arithmetic; Yuktāhār's own curve is unmeasured). At Bakul the manager checks about every 30 minutes and phones Hafizpet with a rough quantity, which is cooked and driven over: 20 minutes early in the morning, 40 in normal traffic, total lead time unstated (single-sourced).

The delay decides the behaviour. On site, a top-up requested when the final surge becomes visible lands at or after the 9:30 close, so the loop cannot track the crest and can only be pre-empted by a buffer cooked in advance, which is L14's hedge. Off site the delay is longer, and a call, a cook and a 20 to 40 minute drive cannot land late in the window. The only hedge against late arrivals is food already held, which is why the Bakul manager aims to have food for about 50 people at 9:20 against the 10 to 15 he says arrive in the last ten minutes. As a candidate reading, that reserve is the structural consequence of the transport delay, an insurance premium paid in surplus. Bakul is the clearest case of a balancing loop whose delay is long enough that the kitchen stops relying on it at the end of the meal.

The loop is also asymmetric. Cooked food cannot be un-cooked and lasts three to four hours, so L12 corrects only upward. Surplus inside the meal is carried forward, held and reheated at Kadamba (the open branch of L13), or held hot at Bakul in insulated boxes at about 60 to 72 °C, where the reserve for the latest diners is also the food held longest and likely the softest (candidate). L12 protects against shortage and cannot reduce surplus. That job falls to the slower L11 where it exists, and otherwise to nothing.

L4: Menu Rotation Governance Response

This is the governance loop in which every link traces to a direct account. A single fixed menu for the whole semester produced menu fatigue. The feedback was intense enough to reach the CDS Student Council, which escalated it to the Student Parliament, and the CDS Committee approved a biweekly rotation. Variety rose and, by the CFS Chair's account, fatigue complaints fell. The two-week cycle is confirmed in the operators' records.

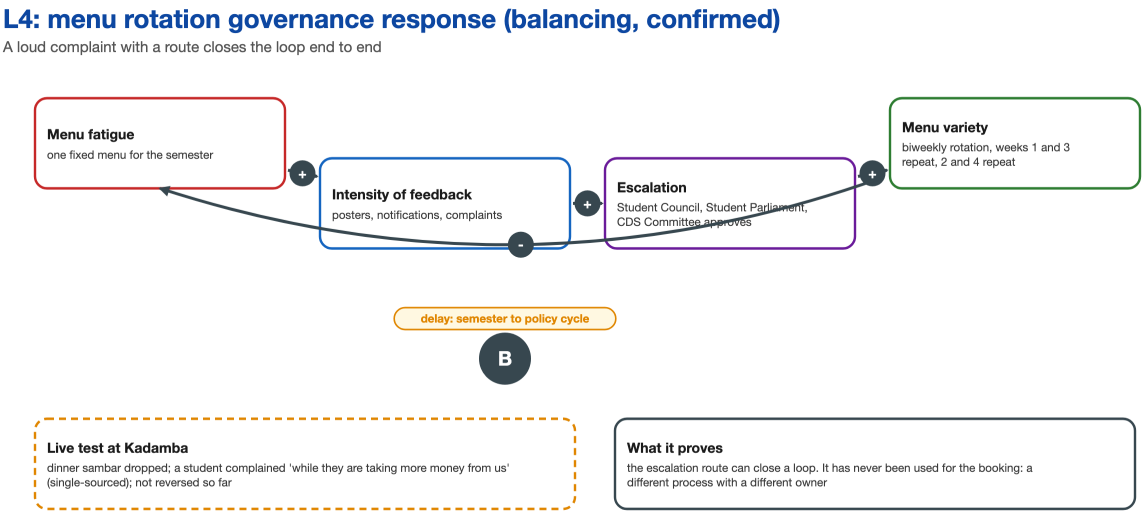


Figure 4. *L4: the menu-rotation loop, closed end to end by a loud complaint with a route.*

The loop shows that the escalation structure can close a feedback loop when a complaint is loud enough, which makes the gaps elsewhere diagnostic. The registration-attendance gap stays open because it never produces the loud, immediate signal this pathway answers. A live test is under way at Kadamba, where dinner sambar was dropped and a student complained it was removed ”while they are taking more money from us” (single-sourced); the change has not been reversed. Menu governance and registration governance have separate owners, so L4 proves the mechanism exists, not that it has ever been applied to the booking.

Candidate Loops

Each of these would close if its last link held, but that link is assumed, so each is drawn dashed and not counted as a loop.

L1: Speculative Registration / Resale Safety Net

A student registers without firm intent, producing an unused registration that feeds the Mess Cell resale market. Two respondents show the first links: ”Kadamba veg breakfast costs around Rs. 48 and I book it every day and never avail it,” and ”I have both bought and sold registrations.” Kadamba's manager describes the transfer from the kitchen side: students register ”even if you are in the village” and a friend eats in their place (single-sourced).

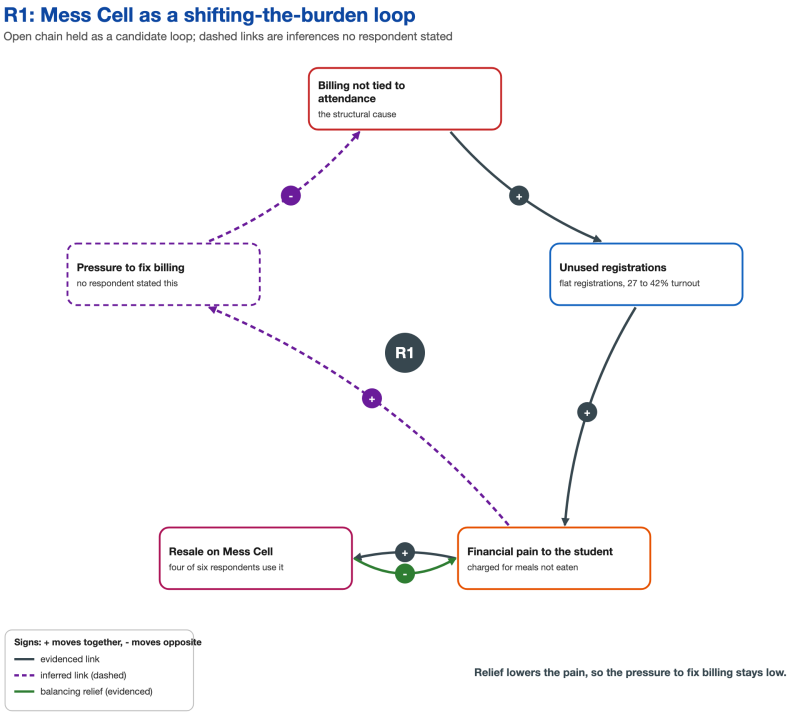


Figure 5. L1: Mess Cell as a shifting-the-burden loop; dashed links are inferences no respondent stated.

The first three links are confirmed. The closing link, that a lower net cost causes more speculative registration, is stated by no respondent, and the strongest student evidence points toward securing a scarce slot: ”I do register for breakfast for the whole month because Kadamba is hard to get and it gets full.” Registration also needs no motive, since it is automatic, so Kadamba's flat count of about 1,070 a day is what the default produces whatever anyone intends (L13). L1 is a behaviour on top of the default, not a driver of registration.

L9: Waste Absorption

Surplus passes through five routes before the bin. Yuktāhār reuses leftovers: idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge and surplus milk is set as curd (confirmed). Kadamba cooks about 30 extra portions for 30 to 50 outsiders such as night-shift security, and about half its own staff eat from the same food (single-sourced); whether these portions sit inside or on top of the 70% first batch is unconfirmed. Food not eaten in the first round is held and reheated into later service (the team's judgement). Latecomers at the back door take a further share (L15). Kadamba's waste then goes to a dustbin emptied by a garbage contractor with no composting (single-sourced), though the CFS Chair describes an outside compost facility and the two accounts are unreconciled. At Bakul some leftovers return to Hafizpet and the rest is thrown.

The size of the absorption is where the evidence strains. Kadamba's operator puts waste at about 5 to 10% a day without saying of what (single-sourced). April averaged about 1,072 registered and 398 eaten a day (confirmed), so

a 70% first batch of about 750 leaves roughly 350 portions before any top-up (candidate arithmetic). By line, veg averaged 757 registered and 245 eaten, so a 70% batch is about 2.2 times what was eaten, and non-veg 315 and 153, about 1.4 times. A 5 to 10% figure fits only if outsiders, staff and held food absorb most of the surplus, if the batch is smaller in practice than the ratio implies, or if the figure uses another base. The evidence does not say which. Waste is visible to operators, per item in kilograms at Yuktāhār. What is invisible is the route from that record to anyone who sets the rules. The two links that would close the loop, that absorbed surplus suppresses pressure and that suppressed pressure permits more unused registration, have never been seen to fire. The better-grounded version is L17.

L17: Kitchen Competence Shifts the Burden

Operators discounting registrations and topping up live (L11, L12) relieve the gap's visible consequences every morning. Queues clear, shortages are re-cooked, and surplus is reused, fed to outsiders and staff, held or binned. Because each morning is handled locally, no signal accumulates upstream, which plausibly reduces pressure for the fundamental fix, a booking that tracks the daily decision to eat.

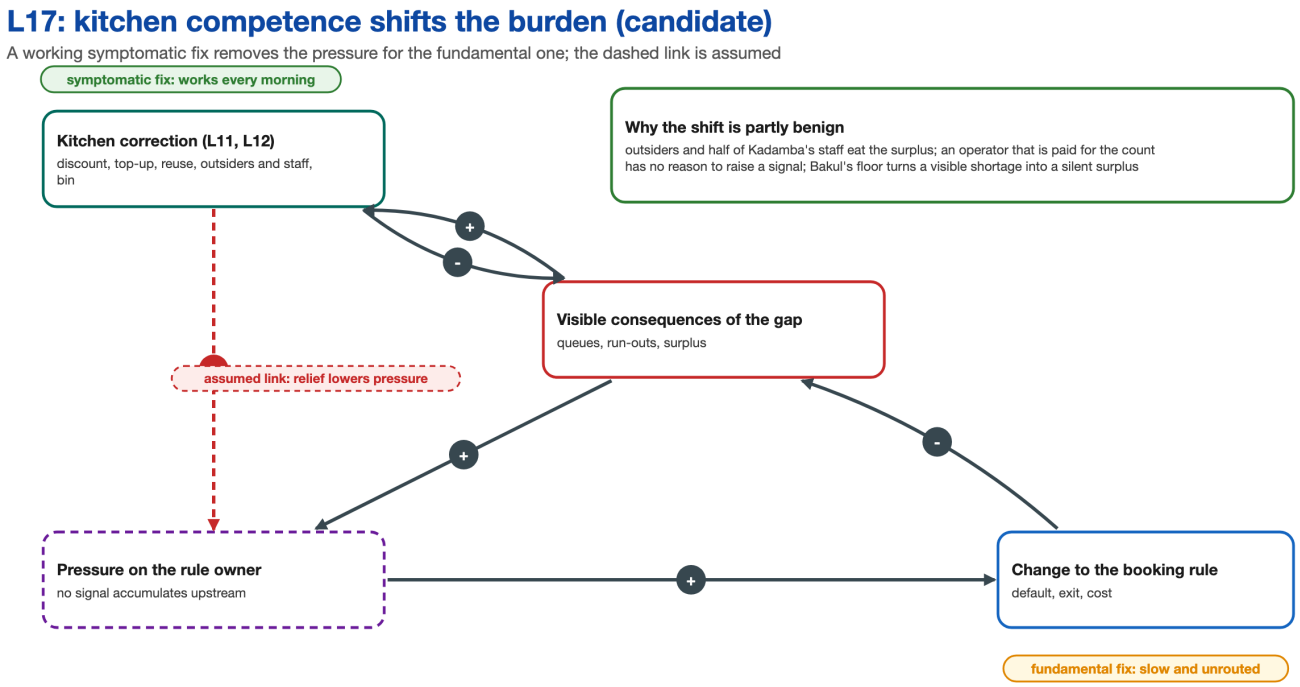


Figure 6. *L17: kitchen competence relieves the symptom each morning, so the fundamental fix never gets pressure.*

Two facts make the shift partly benign for those inside it. The surplus has beneficiaries, since night security and half of Kadamba's staff eat from the same food, so cutting the first batch takes their meals unless it plans for them. And an operator that treats the count as an obligation it is paid to meet has no motive to raise a signal upstream. Bakul's floor turns each low morning's risk of a visible shortage into a silent surplus, and if Bakul is paid on plates served, as its manager reports (L8), it bears that surplus itself and has accepted it (candidate). The symptomatic side is confirmed and the pressure-reduction link assumed, but it is better grounded than L9's version because the correction routes are observed. It is the clearest statement of why the system is stable and not broken.

L3b: Kadamba Demand Lock-in

Kadamba's reputation plausibly drives scarcity, and scarcity drives slot-locking for the respondent who registers all month because Kadamba fills up. April is consistent with that, but automatic registration predicts the same

pattern (about 1,070 a day, 62.9% unused, veg least used at 32.4%), so the aggregate supports the candidate less than it first appeared to. The Bakul operator's belief that Kadamba gets 90 to 95% turnout is contradicted for breakfast by the records. The closing link, that locking reinforces the reputation, is the weakest, and the only direct testimony describes a crowded Kadamba as a negative experience. It is retained as a plausible reason Kadamba stays saturated while Bakul stays under-registered, and is unconfirmed.

L15: Soft Close

At Kadamba, when the front door closes at 9:30 people come in by the back door and the mess cannot refuse them (single-sourced). The operator's wording covers dinner as well as breakfast, so breakfast back-door entry is a candidate, supported by the records: 9.2% of April scans came after 9:30, the latest at 10:07 (confirmed). If students learn the close is soft they arrive later, which builds the late crest and more latecomers, a cycle with no negative link. Only the first link is evidenced.

Causal Chains That Do Not Close

These are real, evidenced mechanisms, and none closes. For the first two that is the diagnostic point.

L13: Standing Booking to Fixed-Ratio Over-Preparation

This is the core chain of the Breakfast Paradox. Registration is mandatory and automatic, running whether or not the student is on campus. The only exit is five cancellations per meal type per month, each at least four days ahead, and beyond them the charge cannot be removed (confirmed). The registered count therefore stays flat across the week, 928 to 1,123 a day at Kadamba in April, about 1,070 on every weekday. Turnout did not. It ran from 42.1% on Tuesdays and 41.1% on Mondays to 34.8% on Saturdays and 27.3% on Sundays, and across the month 62.9% of bookings went unused. At Yuktāhār breakfast ran 17.7% to 40.2% while lunch under the same booking ran 56% to 89% (all confirmed). Whether the cycle is weekly, as Kadamba's operator describes it, or monthly, as students experience it, is unconfirmed.

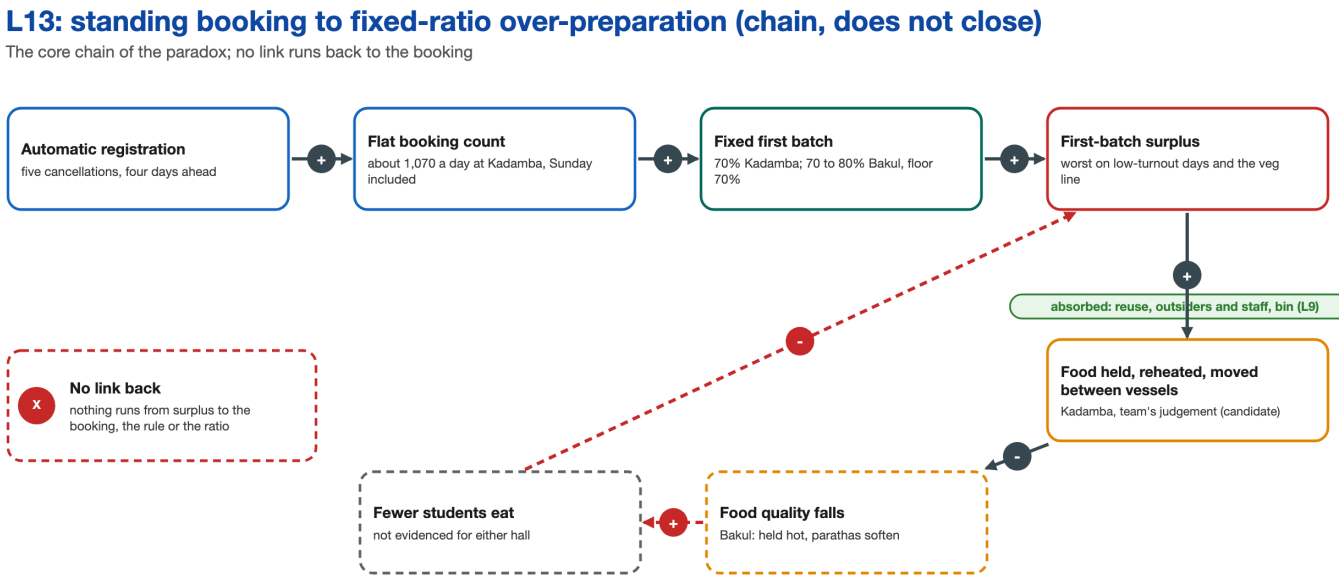


Figure 7. L13: the core chain. A flat booking meets a fixed ratio; the quality branch is dashed because its closing link is not evidenced.

The kitchen converts the flat count into a first batch under one of three regimes. Kadamba cooks a fixed 70%, veg and non-veg planned separately; Bakul cooks 70 to 80% by item and day, kept near 70% on low weekends; Yuktāhār uses no fixed ratio and plans from its books (L11). Kadamba and Bakul top up reactively. On Kadamba's numbers,

as candidate arithmetic, the first batch is about 1.9 times what April turnout needed, 2.6 on Sundays and 1.7 on Tuesdays, and about 2.2 times for veg against 1.4 for non-veg. A fixed ratio on a flat count over-prepares most on the days and the line where turnout is lowest. At Bakul the manager's own ceiling of about 60% sits below both his first batch and his floor, so over-preparation is built in on his numbers (candidate; no record to size it). Even full preparation runs short when portions drift: Kadamba's biryani lunch is cooked at 100%, two pieces of 150 to 200 g with unlimited rice, and when some diners take three, later ones go without. At breakfast bonda, puri, vada and idli ”float” above the gram standard (single-sourced).

Four delays separate the decision from the meal. A student who wants out must cancel at least four days ahead, which decouples the daily decision to eat from the bill. Quantity is set the day before, against procurement tiers: at Kadamba dry goods a week ahead, some vegetables two to three days ahead and perishables the day before (single-sourced). The chef arrives at 5:30, live items start at 7:00, and Bakul's labour-heavy items may start at Hafizpet around 3 AM for a 7 AM arrival. The first batch is thus fixed three and a half to four hours before the crest, after a food-safety sequence that must finish before any turnout signal exists (L8). No link runs from the surplus back to the booking or the rule. Operators control quantities, not rules, nothing routes their registers to the rule owner, and Kadamba treats the count as the obligation, so none runs back to the ratio either. That is the structural reason the gap persists: the only loops that act on it, L11 and L12, act on the kitchen's conversion and never on the booking. The same structure gives 56 to 89% at lunch, which is why the problem is breakfast's.

The surplus has one more consequence at Kadamba, an open branch. In the team's judgement (candidate), the main quality problem is food cooked ahead, reheated repeatedly and moved between vessels, one of them the display plate. More surplus means more holding and lower quality. If lower quality led to more skipping, lower turnout would raise the surplus and the branch would close into a reinforcing loop, but that link is not evidenced, so it is dashed. Bakul has a similar branch driven by distance: food cooked hours ahead and held hot softens, as hot parathas steam in the box (single-sourced, one item, operator only), and earlier dispatch or a bigger reserve lengthen the holding (candidate). Whether texture changes taste as students judge it, and whether taste moves turnout, is untested with any Bakul resident.

L14: Closing Rush to Just-in-Case Over-Preparation

Arrivals at Kadamba crest at close. In April 26.5% of scans fell between 9:15 and 9:30 and 13.5% between 9:00 and 9:15, the rate reached about nine a minute at 9:29, about 79 scans a day fell between 9:20 and 9:29, and the latest scan came at 10:07 (confirmed). Yuktāhār's site head says the rush is always at closing time (confirmed). Kadamba's rule of thumb, about 150 diners in the last ten minutes, matches the shape, and the records remain the measurement. Bakul's manager describes an earlier curve, a main rush at 8:00 to 8:30 with 10 to 15 in the last ten minutes (single-sourced; no scans), so curves differ by hall.

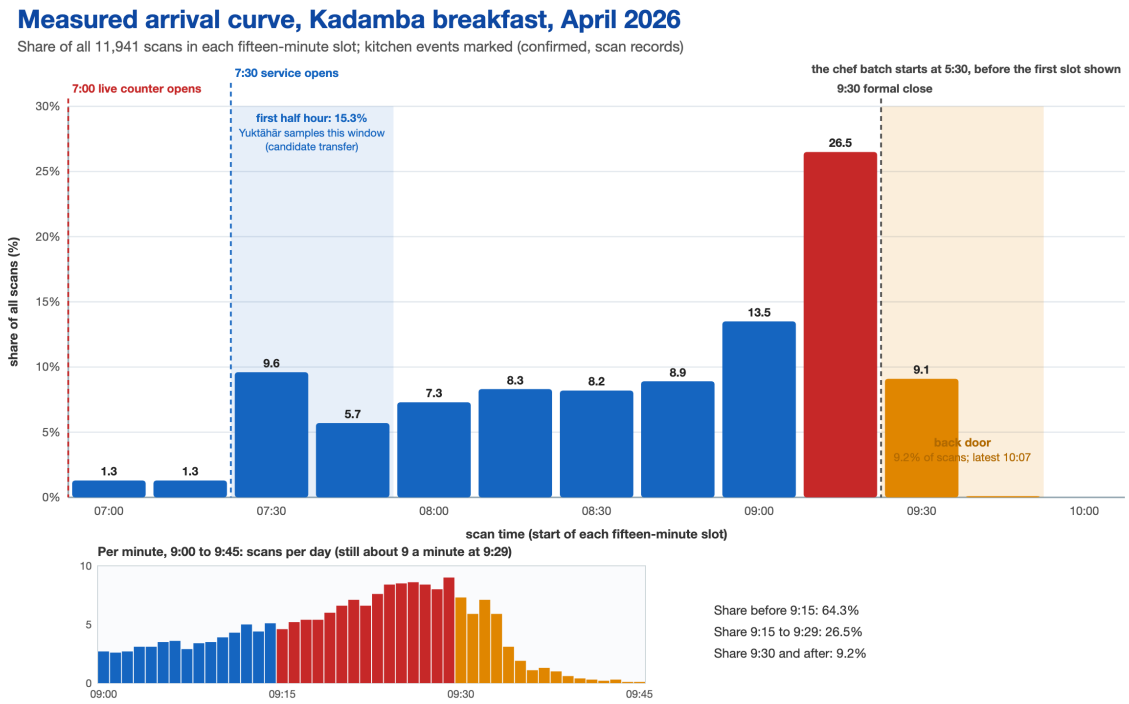


Figure 8. Kadamba breakfast arrivals, April 2026: the crest that a top-up cannot reach.

Where the crest comes at close, demand outruns what a top-up delivers in time, so one morning produces shortage and surplus together. Shortage shows as the 9:00 re-cook at Yuktāhār, back-door service at Kadamba and runs on the floating items. Surplus shows as Bakul's first batch and floor under its no-shortage rule, and Kadamba's raw reserve held for a quick cook. Kadamba's extra portions are not part of the hedge, since they feed outsiders as a planned commitment.

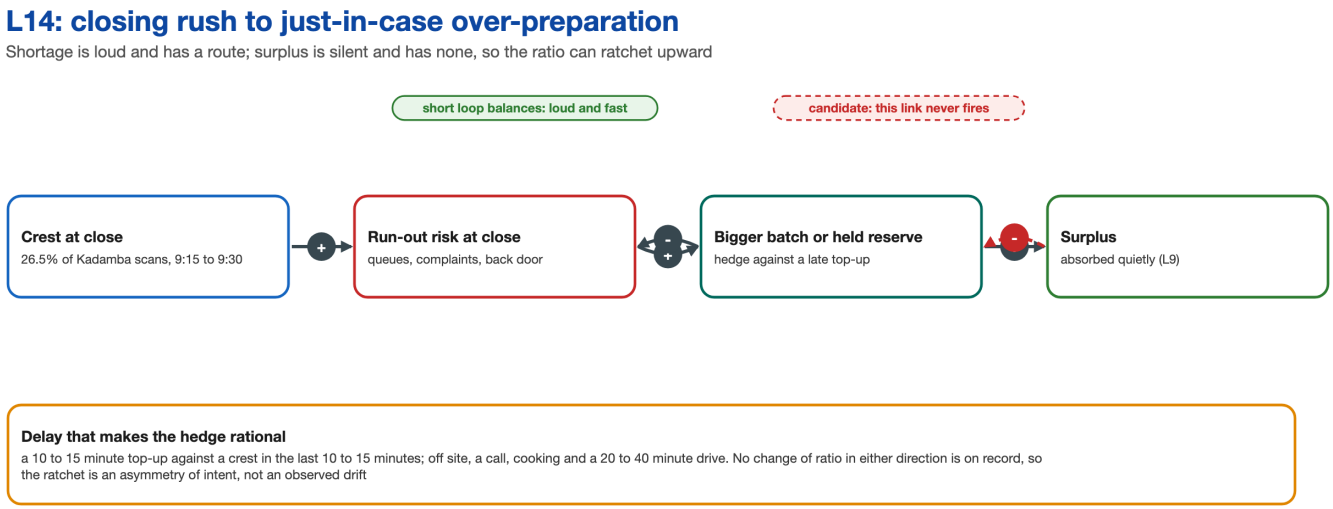


Figure 9. L14: shortage is loud and surplus is silent, so the ratio can ratchet upward. The dashed link is a candidate.

A candidate reinforcing closure sits on the chain. Shortages show as queues, complaints and back-door arrivals, while surplus is absorbed quietly. If an operator raises its ratio after run-outs and never lowers it after surplus, the ratio ratchets upward. No change of ratio in either direction is on record, so this is an asymmetry of intent, not an observed drift, and Bakul states its premise as a rule: shortage outweighs waste, and the weekly review stops at a

floor (single-sourced). Kadamba's operator also perceives the crest as larger than measured, about 43% of diners in the last ten minutes against about 20% on the records, a candidate reason a hedge would be set high. The chain matters without the ratchet. Part of the over-preparation is a rational hedge against a delay at close, not only a misreading of registration.

L2: Skip Meal: a Designed Mechanism That Does Not Close

Skip Meal lets a student who knows they will not attend tell the kitchen to prepare less. Its first link, use by students with real non-attendance intent, is confirmed and close to zero: four of six respondents who addressed it had never used it, one because ”it doesn't return the money.” Skip Meal tells the kitchen but does not remove the charge.

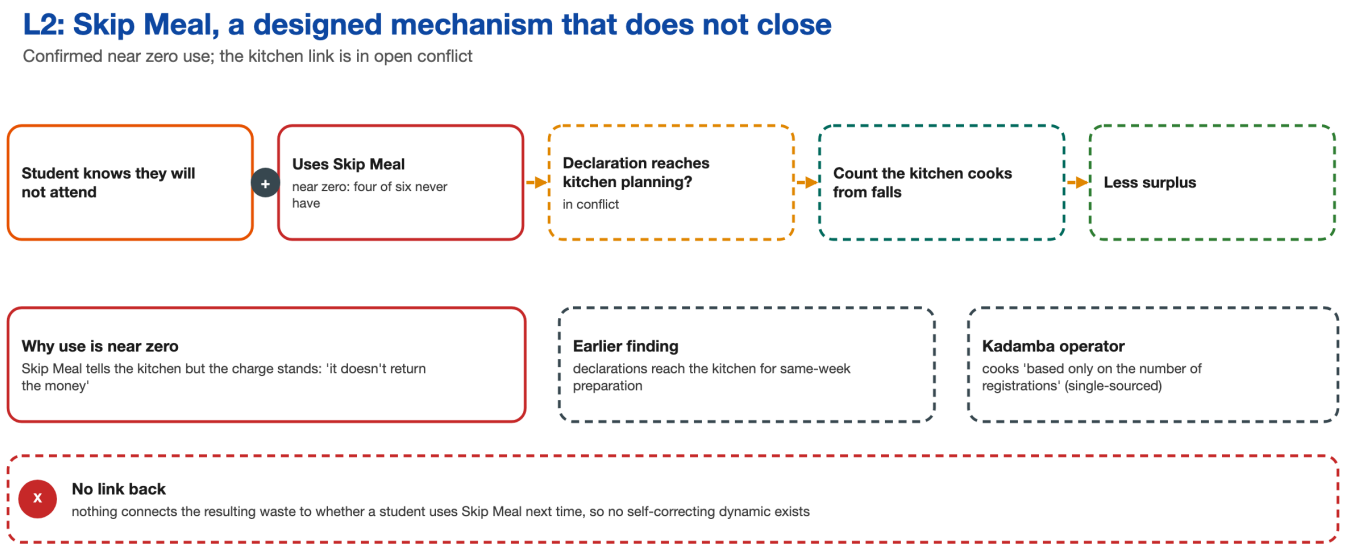


Figure 10. L2: Skip Meal. The student link is confirmed and near zero; the kitchen link is in open conflict.

The kitchen link is in open conflict. An earlier finding says declarations reach the kitchen for same-week preparation. All three operators plan from the portal registration count and none named Skip Meal as an input, and Kadamba's operator says the mess cooks ”based only on the number of registrations” and ”doesn't care if people don't want to eat” (single-sourced). The two fit together only if a declaration lowers the count the operator sees while the operator treats whatever count it sees as the obligation, which has not been checked. Even if the link holds, nothing connects the resulting waste back to whether a student uses Skip Meal next time, so no self-correcting dynamic exists to call a loop.

L5: Menu Transparency Effect

The rotation that closed L4 made the menu predictable, so a student can decide in advance not to come on a day they will dislike, a side effect the CFS Chair raised unprompted. Turnout does depend on the item: Yuktāhār's site head reports longer queues on aloo paratha and dosa days and 92% for Sunday-night paneer, Kadamba non-veg breakfasts ran at 48.5% against 32.4% for veg (both confirmed), and Bakul's manager says pav bhaji draws more students and muesli and Chocos go readily (single-sourced). Whether advance knowledge, rather than same-morning preference, causes skipping is untested. If the effect exists it flows forward into unused registrations on particular days, feeding L13.

L6: Shock-Adaptive Registration Relaxation

During an LPG shortage, and separately during some holidays and long weekends, the institution made registration non-compulsory. Students could walk in and scan, or register a day or two ahead, and were billed only for what they consumed. The trigger and the walk-in mechanism are confirmed, and Kadamba's operator describes cutting quantities and buying some items outside during the shock (single-sourced). The relaxation was episodic: students

say they stay auto-registered through ordinary holidays, so which holidays were relaxed is unconfirmed.

A first reading counted this as a self-limiting balancing loop in which relaxed billing raises procurement uncertainty and lowers institutional confidence in the model. Every link was assumed, and kitchens set cooked quantity the day before and buy dry goods that keep, so the first step is weak. The simpler reading fits better: the model reverts because the shortage or holiday ends, a calendar switch with no evidenced feedback. The candidate reasons it is never left on lie outside the kitchen, in whether per-plate payment counts registrations and in the convenience of an automatic default. Either way an attendance-tracked alternative has been built and has worked more than once. Yuktāhār's June Wastage Book pages add a candidate observation: two Saturday breakfasts with 37 and 34 registered record 33 and 31 diners, but those totals include unregistered and paying diners, and registered students alone numbered 22 and 21, about 60%, roughly double September's 28%. The regime in force in June is not known.

L7: Late-Night Workload to Skip to Crash

Staying up late is the best-evidenced single cause of skipping in the project: ”If I sleep around 3:00 or 4:00 a.m. or later, I don't tend to wake up. I prefer sleep over breakfast.” The records match. Sunday is Kadamba's lowest breakfast at 27.3% and among Yuktāhār's lowest at 17.9% to 24.1%, and Kadamba's median scan falls between 8:51 and 9:11 with the crest at close, consistent with late waking (confirmed, as a population pattern, since the export carries no student identifier). Yuktāhār's site head names hackathons and exams as reasons numbers fall. Bakul's manager gives two reasons, not waking and taste or menu. The first agrees with students, and the second is his hypothesis, which the six interviews rarely support and none of whom is known to live at Bakul. What follows a skipped breakfast, an energy crash, a compensating purchase or a change in that night's sleep, has never been asked, so the rest of the chain is untested. Late waking is also a candidate contributor to the closing crest.

Absent and Disconfirmed Mechanisms

A missing or disconfirmed loop is as much a finding as a working one.

L3, capacity redistribution, is strongly contraindicated. The plausible story, that scarcity at Kadamba pushes students to Bakul or Palash, runs against the evidence: scarcity makes students lock Kadamba down harder, as the respondent who registers all month says directly. Bakul's low registration is better explained by a standing preference gap. This rests on one quote corroborated by the aggregate pattern, hence ”strongly contraindicated” and not disconfirmed, and Bakul and Palash have no attendance records.

L8, vendor quality discipline, is unresolved. Student billing tracks registration (confirmed). The vendor's payment is a different link and the evidence on it conflicts. Bakul's manager says payment is a monthly lump sum with deductions, based on plates served, but ”served” is the synthesis's wording of one remark and may mean plates billed. Kadamba's ”they are getting paid for it” is a hint that payment follows registration, not a confirmation. Contracts may differ by operator and none has been seen. If vendors are paid per registered plate, falling attendance never touches vendor revenue and a fixed ratio turns part of the gap into margin. If per served plate, the vendor has a direct stake in turnout and the loop could close. Yuktāhār shows a third route: a vendor that ties waste reduction to profit gains from cutting waste whatever the plate basis. Operators already respond to item-level signals. Kadamba cut ragi idli and raised plain idli, Bakul switched paneer supplier twice after complaints, and Yuktāhār changes supplier when a grams-of-debris norm is exceeded (all single-sourced).

Quality control before service is a working balancing structure at Kadamba, on taste and safety, not quantity. Each meal has its cooking temperature checked, a sample kept for 72 hours, a wrapped display plate set at the entrance, serving temperature measured per item, and a tasting by the facility team with two Prism staff, then corrections (display plate and tasting confirmed as observed, the rest single-sourced). QC inspects weekly and water TDS is read daily. Nothing in the sequence tests first-batch size. Complaints run down a chain from CFS to CDS to Prism, an operations manager, a manager, a supervisor and the service staff, by director's email into a WhatsApp group at Kadamba and from the central dining team to the manager, the kitchen and the managing director at Bakul. Two different people are on record as head of CDS, and whether one name is wrong or they hold two roles is open.

L16, operator waste to menu choice, is absent. Operators see which items waste: at Kadamba uttapam, set dosa and upma consistently, while puri, bonda, vada and idli move (single-sourced). All three say students or the

committee decide the menu, and Bakul's manager is asking for input. He holds knowledge the menu process does not use, which items go (pav bhaji, muesli, Chocos) and which pairings work: kachori with chole, and Raj Kachori with aloo curry rather than the chana it is listed with, from about 20 years with North Indian food. That is his view of the students' reasoning, not theirs, and whether Bakul residents share it is untested. Decision rights sever the loop, since operators hold the data and the committee holds the choice. It is the counterpart of L4, which closes because complaints are loud and have a route while waste is silent and has none.

L10, the measured-but-unrouted gap, is a static condition and not a spiral. The gap is measured in at least three places: the CDS scan export (62.9% no-shows for Kadamba breakfast), Yuktāhār's register (roughly 72% of registered breakfasts unused), and Yuktāhār's other books, which are a Wastage Book, a wastage board, a grocery issue book, a store and ingredient book, a stock register, a vegetable book and indent, a milk, fruit, gas, electricity and water register, and fridge logs (confirmed). At Bakul the measurement is the manager's weekly review and may be unwritten. The CFS Chair's 35 to 40% breakfast figure is now confirmed for Kadamba at 37.1%. Predictable skipping is noticed and acted on in the quantity of food and never in the registration rule. The failure is a missing route, not a missing instrument, and the instrument is hard to route even where it exists, since Yuktāhār's records are on paper, digitised later by office staff. The chain from CFS through CDS to Prism runs downward, and whether any operator data runs back up it is unknown. Operators are data-rich and authority-poor: even information they need reaches them informally, as the site head says of events and exams, ”We don't come to know... until we sit in the meeting.” The condition stays static because the kitchen loops absorb each morning's consequence locally, and nothing tracks the gap's visibility to decision-makers over time, so there is no basis for calling it reinforcing.

Unintended Consequences

Several mechanisms are side effects of fixes introduced for other reasons, and two run in a helpful direction.

Rule or action	Intended effect	Unintended consequence	Status
Biweekly menu rotation (L4)	End menu fatigue	Students can plan to skip disliked days (L5); the matching-day Wastage Book lookup becomes possible (L11)	Lookup confirmed; skipping untested
Automatic registration, exit four days ahead	Procurement certainty	A default that runs while students are away (L13), feeding resale (L1); perishables are ordered the day before anyway, so the lock buys little	Rule confirmed; tiers single-sourced
Back-door entry after 9:30	Serve latecomers	May teach students the close is soft (L15)	Practice single-sourced; consequence candidate
Bakul's shortage floor	No shortage	More surplus on every low morning; a review that cannot correct below the floor (L11, L14)	Rule single-sourced; consequence candidate
Off-site cooking and hot holding (Bakul)	Hot, safe food after 8 km	Parathas soften; the late reserve is the food held longest (L12, L13)	One item single-sourced; consequence candidate
Fixed 70% batch with reactive top-up (Kadamba)	Avoid shortage	Surplus held, reheated and moved between vessels (L13)	Candidate
Food-safety sequence	Safety and taste	Fixes the first batch hours before any turnout signal	Plate and tasting confirmed
About 30 extra portions for outsiders; staff eat from the same food	Feed night security and staff	Surplus has beneficiaries, so calibrating the batch has losers (L9, L17)	Single-sourced; consequence candidate
Waste tracking at Yuktāhār	Reduce waste	Staff resistance that became ownership once weighing was blame-free	Self-report
Trying and dropping a paid ERP	Stay close to own data	Learning stays on paper, hard to route beyond the kitchen (L10)	Single-sourced
Skip Meal with no refund	Let students signal absence	Near-zero use, so the kitchen gets no signal (L2)	Confirmed

Structural Causes and Systemic Tensions

The three structural causes from the systemic problem analysis explain much of what closes and what does not, and one intermediate mechanism explains the kitchen half.

Registration is an automatic, mandatory standing booking with an exit by five early cancellations. It cannot follow the morning decision to eat, which keeps the count flat in L13 whatever the motive behind each booking, and which is why the problem is breakfast's. Student billing fires at the booking, not at consumption. That allows L1's cheap speculation, underlies L3b and L9, and is what L6 shows the institution can suspend; whether vendor payment follows the same rule is open (L8). The gap is measured but unrouted, because decision rights over menu, kitchen quantity, academic scheduling, registration and billing sit with different actors and none holds more than one or two. That fragmentation is why L4 can close, since menu has a real owner and an escalation path, while L10 and L16 cannot.

Beneath the kitchen half, three kitchens convert registrations into a first batch under three regimes, and every kitchen can correct only upward within a meal. Kadamba applies a fixed ratio with no learning record, and its gram standards fix portions, not quantities. Bakul adjusts weekly by judgement down to a floor and tops up by phone from 20 to 40 minutes away. Yuktāhār calibrates from its books and projects from the first half hour, yet anchors on a wrong belief that turnout is about half. This sits downstream of the three causes, inside authority the operator already holds. It explains why L13 produces surplus on low-turnout days, why L12 cannot remove it, and why L14's hedge can ratchet.

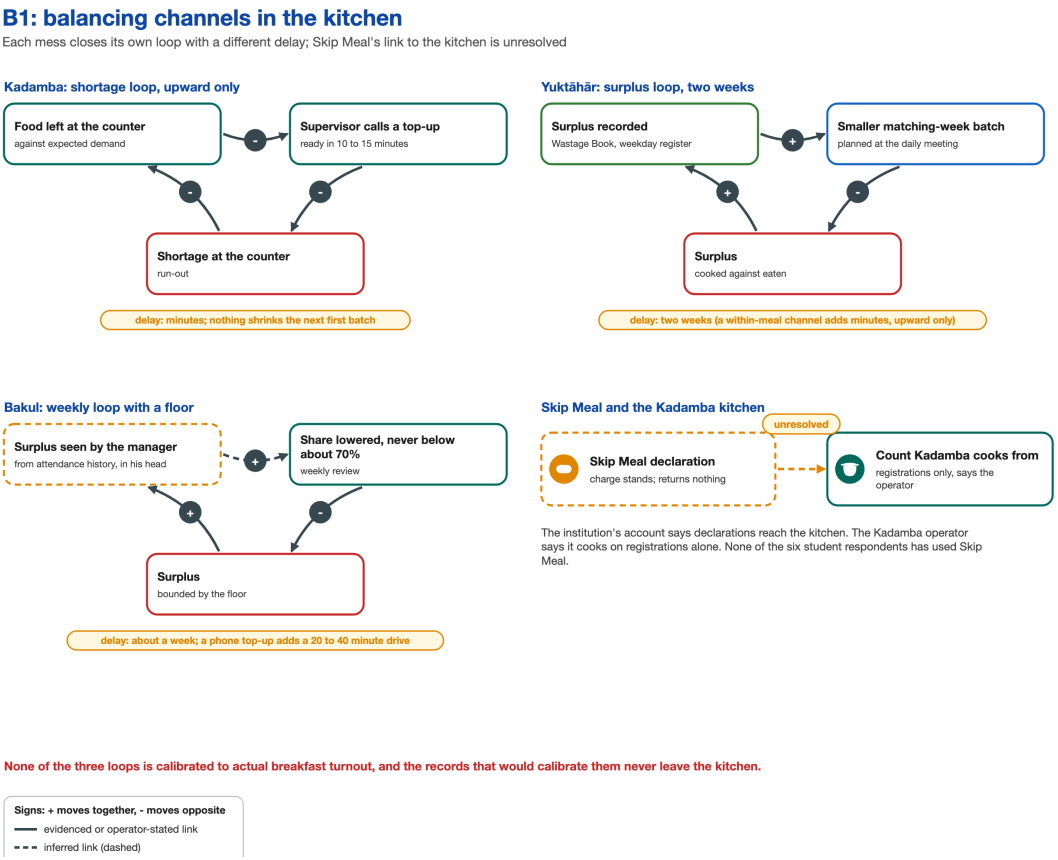


Figure 11. *The three kitchens close different loops with different delays.*

Delays set the timing. Raw surplus is largely recoverable, since Yuktāhār buys groceries fortnightly, vegetables two or three times a week and milk daily (confirmed), while cooked surplus is not. The binding delay is between the cooked first batch and the late crest, not the four-day cancellation lead. Yuktāhār's learning loop runs on a lag of one to two weeks and its culture change took about four months. The rest do not close for narrower reasons: L2 because Skip Meal pays the student nothing, L5 because item-level turnout is noticed only in the kitchen, L7 because nobody has asked what follows a skip. The evidence runs out at three points: the plate basis that decides L8, the absence of records for Bakul, Palash and for lunch and dinner at Kadamba, and the base of Kadamba's 5 to 10% waste figure.

Eight tensions follow, in three groups. Between timing and certainty: the four-day lead pulls against perishables ordered the day before, and operators themselves challenged the cut-off; avoiding shortage at close pulls against

avoiding waste after it (L14), a trade Bakul settles by rule for no shortage. Between quality and batch cooking: aloo paratha is best eaten at once yet may start around 3 AM for a 7 AM arrival, so holding keeps temperature and loses texture; Yuktāhār's head argues against batching because ”student fluctuation is very high” while cooked food lasts three to four hours; and at Kadamba food is held, reheated and transferred until it degrades (L13). Between who holds what: operators hold the data while others hold the authority (L10, L16), cutting waste pulls against feeding the people who eat the surplus (L9, L17), and one registration rule leads one operator to cook to the count as a paid obligation, another to learn waste away as lost profit and a third to accept waste as the price of never running short.

What This Means for the Reframe

Two kinds of closed loop are now demonstrated. L4 shows governance closing a loop end to end when a complaint is loud enough, and L11 and L12 show the kitchen closing loops every day, within the meal and across weeks. Neither touches registration. L1 and L6 fit a stable equilibrium but neither closes on the evidence, since L1's closing link is assumed and L6 is a calendar switch.

The steadier support for the equilibrium reading sits in two places. The first is the absorption layer: resale and informal transfer, the operator's discount and live top-up, reuse at Yuktāhār, outside diners and staff at Kadamba, held and reheated food (candidate), back-door service, outside venues, and finally the bin. The second is the asymmetry of attention. Shortages are loud and trigger fast correction, while surplus is silent, can ratchet the ratio upward (L14) and never reaches the menu owner (L16). As a candidate reading, Bakul's floor is the kitchen-level form of the same attention to what is loud, and the kitchen's competence at absorbing the gap is the plausible reason no signal accumulates upstream (L17). Under the same booking rule, operator culture sets how much of the gap becomes surplus, so the kitchen can learn, locally, and at Bakul only down to a floor. A precedent for solving the core problem exists in L6, and two designed corrective mechanisms do not connect back to the behaviour they were built to correct: L2, and L8 if vendors are paid per registered plate. Read together, the evidence supports organisational inertia around a problem that is measured, locally managed and previously solved at least as strongly as an actively self-reinforcing spiral.

What Remains Untested

Several links can be settled with one question or one record.

- The plate basis per operator, from the contract or CDS, which decides L8 and whether a fixed ratio turns the gap into vendor margin.
- Whether a Skip Meal declaration lowers the count Kadamba cooks against (L2).
- Whether any operator has ever changed a standing ratio, up after run-outs or down after surplus, which tests the L14 ratchet.
- The base of Kadamba's 5 to 10% waste figure and whether a record exists; whether the roughly 30 outside portions sit inside or on top of the 70% batch; how many people ”half the staff” is; and whether staff meals are an entitlement or a norm.
- Whether held surplus at Kadamba is reheated into later service, and whether reheated or softened food drives complaints or skipping, including whether holding changes how Bakul residents judge the food (L13's quality branches).
- Whether Yuktāhār uses its first-half-hour projection at breakfast and adjusts for the crest, whether its issue quantities fall over successive matching weeks, and which of its two 3 September diner records is right.
- Which registration regime applied in Yuktāhār's June breakfasts and which holidays were relaxed (L6); whether the cycle is weekly or monthly; and who heads CDS.
- Whether back-door entry happens at breakfast and shifts arrival later (L15).
- Whether Bakul's weekly review writes anything down, the real Hafizpet top-up lead time, whether the 9:20 reserve rests on data or judgement, Bakul breakfast scans to test the 60% estimate and the 8:00 to 8:30 rush, whether Palash shares the kitchen, and where Yuktāhār's unreused surplus finally goes.
- What follows a skipped breakfast (L7), whether advance knowledge of the menu causes skipping (L5), and why

Bakul residents skip, asked of them and not inferred from the manager.

Systemic Problem Analysis

IIIT Hyderabad Breakfast Mess System

Contents

Overview 1

1. Events, Patterns, Structures and Mental Models 2

Chain 1: Students register and don't show up 2
Chain 2: Food runs out before the mess closes 3
Chain 3: Complaints get ignored until something forces a response 5
Chain 4: A working fix already exists and gets switched off anyway 6
Chain 5: A real fix, and a shadow it cast 7
Chain 6: A meaningful slice of this isn't a problem at all 7
Chain 7: The kitchen cooks to a guess, then absorbs the difference 8

2. Structural Causes: What Keeps Repeating and Why 9

3. Reinforcing and Balancing Feedback Loops 11

4. System Archetypes 12

5. Unintended Consequences 13

6. Systemic Tensions 14

7. Power Asymmetries 15

8. Workarounds: What Students and Operators Have Built Around the System 16

9. Synthesis 16

Overview

Students at IIIT Hyderabad register for breakfast in large numbers and most of those registrations go uneaten. Kadamba took 32,162 breakfast registrations in April 2026 and scanned 11,941 at the counter: 37.1% turnout, 62.9% no-shows. Yuktāhār's September register is legible on thirteen days, and breakfast turnout on those days ran from 17.7 to 40.2%, about 28% on average, while lunch in the same hall ran from 56 to 89%. Both figures are confirmed from the halls' own records. Where lunch is recorded too, the gap belongs to breakfast.

The analysis works in layers. An event is one thing that happened. A pattern is what keeps happening across events. A structure is the rule or design that produces the pattern, and a mental model is the belief that keeps the structure standing. Read from the bottom, a belief shapes a rule, the rule produces a habit, and the habit shows up as an incident you can point to.

Claims carry four labels. Confirmed means read off a record, seen on site by a team member, or stated clearly by a named source. Single-sourced means one account. What the Kadamba operator told a team member who heard it directly is single-sourced, and the figure is not in doubt, though it remains the operator's own account of its practice. The Bakul manager's account reaches us through a team member's synthesis of the recordings, and no Bakul record of any kind has been seen, so every Bakul number is the manager's statement or estimate. Candidate is a reasonable reading not yet checked, and assumed is a systems judgement offered without proof.

Two findings carry the analysis. First, the institution has already run a version of breakfast where billing follows who shows up. During a gas shortage, and on holidays when students leave, registration stopped being compulsory, and a student who neither registered nor came was not charged. The model was switched off both times when the trigger passed, and nobody seems to have asked whether any of it should stay. Second, the gap is measured twice and reaches no one who can change the rules. The central dining system logs every scan against every registration. Yuktāhār's operator, ABC Hospitality Services, records per meal and per item how many registered, how many ate,

what was cooked, what was left and when each item ran out. Meanwhile three kitchens plan three ways, and in the two halls with records the breakfast gap is about the same size under opposite philosophies. The kitchen is rich in data and poor in authority. The offices that own registration and billing hold the authority and not the data.

1. Events, Patterns, Structures and Mental Models

Seven areas were traced from event up to belief. Each keeps its own layers, because the point is to see how many different rules and beliefs sit under one visible symptom.

The seven chains at a glance

One visible event in each chain, traced down to the belief that holds it in place

	Event	Pattern	Structure	Mental model	Root cause
1 Registered, not showing up	Sunday 6 Sept: 267 registered at Yuktāhār, 50 ate	Flat 1,070 a day at Kadamba; turnout 27 to 42%	Automatic default; five cancellations; billed at booking	"We have no opportunity to nullify it"	Billing tracks registration, not attendance
2 Food runs out at the close	Idli and puri gone by 9:20; Kadamba last scan 10:07	26.5% of scans in 9:15 to 9:30; Sunday latest	First batch fixed before any signal; top-up only upward	"The highest flow is in the last 10 minutes"	Late one-way top-up against a crest
3 Complaints ignored	Cap tightened Dec 2024; water complaints unheard in 2023	A loud problem is fixed, a quiet one is not	Menu, kitchen, calendar, billing with different owners; oversight on safety and taste only	Consultation is optional; workload is the reason	Nobody owns the whole picture; action follows loudness
4 A fix switched off	Gas shortage and holidays: no compulsory registration	Worked every time; reverted every time	No review exists; the cap has been moved before	An automatic count beats matching demand	The institution solved it twice and did not keep it
5 A fix and its shadow	Fortnightly menu rotation replaces a fixed semester menu	Fatigue fell; turnout moves with the item	Student Council, Parliament, Committee path worked	Menu variety is worth a policy change	Loud problems get fixed; visibility also enables avoidance
6 Not a problem at all	A student fasts on purpose	Late sleep, Sundays lowest, ceiling on variation	Default registers the absent and fasting as no-shows	Sleep outranks a booked breakfast	A real slice, too small to explain the gap
7 Cook to a guess, absorb the rest	Site head sees 18.7%, still says about 50%	70% batch is about 1.9 times turnout; surplus routes	Three ratio regimes; records stay in the kitchen	"We are paid for the registration"; "waste is money"	Cooked food cannot be shrunk; the booking is upstream

Figure 1. The seven chains at a glance: from a visible event down to the belief that holds it in place.

Chain 1: Students register and don't show up

Layer	What the evidence shows
Events	A student books Kadamba veg breakfast at Rs. 48 every day and skips most days; they cancel about five times a month and have never used Skip Meal because "it doesn't give any money back". Another books the whole month because Kadamba "fills up", and resells only when a slot fetches a good price. On Sunday 6 September 267 students were registered at Yuktāhār and 50 ate (18.7%). On Sunday 5 April 1,077 were registered at Kadamba and 273 scanned in (25.3%); only 20.5% of the vegetarian registrations that morning were used (confirmed).
Patterns	Registration beats attendance every day, far more at breakfast than lunch. Kadamba's April turnout was 37.1% (32.4% veg, 48.5% non-veg); Yuktāhār's September breakfast averaged about 28%, below the 35 to 45% band the CFS Chair gave. Registrations are flat across the week, about 1,070 a day at Kadamba with Sunday included, while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed). Flat registration is what a default looks like in the records. Bakul's manager estimates at most about 60% turn up (single-sourced, no records); if Bakul resembles the two recorded halls, that is an upper bound (candidate). Nobody interviewed has ever got anything from Skip Meal, and four of six use Mess Cell regularly.

Layer	What the evidence shows
Structures	Billing counts registrations, not eaters. Every student is registered automatically, on campus or not, holidays included, and the only exit is five cancellations per meal type per month, each at least four days ahead; beyond them the charge cannot be removed (confirmed from the dining poster and a team member's own experience as a registrant). The four-day close applies to registration. Kitchens fix quantities the day before. Walking in unregistered costs about 70% more (Rs. 81 against Rs. 48 for vegetarian), and walk-ins stay small: none to twelve a morning at Yuktāhār. Mess Cell lets a student sell a slot they will not use. Skip Meal tells the kitchen and pays the student nothing, and whether it changes what Kadamba cooks is open: the institution's account says it reaches the kitchen, while the Kadamba operator says it cooks on registrations alone (single-sourced).
Mental models	The mess office treats registration totals as a good enough stand-in for demand, and keeps billing that way although a working alternative has been run. It is the strongest belief in the analysis because it survives contrary evidence. Students hold that the booking is not theirs to manage: ”We don't have any opportunity to nullify it.” For one student, resale pays and Skip Meal does not, so Skip Meal is pointless. Kadamba's operator treats the count as the obligation it is paid for and cooks ”because it is a responsibility and they are getting paid for it”; Yuktāhār's treats waste as a cost and a profit lever (both single-sourced). Neither has a route to the registration rule.

The Kadamba manager also believes a friend often eats in an absent registrant's place (single-sourced). The scan export carries no student ID, so the records cannot confirm it.

Root cause. Billing tracks registration on a booking the student never actively makes. The counterfactual has happened: the CFS Chair says the no-show problem eased when billing followed attendance during the shortage and holidays. No attendance record from those periods has been seen, so the size of the effect is unmeasured. The same gap, 37% at Kadamba and about 28% at Yuktāhār, appears under opposite kitchen philosophies, which puts the cause upstream of the kitchen.

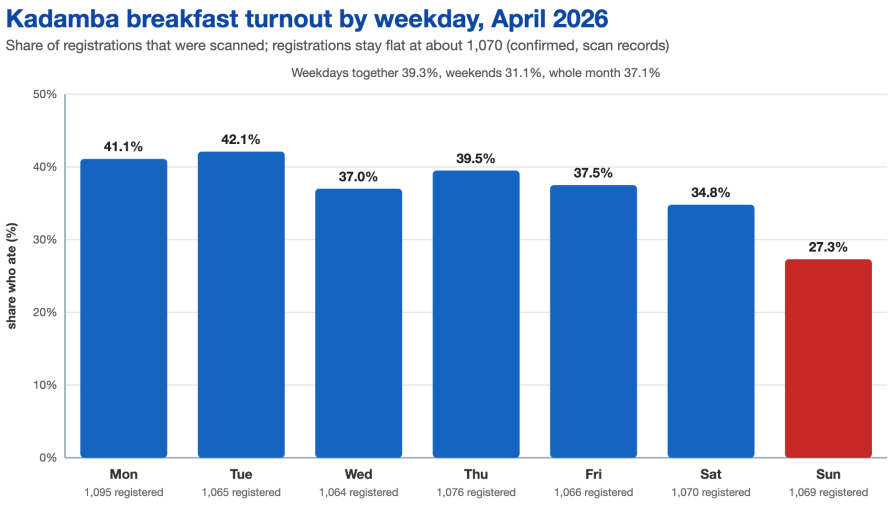


Figure 2. Kadamba breakfast turnout by weekday against flat registrations, April 2026.

Chain 2: Food runs out before the mess closes

Layer	What the evidence shows
Events	Around 9:20 most mornings idli, puri and bhatura are gone at Kadamba, fruit is not restocked, and staff promise more before the 9:30 close that usually does not come. On Thursday 3 September Yuktāhār re-cooked a kilogram of semiya upma at 9:00; that morning's production sheet shows 77 of 278 registered eating (28%) and the attendance register 110 (40%), two confirmed records that disagree. On Friday 3 April Kadamba's last breakfast scan was at 10:07.

Layer	What the evidence shows
Patterns	Crowding builds to one crest. At Kadamba in April 26.5% of scans fell between 9:15 and 9:30 and 13.5% between 9:00 and 9:15, with a small burst of 9.6% at 7:30; the rate reaches about nine a minute at 9:29 (confirmed). The Yuktāhār site head describes the same closing rush. Kadamba's operator puts it at about 100 diners early, 100 in the middle and 150 in the last ten minutes, which matches the shape. Bakul's manager describes a different curve, a main rush at 8:00 to 8:30 and 10 to 15 in the last ten minutes (single-sourced, no scans), so curves differ by hall. The crest is latest on Sundays, when 45.9% of Kadamba's scans come after 9:15 against about 32% on Tuesdays, and with no classes that day it is partly a sleep pattern (candidate). Every April day has scans at or after 9:30, 9.2% in all, about 37 a day; latecomers use the back door and ”mess can't say no to them” (single-sourced). One morning therefore holds a run-out at the counter and surplus in the pot.
Structures	The first batch is fixed before any turnout signal. Kadamba cooks 70% of registrations, veg and non-veg planned separately; Bakul 70 to 80% by item from its manager's weekly review; Yuktāhār estimates each meal from the matching week in its books (conversion regimes are compared in section 2). Top-up exists everywhere and only moves upward. Kadamba's supervisor tells the kitchen how much more to cook against gram standards, ready in 10 to 15 minutes. Bakul's manager checks about every thirty minutes and phones the Hafizpet kitchen; the drive alone is about 20 minutes early and 40 later, the total lead time is unstated, and he aims to hold food for about 50 at 9:20 against 10 to 15 late arrivals, a reserve that a top-up too slow for the crest would explain (candidate). Yuktāhār checks every thirty minutes from about 8:45 and projects the rest of the meal from the first half hour, which carried about 15% of Kadamba's April scans against 26.5% in the last fifteen minutes. Before service every Kadamba meal passes cooking temperature, a 72-hour sample, a wrapped display plate, serving temperature and a tasting with two Prism staff (plate and tasting observed, confirmed), so the batch must exist, cooked and checked, before any diner. Cooked food lasts three to four hours and a cooked batch cannot shrink. Bakul's labour-heavy items may start around 3 AM for a 7 AM arrival and sit in hot-water holding, which keeps temperature and softens aloo paratha (single-sourced). Portions are set per item: every Kadamba dish has a gram standard, and the items diners take above it (bonda, puri, vada, idli) are the ones students say run out first (candidate). Kadamba buys dry goods a week ahead, some vegetables two to three days ahead and perishables the day before.
Mental models	The Yuktāhār site head carries a breakfast figure of ”about 50%” (”at most 50%” on another occasion) against a September register of 18 to 40%, about 28 on average (confirmed); the operator with the best records carries a headline above them. Kadamba has ”no idea” of forecasting beyond ratio and gram standards: ”it was wasted many times”. The Bakul manager believes Kadamba sees 90 to 95% turnout, which Kadamba's own records contradict for breakfast. ”Student fluctuation is very high”, so Yuktāhār cooks once. Low Sundays and exam weeks ”cannot be helped”. Bakul's rule is ”there should not be a shortage of food, even if some wastage occurs”. Kadamba's operator reads the crest correctly: ”the highest flow is in the last 10 minutes.”

Root cause. The kitchen can react to a same-day spike, but only upward and late: ten to fifteen minutes on site at Kadamba against a crest of nine scans a minute, and a call, a cook and a drive at Bakul. Students' reports that the restock never arrives fit a top-up that lands after the crest. Yuktāhār's first-half-hour projection is the one forward correction and it samples the calm opening. The same morning therefore produces a shortage at the counter and a surplus in the pot.

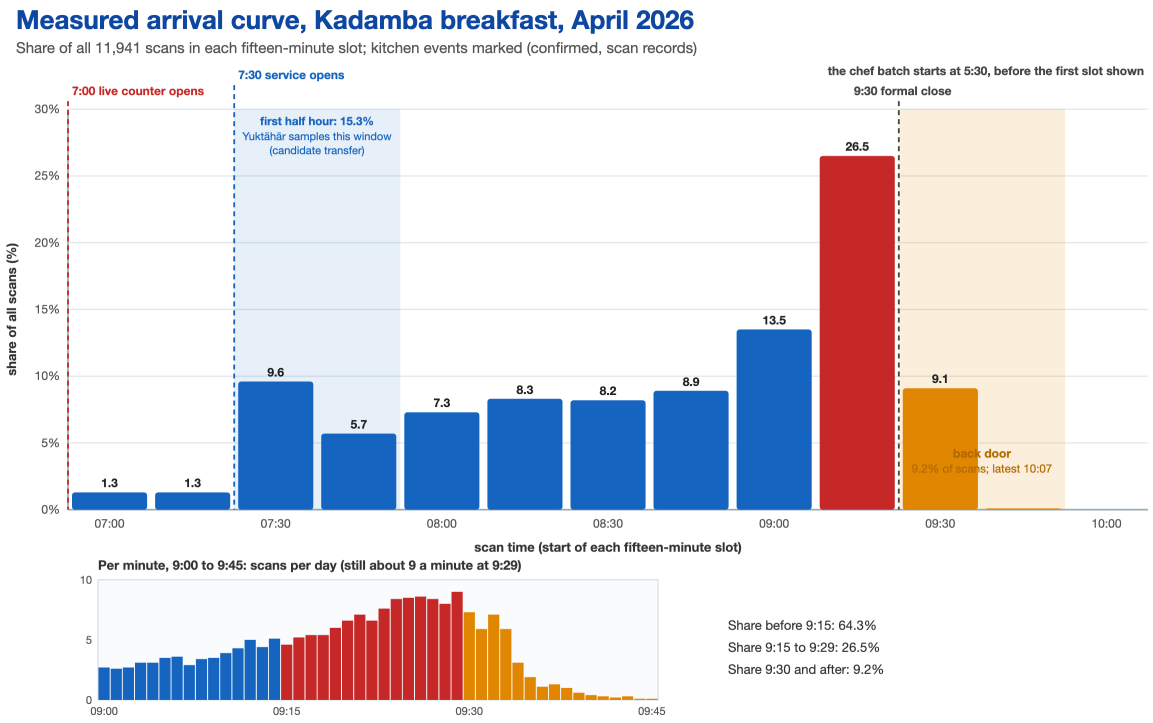


Figure 3. *Measured arrival curve, Kadamba breakfast, April 2026.*

Chain 3: Complaints get ignored until something forces a response

Layer	What the evidence shows
Events	In November 2024 a frog was reported in Kadamba's chicken biryani and non-veg meals were pulled without an official reason. In December 2024 the cancellation cap was tightened after a Kadamba renovation added to Mess Office workload, with no record of committee sign-off; a Ping survey found most respondents unhappy, and Parliament representatives met the Mess Office twice before it went ahead anyway. In 2023 complaints about drinking water went unanswered for months until more than 38 confirmed typhoid cases forced a response. In August 2025 Yuktāhār staff feared blame when waste weighing began (”they are catching us”), and by December they reported waste had fallen (confirmed). In September 2026 a Kadamba student complained that dinner sambar had been dropped ”while they are taking more money from us”, and the operator answered that students or the committee decide the menu.
Patterns	A complaint sits until a larger incident forces action. When the Mess Office wants a quick change it tends to act alone. Complaints reach the operator, who points back to whoever owns the rule.
Structures	No confirmed rule requires committee approval before an operational change. Menu, kitchen execution, calendar and billing sit with different owners. Operators see which items waste (uttapam, set dosa and upma at Kadamba) and which move (puri, bonda, vada, idli) and do not choose the menu; the Bakul manager asks for a say, and his objection is to pairings, not dishes (kachori belongs with chole), untested with Bakul residents. Oversight of Kadamba runs CFS, CDS, Prism, operations manager, manager, supervisor, service staff, and covers safety and taste closely (daily tasting, weekly inspection, daily water reading) while no check asks how much was cooked against how many ate. The operator and an earlier institutional interview name different people as head of CDS, and the facility team's identity is also open. Complaints travel to the kitchen, never to the rule owner: a director's email to the manager and a WhatsApp group at Kadamba, the central dining team to the manager, kitchen and managing director at Bakul. Three student channels (in-app rating, email, Ping) have no confirmed case of changing registration, billing or hours. Change happens where the measurer also holds the lever: Yuktāhār is flat, with two project managers and peer cross-checking, and Kadamba cut ragi idli and raised plain idli from what it saw at the counter.
Mental models	The Mess Office appears to see these decisions as its own, with consultation optional under time pressure, and its stated reason for the December tightening was workload. At Yuktāhār waste is something to learn from, not punish, and staff treat each other ”like family”.

Root cause. Nobody owns the whole picture, and at the level of registration, billing and menu the system reacts only when something is loud. Oversight follows the same logic: intense on safety and taste, where failure is loud, and absent on quantity, where surplus is quiet. The kitchen mirrors it. At Bakul a run-out has a route to the managing

director and a surplus has none, and the operator's rule is to avoid the first at the cost of the second (candidate reading as the kitchen's answer to loudness). Inside a kitchen quiet numbers already work, because the person measuring also decides.

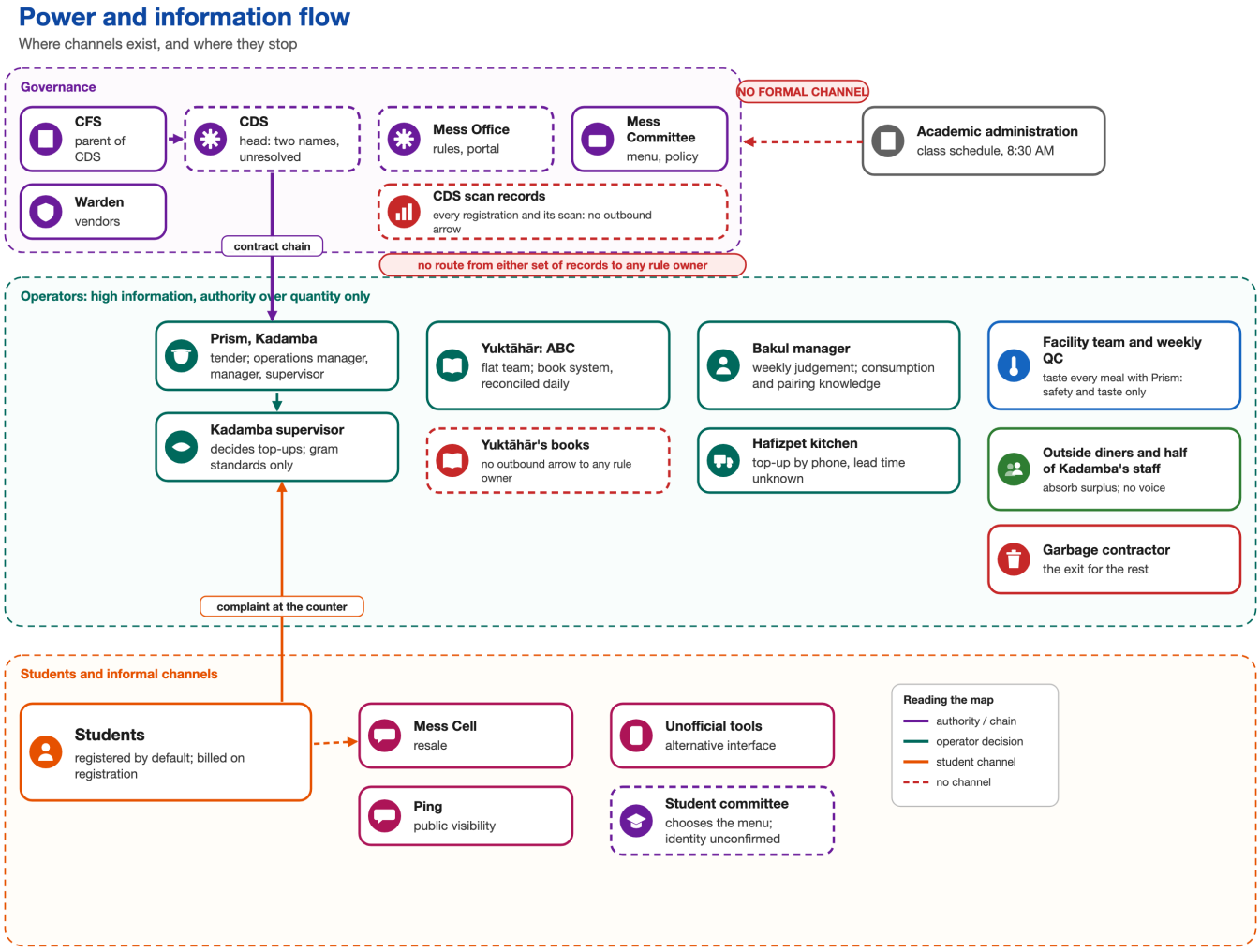


Figure 4. Power and information flow: where channels exist and where they stop.

Chain 4: A working fix already exists and gets switched off anyway

Layer	What the evidence shows
Events	During a gas shortage the mess cut to one item at the same price and the cancellation cap doubled from five to ten a month; Kadamba says it cut quantities and bought some items outside (single-sourced). On holidays and long weekends, Felicity being the named example, registration stopped being required: students could walk in and scan, or register a day or two ahead at any of the four messes, and anyone who neither registered nor came was not billed.
Patterns	By the CFS Chair's account it worked every time, and every time it was switched back off when the shortage or holiday ended. There is no sign of any review of keeping part of it.
Structures	The kitchen's own horizon for anything perishable is one day. Quantities are fixed in a meeting the day before at Yuktāhār, and Kadamba cooks a fixed share and tops up. Kadamba also cooks about 30 extra portions for 30 to 50 people outside the registration, such as night-shift security; an earlier noisy figure of 30 to 40 extra portions for paying walk-ins may be the same allocation, unconfirmed. Nothing exists to judge whether a temporary fix should become standing, and the cancellation cap is a number the institution has already shown it will move.
Mental models	Someone in registration and billing values an automatic, pre-committed count over matching food to demand, as a default and not a case-by-case judgement (candidate). Administrative workload is the likeliest source. Attendance-based billing seems acceptable only as an emergency measure.

Root cause. The counterexample to chain one: the institution has solved the student half twice and chose not to keep it. The kitchens do not need the four-day close. Several reasons remain unchecked. The per-plate payment may count registrations, so a walk-in model would cost the operator revenue (section 2 sets out the contested evidence). An automatic default is less administrative work. Or the relaxed model simply reverts with the calendar and nobody decides anything.

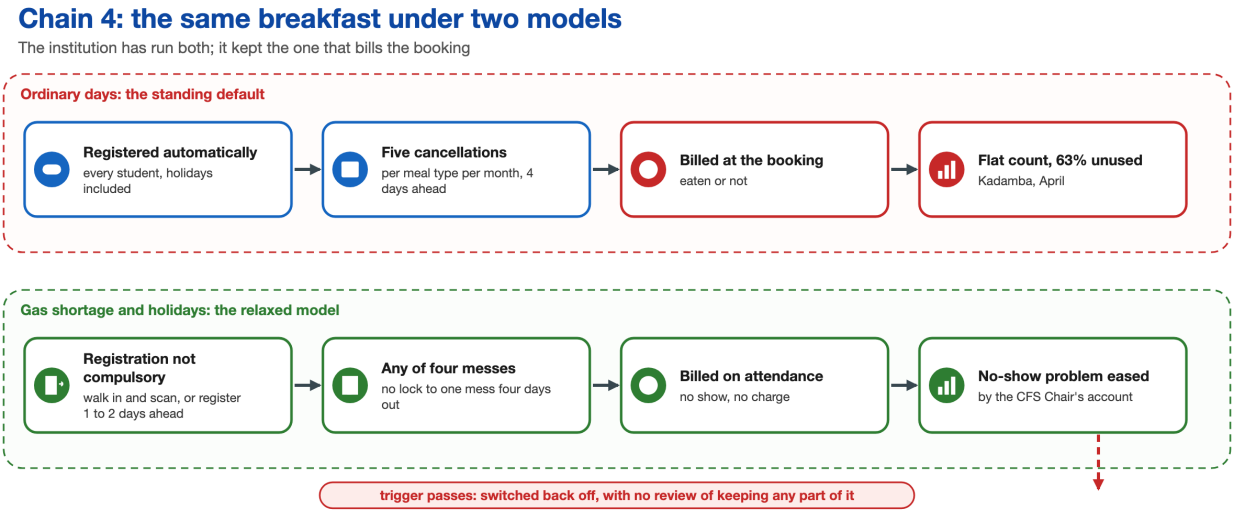


Figure 5. The same breakfast under the standing default and under the relaxed shock model.

Chain 5: A real fix, and a shadow it cast

Layer	What the evidence shows
Events	CDS convinced the Student Council, which convinced the Student Parliament, to replace one semester-long menu with a fortnightly rotation. Unprompted, the CFS Chair worried that a menu known in advance lets students decide not to come on days they dislike. On Thursday 3 September Yuktāhār planned breakfast from a sheet headed ”Thu wk 1&3”: the Thursday of weeks one and three share a menu and a sheet (confirmed).
Patterns	Menu-fatigue complaints dropped after the rotation, by the CFS Chair's account. Turnout moves with the item: non-veg registrations at Kadamba were used at 48.5% against 32.4% for veg (confirmed), biryani days reach about 700 registrations, and some items waste consistently while others run out. Bakul's manager names pav bhaji as a draw and muesli and Chocos as items that go. Whether students skip known bad-menu days has not been checked. Where a kitchen expects full turnout it drops its discount: Kadamba cooks biryani lunch at 100% of registrations, two pieces of 150 to 200 g with unlimited rice, and when some diners take a third piece later ones run short (lunch, single-sourced).
Structures	A real path runs from student complaint to policy: Student Council, Parliament, Committee approval, and this time it worked. Feedback is sorted by loudness, with low-intensity items going to the CFS head and high-intensity ones escalating. The menu is posted two weeks ahead and also changes monthly and seasonally, and the two-week cycle is what lets Yuktāhār plan each meal from the matching day in its Wastage Book.
Mental models	CDS believed menu variety was worth a policy change and acted on it, and is now questioning its own earlier belief that more menu transparency is automatically good. In the Bakul manager's account students choose combinations for balance, protein and preference while he reasons from traditional pairings and what is eaten, after about 20 years with North Indian food; that is his account of their reasoning, and whether familiar pairings matter to residents is untested.

Root cause. The opposite of chain three: proof that the system responds when a problem is loud enough. The new risk comes from the same structure that made the fix possible, since visibility that cures fatigue also enables avoidance. The rotation also gave kitchens a repeating unit to calibrate against, which a menu change breaking the two-week cycle would remove.

Chain 6: A meaningful slice of this isn't a problem at all

Layer	What the evidence shows
Events	One student fasts on purpose and stopped cooking breakfast years ago. Another says a friend's absence would not change whether they go, and a third says friends are a reason to go but never the deciding one.
Patterns	Late sleep is the reason given most often, ahead of class timing or food quality, and the records agree at pattern level: Sunday is the lowest day in both halls (27.3% at Kadamba, 17.9 to 24.1% on Yuktāhār's three legible Sundays) and has the latest arrivals (confirmed). The Yuktāhār site head adds that hackathons and exams push numbers down. Bakul's manager describes weekends as very low and gives two reasons, students not waking and students disliking the taste or menu; the first matches what students say and the second is his own hypothesis, since the six students interviewed rarely name food quality and none is known to live in Bakul. Legitimate variation has a ceiling: Kadamba's daily turnout ranged from about 25 to 48% and never neared lunch levels, so weekday and weekend variation cannot explain a 63% no-show rate (candidate).
Structures	Breakfast draws less company than lunch and sits outside a student's daily path. Only Yuktāhār offers a Jain line and counts it separately, about eight to fifteen Jain registrations per breakfast in September. The system registers everyone by default, so a legitimate absence, a fast or a trip, is counted as a no-show.
Mental models	For at least one student sleep outranks a booked breakfast whenever the two clash. The Bakul manager believes students skip because they may not wake or may not like the taste. A meal you have stopped cooking stops feeling like something you miss. The system assumes everyone should default into breakfast.

Root cause. This slice sits outside the billing and governance story. Some of what looks like the paradox is how a varied student body eats, and treating all of it as dysfunction would be a mistake. It is also too small to explain the gap.

Chain 7: The kitchen cooks to a guess, then absorbs the difference

Layer	What the evidence shows
Events	Shown the 18.7% Sunday row, the Yuktāhār site head said the kitchen would adjust ”not even next month, next week” (confirmed). Shown his own register, he still called breakfast ”about 50%”. On one biryani day 25 students arrived after close and the door was reopened (single-sourced; breakfast or lunch is unclear).
Patterns	The fixed ratio sits well above recorded turnout. Kadamba's 70% first batch against 37% actual is about 1.9 times what is eaten before any top-up. At Bakul the 70 to 80% share and the weekend floor of about 70% both exceed the highest turnout the manager himself expects, about 60%, so over-preparation is built in on his own numbers (candidate, no record to size it). Surplus leaves by several routes, none counted as institutional waste. Yuktāhār reuses it: leftover idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge and surplus milk becomes curd (confirmed). At Kadamba about 30 extra portions feed 30 to 50 outsiders and about half the operator's own staff eat from the same food (single-sourced); whether the outsiders appear in any record the institution sees is unknown. At Bakul some leftovers return to Hafizpet and the rest is thrown. The rest goes to the bin. At Kadamba the team's observer judged that food is cooked ahead, reheated and moved between vessels until it degrades (candidate), so over-preparation costs quality as well as waste; at Bakul the equivalent cost is softening in transit. Waste is measured daily inside the kitchen: Yuktāhār weighs plate and kitchen waste per meal and logs percentage waste per item, and Kadamba reports about 5 to 10% daily with no stated base. In April Kadamba averaged about 1,072 registrations and 398 scans a day, so a 70% batch is about 750 portions and a surplus near 350 before any top-up (candidate arithmetic). A 5 to 10% waste figure fits that only if outsiders and staff, reheating, smaller top-ups or a different measurement base absorb most of it, and which is not known.
Structures	Yuktāhār plans at a twice-daily meeting from portal registrations (regular plus Jain plus cash meals), the Wastage Book entry for the matching day, the weekday attendance register, the month's pattern, and events, exams and holidays only if someone raises them (”We don't come to know until we sit in the meeting”). The Wastage Book works as a rule: ”If it runs out at 1:30 we re-cook; if at 2:20 we don't.” It runs a full set of paper books (Wastage Book, attendance register, wastage board, grocery issue book, store book, stock register, vegetable book and indent, utilities register, fridge logs, confirmed), reconciled by a two-book rule that the store's issues must match the kitchen's cooking, and a mismatch goes to the daily meeting. Kadamba sets 70% of registrations, veg and non-veg separately, with gram standards and a supervisor calling top-ups, and keeps no learning record. Bakul's manager lowers a share after surplus only down to his floor, and whether the weekly review rests on anything written is unknown. Operator records stay in the kitchen: nothing routes the Wastage Book or attendance register to whoever sets registration, billing or menu rules, and nothing routes the central scan records back to the kitchen. Kadamba's waste goes to a dustbin collected by a garbage contractor, with no composting.
Mental models	”Breakfast is about 50%” at the best-recorded mess, and ”at most about 60%” at Bakul with no record to test it. Three stances under one booking rule: Kadamba cooks to registrations as the job it is paid for, Yuktāhār treats waste as money and spends part of its margin on rock salt, cold-pressed oil and mill visits, and Bakul treats a shortage as the failure to avoid and waste as its accepted price. ”We don't have that idea” at Kadamba. At Yuktāhār, ”Prediction will be based on our data only... I want to remain close to the data”: it tried a commercial ERP and dropped it, since paying lakhs of rupees for prediction made no sense when the data sits in its books and its staff are not digitally inclined.

Root cause. Once cooked, the conversion from booking to plate cannot be reversed, and no kitchen can change the

number it starts from. In the two halls with records the breakfast gap is about as large under opposite philosophies, because the cause sits in the booking: calibration cuts waste and run-outs, not no-shows. The kitchen measures the gap in more detail than anyone and controls only quantities.

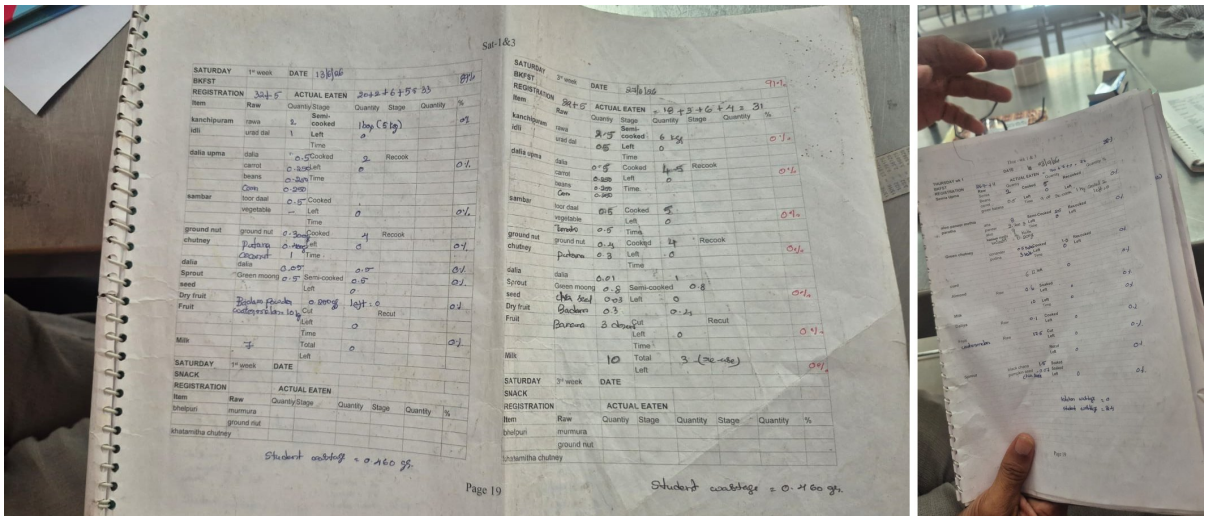


Figure 6. Yuktāhār's production and waste sheets for breakfast: issued, cooked, left, run-out time, re-cook and plate waste.

2. Structural Causes: What Keeps Repeating and Why

Across the seven chains three structural causes recur, and the problem splits into two halves: students not showing up for what they booked, and kitchens cooking more than is eaten.

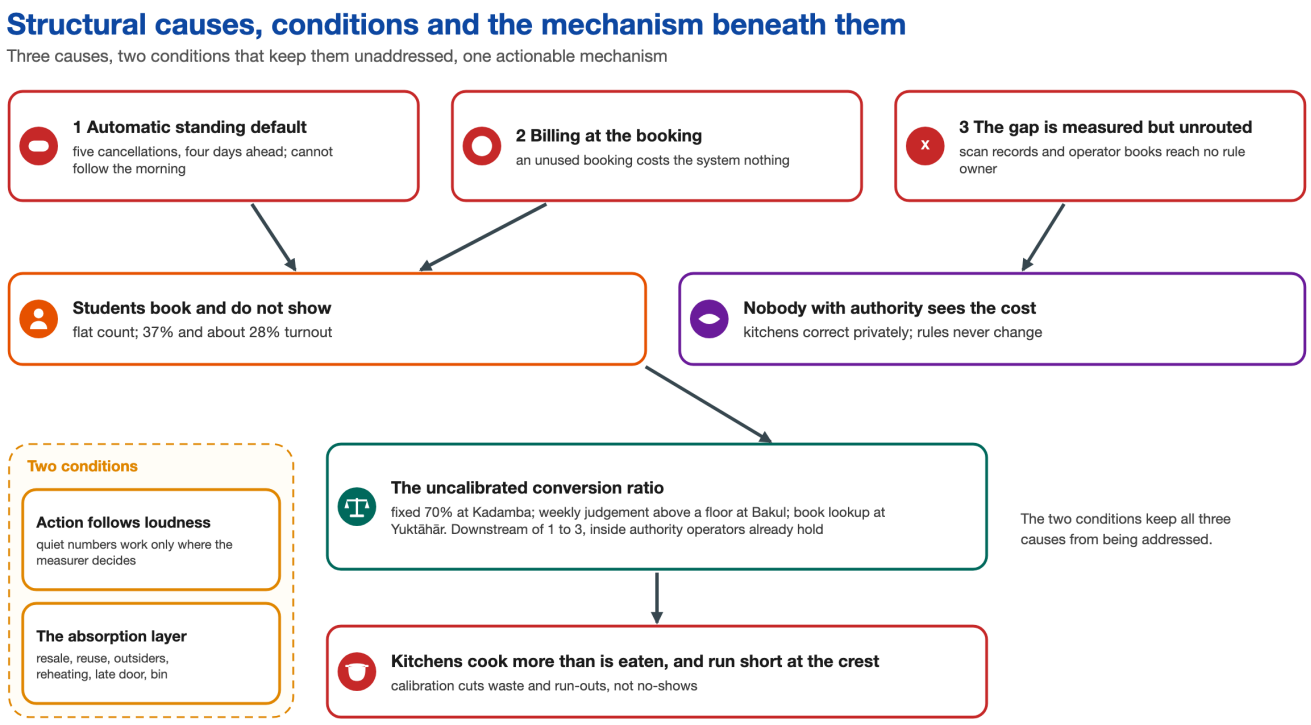


Figure 7. The structural causes, the two conditions that keep them unaddressed, and the conversion ratio beneath them.

Registration is an automatic standing default. With a narrow, early exit it cannot follow the decision each student

makes on the morning, and the records show its signature directly in the flat count. It also makes the problem breakfast's. Lunch runs under the same booking and mostly works, because the lunch decision is steadier.

Billing fires at the booking, not at consumption. This single rule explains the incentive to book broadly, why Skip Meal goes unused, why a cancellation cap is needed and why Mess Cell exists. It has been tested: billing followed attendance during the shortage and the holidays, and the no-show problem eased by the CFS Chair's account. It does not explain why kitchens cook above turnout.

The gap is measured but unrouted. Menu, kitchen quantity, calendar, registration and billing each sit with a different owner. The central system holds a timestamped scan for every registration, Yuktāhār holds per-item turnout and waste, and neither reaches a rule owner. The institution shows no sign of knowing the operator holds this data, and oversight of the kitchen checks safety and taste every day and never quantity. That is also why a fix that worked twice was never evaluated for keeping.

Two conditions keep all three unaddressed. **Action follows loudness at the institution level, while inside the kitchen quiet numbers already work.** Menu fatigue was loud and got fixed. The much larger registration gap is quiet and never travelled anywhere. At Yuktāhār the opposite holds: its own records cut waste between August and December with no complaint involved, because the measurer also decides and the measuring was made blame-free. The same logic shows on the operator side, where a Bakul shortage travels up to the managing director and a surplus goes nowhere. The **absorption layer** is the second condition. Resale and informal transfer, the operator's discount and live top-up, leftover reuse, outside diners and operator staff, held and reheated food (candidate), the walk-in buffer, late and back-door service, outside venues and finally the bin each take up a consequence of the gap before it becomes a signal anyone with authority sees.

Beneath the kitchen half sits one mechanism, **the uncalibrated conversion ratio**, which runs in three regimes.

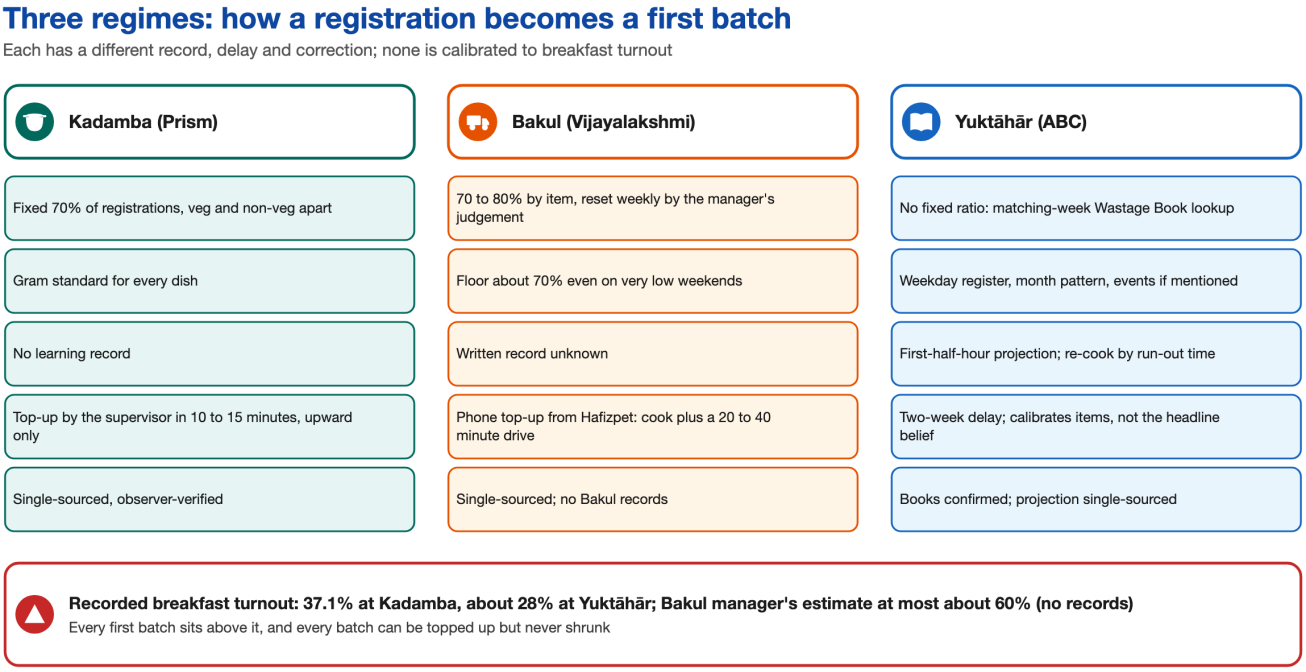


Figure 8. The three regimes that turn a registration into a first batch.

It is downstream of the booking and of the unrouted measurement, so it is not a root cause. It is the most actionable structure in the evidence because it sits inside authority operators already hold, one already calibrates and a second already adjusts weekly.

Each operator is paid monthly as a lump sum on plates, with deductions, and what a plate counts is contested. The Bakul manager says payment is on plates served (single-sourced, from one remark in the synthesis), while the

Kadamba operator's remark that it is paid for the registrations points the other way (candidate, not a contract term). Contracts may differ, one remark may not describe the formula, and no contract has been seen, so it is unknown for any operator whether turnout reaches its revenue. If Bakul is paid on served plates it bears the cost of its own over-cooking and accepts it, which fits a judgement that a shortage costs more than waste (candidate).

3. Reinforcing and Balancing Feedback Loops

Eighteen mechanisms were checked, numbered L1 to L17 plus L3b, the scheme used in the causal loop analysis. A loop counts only if the last step feeds back into the first and is evidenced. Three close, all balancing. Five close only through an assumed last step and are held as candidates. The rest stop partway and calling them loops would mislead.

Balancing loops that close. L4 is menu fatigue and rotation: fatigue drove feedback, feedback escalated through the Student Council, the Committee approved a rotation and variety rose, which lowers fatigue again; every step is confirmed by the CFS Chair and the operators' records. L11 is matching-day calibration at Yuktāhār: the Wastage Book entry for the matching day, with the month's pattern and events, sets the next quantity and turnout estimate, which produces a new leftover to record. Staff first feared blame when weighing began in August 2025, the managers made it blame-free, and the book was still filled in daily in September 2026. It calibrates item quantities, and the site head's headline figure stays above his register, so the headline belief does not drive planning (candidate). L11 is absent at Kadamba, which keeps gram standards and no learning record, and Bakul is a candidate second instance, bounded by its floor. L12 is in-meal top-up: the counter runs low, someone checks what is left against the time, more is cooked. It is confirmed at Yuktāhār, single-sourced at Kadamba and at Bakul, where the phone call and drive make it the clearest balancing loop with a delay. Its weakness is that it only moves upward and lags a crest.

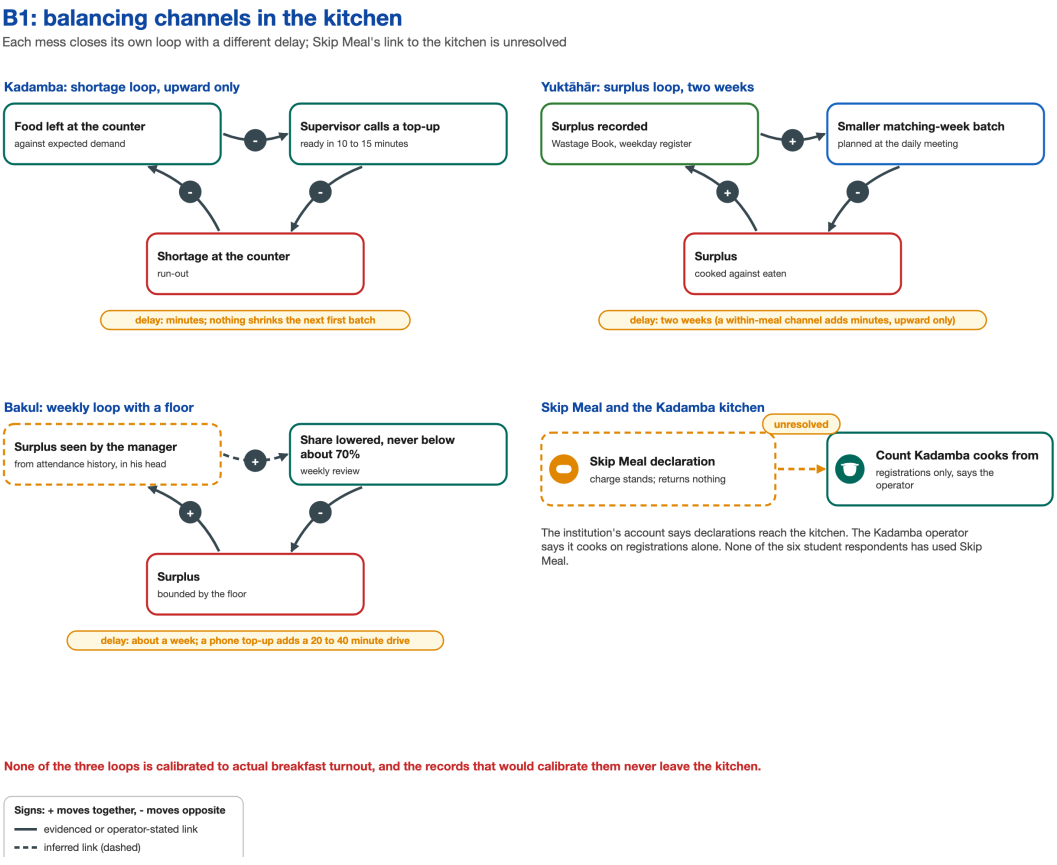


Figure 9. B1: how the three kitchens close their loops, with their different delays.

Candidate loops with an assumed last step. L1, loose registration and resale: registering loosely feeds resale, resale softens the cost of an unused registration, and that probably makes loose registration comfortable again;

most links are confirmed by direct quotes and the last is assumed, and the automatic default now explains flat registration without it. L3b, Kadamba scarcity: popularity drives defensive registration, which keeps it scarce on the platform and reinforces popularity, though at breakfast the scarcity is on the platform and not in the seats, with 1,070 registrations a day against a capacity of 1,200 and only 37% used. L9, surplus absorbed out of sight: reuse, outside diners and staff, held food, late service and the bin keep surplus from becoming pressure on whoever sets registration rules, and the closing step is assumed. L15, late-door tolerance: serving latecomers may teach students that arriving late is fine, which would keep the crest late; the first link is well evidenced and the closing link is assumed. L17, kitchen competence shifting the burden: the operators' local clearing of each morning's consequence plausibly stops any signal accumulating upstream, observed in the first step and assumed in the second.

Chains that look like loops and are not. L13, standing booking to over-preparation, is the core chain of the paradox: an automatic default keeps the count flat, Kadamba cooks a fixed share, over-preparation is worst where turnout is lowest, and nothing runs from the surplus back to the booking. L14, raising the ratio after a run-out, has only its first link evidenced, run-outs at the Kadamba crest, and Bakul's no-shortage rule supplies the premise as a stance and not an observed drift. A chain from surplus to reheated food to quality to skipping has no evidence for its last link, and Bakul has a distance counterpart in softening food. L6, the shock exception, is a calendar switch with no evidenced step feeding it back into its own reversal. L2, Skip Meal, runs one way and almost nobody uses it. L5, menu transparency, is a side effect of chain five. L7, late nights to skipping, has a confirmed first link and an untested rest. L8, vendor discipline, is unresolved until the plate basis is known. L16, operator waste to menu choice, is absent. L3, redistribution to Bakul or Palash, is contradicted: people lock down Kadamba harder when it is scarce. L10 is a stable stuck state and not a spiral.

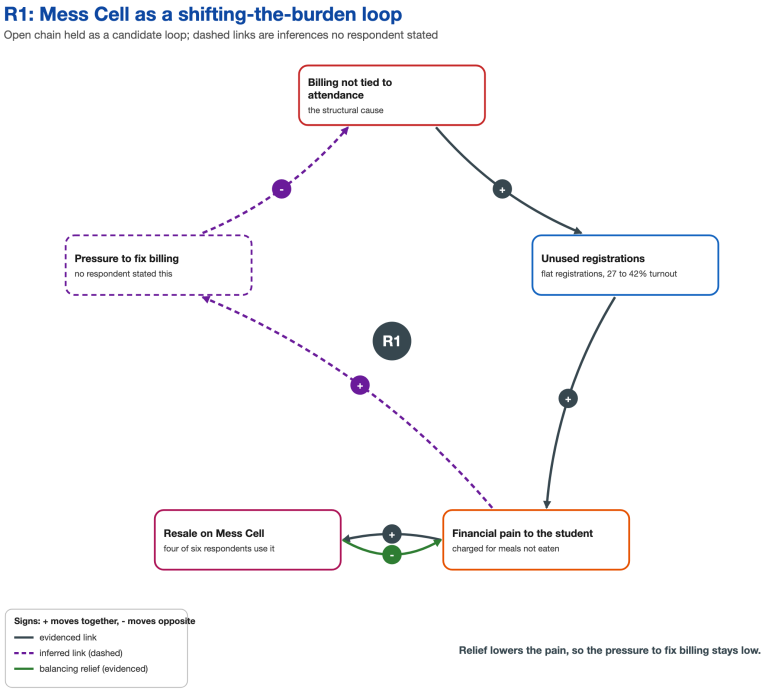


Figure 10. L1: Mess Cell as a shifting-the-burden loop; dashed links are inferences no respondent stated.

4. System Archetypes

Eleven standard archetypes were checked against the system.

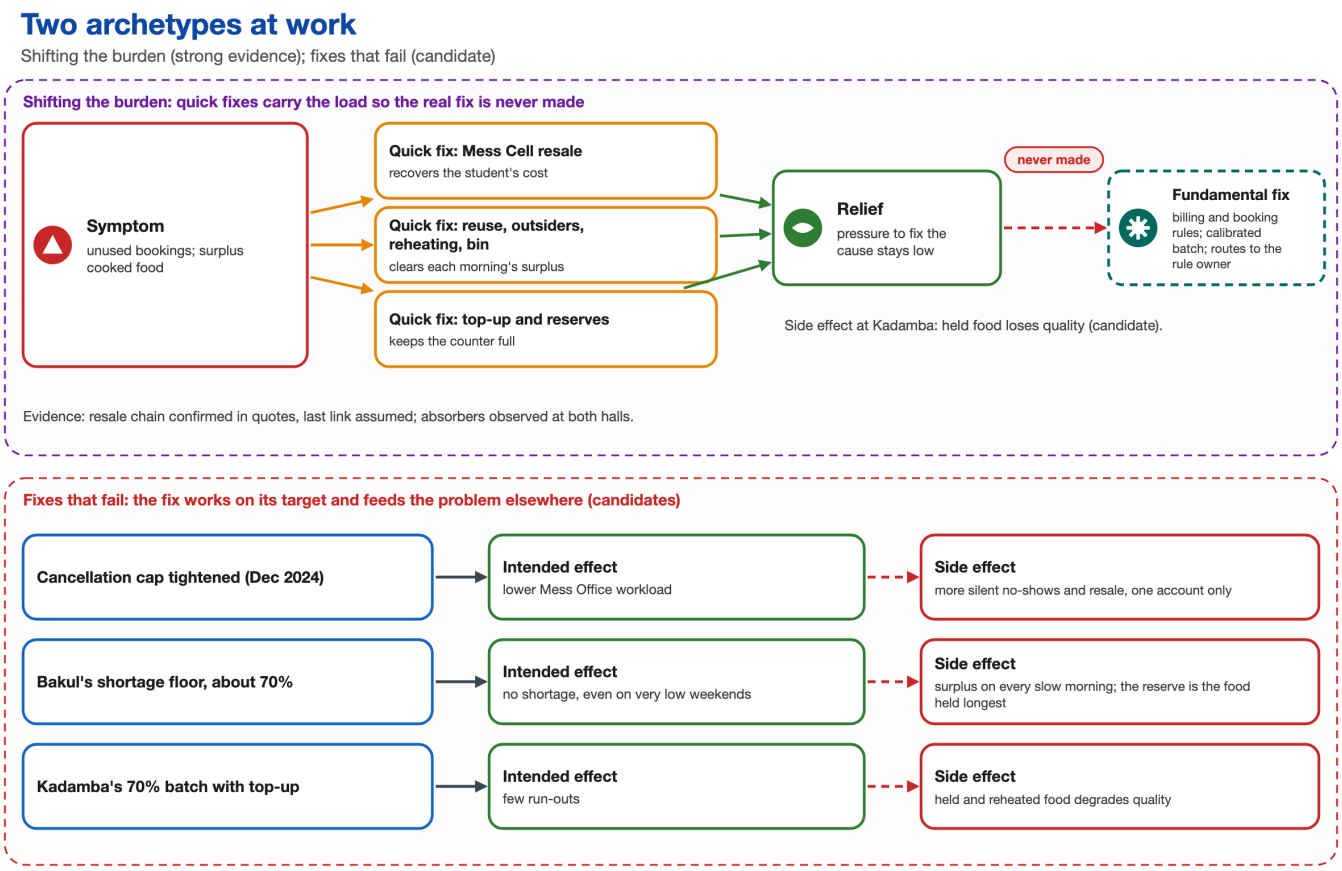


Figure 11. Two archetypes at work: shifting the burden, and fixes that fail.

Five hold up. **A quick fix standing in for a real one** appears twice: Mess Cell resale stands in for fixing billing, and reuse, outside diners and staff, held food and the bin stand in for calibrating the first batch. At Kadamba the substitute may carry a side effect, food held and reheated through service losing quality (candidate). **Growth hitting a ceiling** fits Kadamba's popularity, with Hafizpet's kitchen a candidate ceiling for Bakul and Palash. **Success to the successful** fits Kadamba's scarcity feeding defensive registration while Bakul stays underfilled. **A rule defeated by its own use** fits registering broadly to guard access: a Kadamba slot that is hard to get is used 37% of the time. Two more are real possibilities with too little evidence. **Fixes that fail** appears three times as a candidate: the December cap tightening may have worsened no-shows, though one account without before and after numbers supports it; Bakul's shortage floor solves shortage and feeds surplus on every slow morning; and Kadamba's 70% batch with top-up avoids run-outs and feeds reheated food. Four archetypes do not fit: access to meals is already limited by registration and a priced walk-in allowance, no two parties are shown escalating against each other, no stated target was quietly lowered, and every accidental-adversary candidate turned out to be one office's own trade-off.

5. Unintended Consequences

Billing on registration was meant to give the kitchen a stable number. It produced registrations out of habit with no cost to skipping, and a system that cannot tell a real no-show from someone defaulted into a meal. Automatic registration made the count flat, so the count carries nothing about the weekday pattern in who eats, and Kadamba does not rebuild it. The cancellation cap, meant to limit Mess Office workload, leaves students who exceed it or decide inside four days with no legitimate correction, which likely pushes them to silent no-shows or Mess Cell. Skip Meal, meant to cut preparation for a known no-show, gives the student nothing, so almost nobody uses it.

The fixed first batch was meant to stop kitchens cooking for every registration. It cooks below registrations and still over-prepares at breakfast, because 70% and 70 to 80% sit near Yuktāhār's lunch turnout and not the 28 to 37%

breakfast turnout. The menu rotation fixed boredom, created a new way to plan around a menu known to be disliked, and gave Yuktāhār a repeating unit to calibrate against. The shortage exception got the mess through a supply problem and proved, by accident, that the standing model can be replaced by one that tracks attendance, which no one has revisited. The food-safety sequence keeps the mess compliant and also locks the first batch in before any turnout signal (candidate). Waste does appear as a number at Yuktāhār, and the number stays in the kitchen. Any new measurement risks the blame reaction that weighing first met there. Outsourcing was meant to save money and modernise hygiene. It put the most detailed records with the party that has least say over registration, billing and menu, and produced three stances on waste.

6. Systemic Tensions

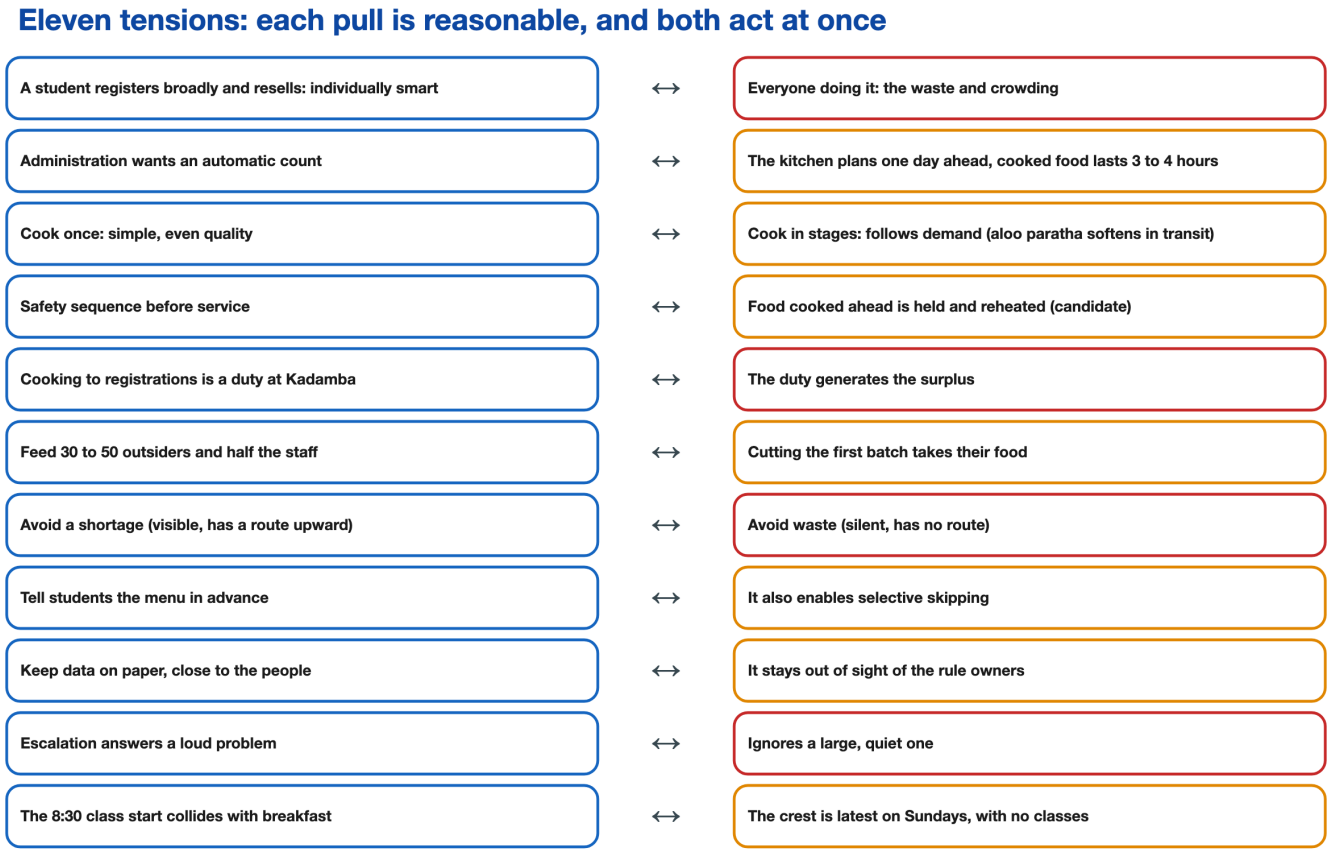


Figure 12. *Eleven tensions: each pull is reasonable, and both operate at once.*

A student registering broadly and reselling the rest is individually smart, and added up it is the waste and crowding the analysis describes. Registration administration wants an automatic count while the kitchen plans one day ahead with three to four hours of cooked life. Both the standing system and the shock exception have been run by the same institution, which shows the tension between a fixed count and real demand is real. Cooking once is simpler and keeps quality even, and cooking in stages follows demand; Yuktāhār cooks once because ”student fluctuation is very high”, and at Bakul aloo paratha started around 3 AM for a 7 AM arrival loses texture in the box. A safety sequence of sampling, display and tasting keeps food safe, and with cooking ahead it leaves food held and reheated through a two-hour service (candidate).

At Kadamba cooking to registrations is framed as duty, and the duty generates the surplus. At Yuktāhār waste reduction is a profit lever the operator spends partly on quality, so profit and quality point the same way. Staying close to the data on paper keeps planning owned by the people who do it, and keeps the data out of sight of the rule owners. Feeding outsiders and staff from the same food is a welfare benefit that absorbs surplus uncounted, and

any cut to the first batch takes their food unless they are planned for. Kadamba's own figures leave an unresolved tension: a 70% batch against 37% turnout implies a surplus near 350 portions a day, while the operator reports 5 to 10% waste on no stated base.

Avoiding a shortage and avoiding waste pull against each other at every first batch, and Bakul has settled it by rule: no shortage even if food is wasted. A shortage is visible and has a route to the managing director while a surplus is silent, which also explains the reserve at 9:20. The menu is chosen for nutrition and preference and, in the Bakul manager's account, eaten by habit and pairing (his view, untested). Telling students the menu in advance fixes boredom and enables selective skipping. A registration system built for one kind of student leaves little room for someone who fasts or eats differently, and folds that into the same no-show figure. Escalation answers a loud problem and ignores a large quiet one. Nobody has asked the academic side to move the 8:30 class start and this document does not suggest the timetable should change. The collision is real but should not be over-read, since the crest is latest on Sundays, when there are no classes (candidate).

7. Power Asymmetries

Students carry nearly all the visible cost: money spent on meals never eaten, crowding and running out. They control almost none of the rules that produce it, and they are registered by default with no way to nullify the charge.

The operator is data-rich and authority-poor. Neither Kadamba, which sets its own ratio, nor Yuktāhār, which holds per-item daily records, has any part in the registration rules that generate its starting number, or in the menu. The Bakul manager holds two kinds of knowledge the menu process does not use, which items are eaten and which pairings work, and his weekly calibration may live only in his head (that this knowledge would improve the menu is a candidate). Inside the operators power splits differently. At Kadamba the person who decides quantities in service is a supervisor several levels below the vendor and further still from CDS; at Yuktāhār the measurers and deciders are one small peer group. The central dining side holds a timestamped scan for every registration and nothing shows anyone analysing it. Students collectively, through Mess Cell, hold a real-time read on who is coming that counts formally nowhere.

The academic side holds total control over the schedule collision that appears in nearly every interview, and faces none of its consequences. When asked whether anyone had raised the 8:30 overlap with them, nobody really had. The calendar reaches Yuktāhār's kitchen only by word of mouth (confirmed). The Committee sets the menu without a view of what is eaten against what is registered, and the dropped dinner sambar reached students as an unexplained cut. The Warden holds authority over vendor contracts and structural change without a confirmed link to the numbers. The facility team and weekly inspection hold authority over safety and taste and none over quantity. CDS's chair, whose office sits directly above the vendor, has not been interviewed. Mess Cell runs with complete flexibility and no oversight, and a student's own registration behaviour (register, cancel, resell, or not show) costs the same either way. Yuktāhār counts faculty and staff diners separately, zero to two per breakfast (confirmed).

8. Workarounds: What Students and Operators Have Built Around the System

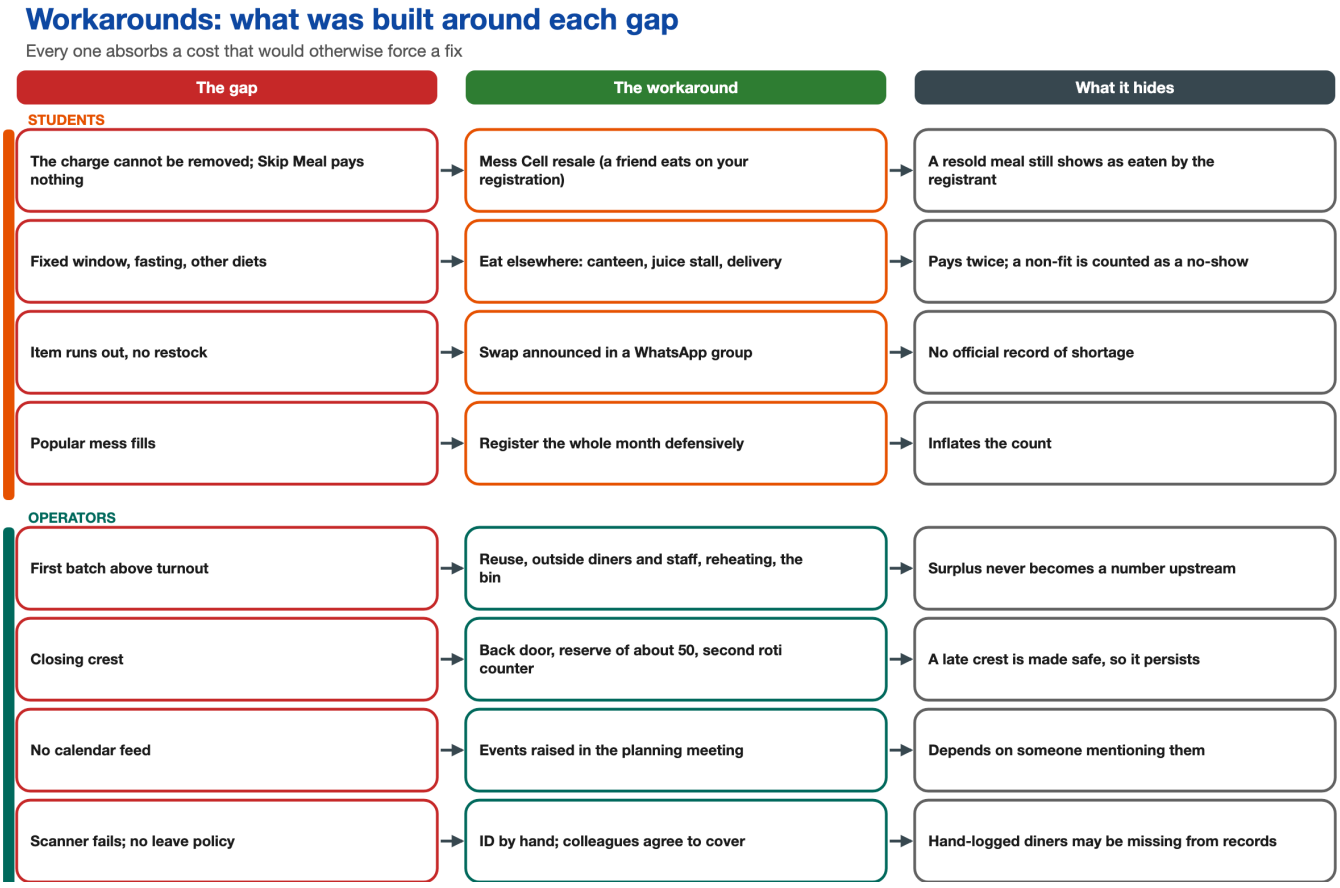


Figure 13. *Where the system leaves a gap, and what people have built to absorb it.*

Every workaround quietly absorbs a cost that would otherwise force someone to fix the real problem. Mess Cell absorbs the financial pain of over-registering, so the pressure to fix billing never builds, and it hides the gap, since a resold meal probably still appears as eaten by the original registrant; the scan export has no student ID, so the records cannot show it. Skip Meal was built to solve this, and students replaced it with Mess Cell because Mess Cell pays. Eating elsewhere (the canteen, a juice stall, a delivery app) gets around a fixed window and a fasting calendar, usually at the price of paying twice. When something runs out, staff post the swap in a WhatsApp group after the fact, which keeps trust without any official record. Students whose dietary or cultural needs the menu does not cover mostly just do not come.

Operators have built their own layer, described in chain 7 and drawn above. Beyond the absorbers already named, Yuktāhār holds raw material to an impurity norm, opens two roti counters at peak and covers absences by agreement in the daily meeting. Kadamba writes ID and name by hand when the scanner fails and bought items outside during the gas shortage. Bakul phones Hafizpet, holds its reserve and switched paneer suppliers twice after complaints. Kadamba does not compost, and where its contractor takes the waste is unknown, as is Yuktāhār's disposal route.

9. Synthesis

Breakfast registrations are used 37% of the time at Kadamba and about 28% at Yuktāhār, against 56 to 89% for lunch. Attendance is low partly because every student is registered by a default that cannot follow the morning, partly because registering costs nothing extra if you do not show and nothing removes the charge, partly because the gap is measured in two places and owned by no one, and partly because a real share is just how a varied student body eats. On the kitchen side Kadamba turns a flat default into a first batch at a fixed ratio above turnout, Bakul adjusts weekly by judgement and never below a shortage floor, and Yuktāhār plans from its books. Each tops up only upward and too late for the crest, and absorbs the rest by reuse, outsiders, reheating, late service and the bin.

The institution has already proven it can fix the student side. It found a way to charge people for what they eat, ran it more than once and did not keep it. The kitchens do not need the four-day close, so what holds the standing model in place is unknown: the per-plate payment, the convenience of an automatic default, or a reversion with the calendar. The trade-off has never been weighed on purpose, because no single actor holds both the data and the authority, and because the system attends to loud problems and not large, quiet ones.

Several things stand out. The fix exists and has worked more than once. The student half comes from an automatic standing booking with billing tied to it, both optional since the alternative was tested. The kitchen half comes from a conversion inside operators' own authority. The two best-recorded halls are a natural comparison, a fixed ratio with no learning record against a book-based system with blame-free measurement, and they produce about the same gap, so calibration reduces waste and run-outs and does not move turnout. A real share of the gap is not a problem to fix, though weekday variation alone cannot explain a 63% no-show rate.

Three questions decide what comes next. What does the per-plate payment count, registrations or scans? Why must Kadamba guess its ratio when the data to calibrate it is recorded and one operator already does? And where does Kadamba's surplus go, given 5 to 10% reported waste against a 70% batch at 37% turnout? Several source conflicts stay open until someone asks directly: whether Skip Meal changes what Kadamba cooks, whether the CDS head named by the operator and the one named earlier are two people or one, whether the Yuktāhār head's 50% is an average or a ceiling, whether Kadamba's outside diners appear in any record, how long a Hafizpet top-up takes, and whether Bakul's weekly review rests on a written record. Until someone asks, this will likely keep repeating, not because the answer is unknown, but because the records that hold it never reach the people who set the rules.

Reframed Systemic Problem Statement

IIIT Hyderabad Breakfast Mess System

Contents

The Problem, Reframed 1

1. Challenging Initial Assumptions: What the Original Question Got Wrong 2

2. Comparing Stakeholder Perspectives: What Each Stakeholder Actually Experiences 3

3. Identifying Contradictions and Tensions: What Holds This in Place 4

4. Reading It Through the Iceberg 7

5. Why This System Holds Steady Instead of Breaking 8

6. Developing Alternative Problem Framings: Other Ways to Frame This, and Why They Fall Short 10

7. Validating the Reframing with Evidence 11

8. What's Still Missing 12

The Problem, Reframed

> The Breakfast Paradox is a stable equilibrium. The system holds the gap between registration and attendance in place because every actor's behaviour inside it is individually reasonable, and every consequence it produces is absorbed before it reaches anyone who sets the rules.
>
> Three structures hold it. Breakfast is booked by default: every student is registered automatically, on campus or not, holidays included, and can leave the booking only through five cancellations per meal type per month, each made at least four days ahead. A default the system assigns cannot follow the decision each student makes on the morning. Billing fires at the booking, not at consumption, and beyond the five cancellations nothing removes the charge, so the inaccuracy costs the system nothing and, if operators are paid on booked plates, costs the operator nothing either. And the gap is measured twice, in the CDS scan record and in the operators' own books, but neither measurement is routed to the people who set registration, billing and menu rules, whose decision rights are split so that none of them holds both the knowledge and the authority to act. Lunch runs under the same booking and billing rules and mostly works, which is what makes this a breakfast problem and not a billing problem.
>
> What follows is quietly absorbed. Resale takes the student's loss. The operator's own discount on the booking, live top-ups, leftover reuse, the outside diners and staff who eat from the same food, and late-door service take most of the physical surplus. Holding and reheating turns some of the rest into lower quality, and what cannot be used leaves in the bin. Outside venues take the dissatisfaction, and informal workarounds take the cultural mismatch. Most of this is useful and none of the useful parts should be removed. Together they keep the gap from producing a signal loud enough to reach a system that responds to complaint intensity.
>
> No single actor feels this as their whole problem. The student feels a charge. One operator measures the gap every morning and has nobody to tell, another cooks to the count as the obligation it is paid for, and a third adjusts by weekly judgement but never lets its first batch fall below a floor held against shortage. The office holds a statistic. The paradox is seen in aggregate only by the actors with the least authority over the rules.

The original question was operational: students register, many do not show up, they are charged anyway and the food is cooked regardless, so why does registration not match who eats? That question sits at the wrong layer.

Beneath the kitchen half of the problem is one mechanism that is downstream of the three structures and inside authority the operator already holds. Each operator converts the booking into a cooked first batch above turnout, and can then top it up but never shrink it. Kadamba does this by a fixed ratio, Bakul by a weekly judgement with a floor, and Yuktāhār from its own books.

The rest of this document makes the case: the assumptions the statement overturns, what each stakeholder sees, the tensions that hold the system, and the evidence that validates it. The Design Opportunity Statement is a separate document that turns the diagnosis into where to intervene. Bakul has no records of any kind, so every Bakul figure is the operator's own statement.

1. Challenging Initial Assumptions: What the Original Question Got Wrong

The original question rested on five assumptions. Six student interviews, a CFS Chair interview, three operator interviews, a team member's first-hand account of two kitchens, Yuktāhār's books and 11,941 timestamped Kadamba scans test each one.

Students not showing up is the problem. It is a symptom, and part of what the turnout number counts as failure is not a problem. One respondent fasts intermittently by choice and does not notice the mess even in the same hostel. Others come from households where a formal breakfast was never the norm. The records show the same variety: at Kadamba in April, 48.5% of non-vegetarian bookings were eaten against 32.4% of vegetarian ones, and Sunday was the lowest day at Kadamba and among the lowest at Yuktāhār (confirmed). Bakul's manager calls weekends very low (single-sourced, no records). The billing record shows a considered choice and a forgotten booking identically, as a paid slot nobody used.

Students are careless or irrational. Most are not choosing a booking at all. As one put it: ”Even if you want to eat breakfast or not, it is automatically registered. We don't have any opportunity to nullify it.” The exit is narrow and early, so it cannot follow a morning decision (confirmed, from the CDS poster and the student's account). Kadamba bookings sat near 1,070 a day on every day of the week in April, Sunday included, while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed). Reselling on Mess Cell makes sense when Skip Meal pays nothing. Deciding at 9:10 whether the walk is worth it makes sense when the mess is about to reach its most crowded and popular items may already be gone. Every individual decision is defensible, which is why this is a systems problem.

Nobody has solved this. The institution has run attendance-tracked billing twice, during a gas shortage and during holidays such as Felicity. Students scanned in and were charged only if they ate. It worked both times and was switched off both times once the trigger passed, with no sign anyone evaluated keeping any of it. The data that model needs is still collected every day, one scan per breakfast eaten (confirmed), with a handwritten fallback when a scanner fails (single-sourced).

Nobody knows how big the gap is. It is measured in at least three places and never reconciled. Of 32,162 Kadamba breakfast bookings in April, 11,941 were eaten, 37.1%, leaving 62.9% unused. Yuktāhār's September register shows breakfast at 17.7% to 40.2%, about 28% on average, against lunch at 56% to 89% on the same days (confirmed). The CFS Chair's recalled 35% to 40% for breakfast holds for Kadamba, and his 70% for general items matches lunch. The project's earlier gap of about 30% is a lunch figure. Beliefs sit above the records: Yuktāhār's site head puts breakfast at up to about 50%, Bakul's operator believes Kadamba fills 90% to 95%, and Bakul's manager estimates at most about 60% for his own hall, an upper bound if Bakul resembles the two recorded halls (candidate). The measurement is also uneven. Yuktāhār weighs every item, Kadamba keeps gram standards and a waste estimate of 5% to 10% a day on an unstated base (single-sourced), and Bakul's lives in a weekly review that may be in one person's head. None of it has reached a registration or billing decision. The gap is known and has not travelled upward.

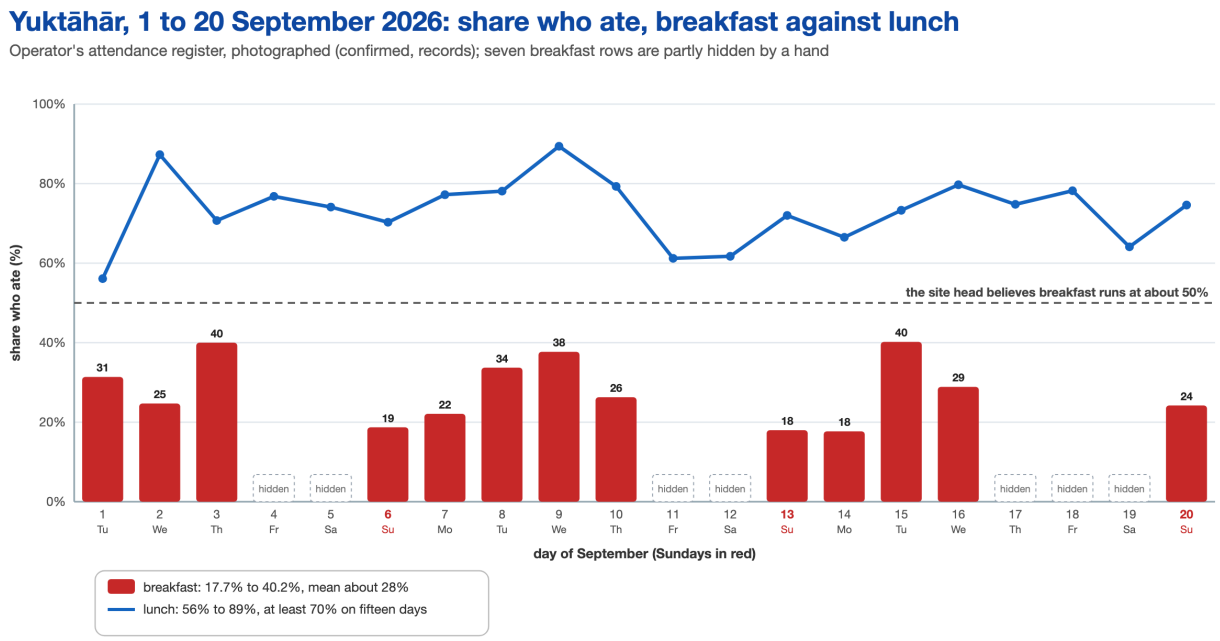


Figure 1. Yuktāhār, September 2026: breakfast against lunch under the same booking and billing rules.

This is one problem. It is three structural causes plus one thing that is not a problem at all, and treating it as one gap produces a single fix aimed at the average of four different things.

2. Comparing Stakeholder Perspectives: What Each Stakeholder Actually Experiences

Ask anyone in the system what their problem is and none of them will describe the Breakfast Paradox. Each feels a smaller piece, and the figure sets that beside what each cannot see.

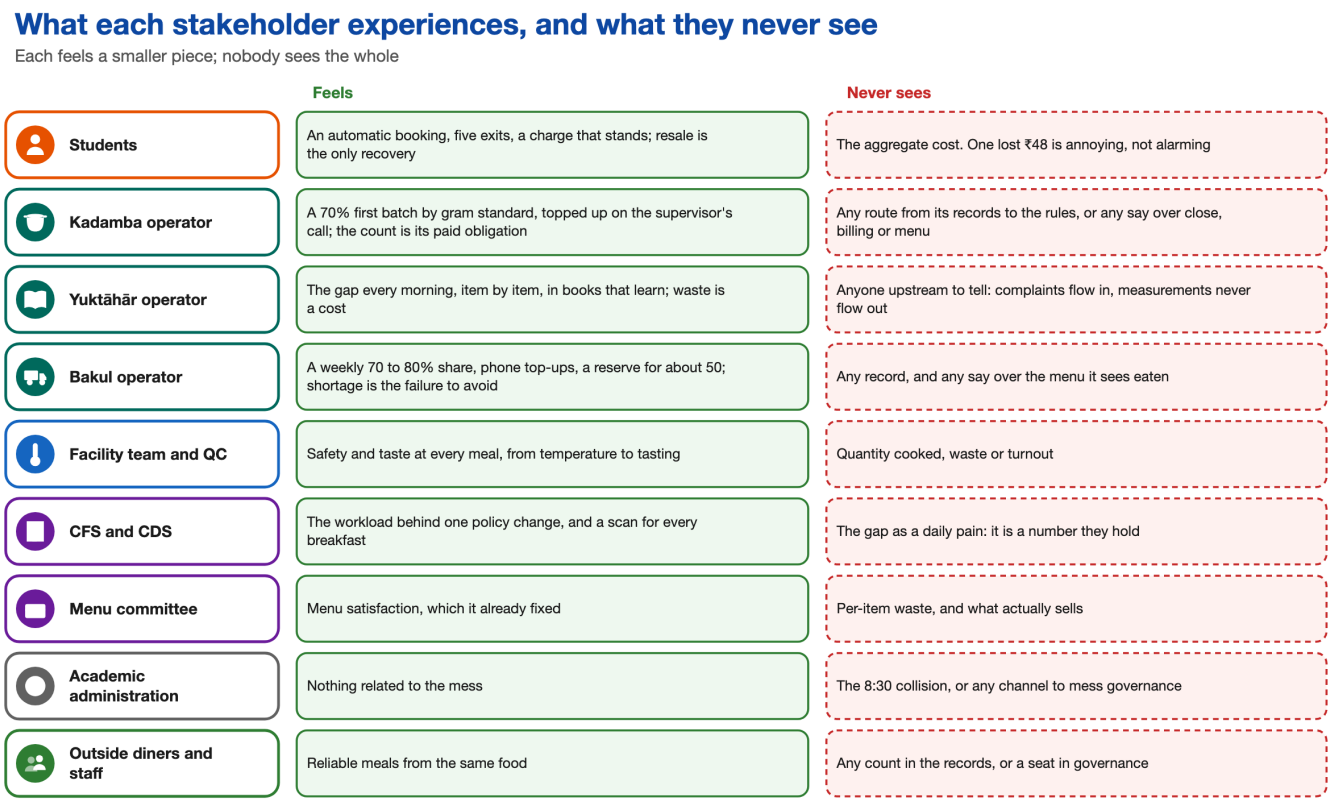


Figure 2. *What each stakeholder experiences, and what stays invisible to them.*

The operators also carry their own reading of the gap. Kadamba's manager says some no-shows are students who book while away in their village while a friend eats in their place. The scan record carries no student ID and cannot test this, but automatic registration of students who are off campus would produce the pattern without anyone gaming it (candidate). Bakul's manager gives two reasons, that students may not like the menu or may not wake up (single-sourced). Not waking matches what the six interviewed students say most often. Taste is not a leading reason in those interviews, and none of the six is known to live at Bakul.

This is the centre of the reframe. The operator sees the gap more completely than anyone, and at Yuktāhār, where it is recorded per item and weekday, holds no authority over any rule that produces it. At Kadamba the operator sees live counter flow and gram standards and keeps no outcome record the team has seen. At Bakul it reads the gap through a goal that puts avoiding a shortage ahead of avoiding waste. The paradox is visible in aggregate only at the bottom of the chain of authority.

3. Identifying Contradictions and Tensions: What Holds This in Place

None of these tensions is fixed by asking one side to behave better. Each side is already behaving sensibly given the structure around it.

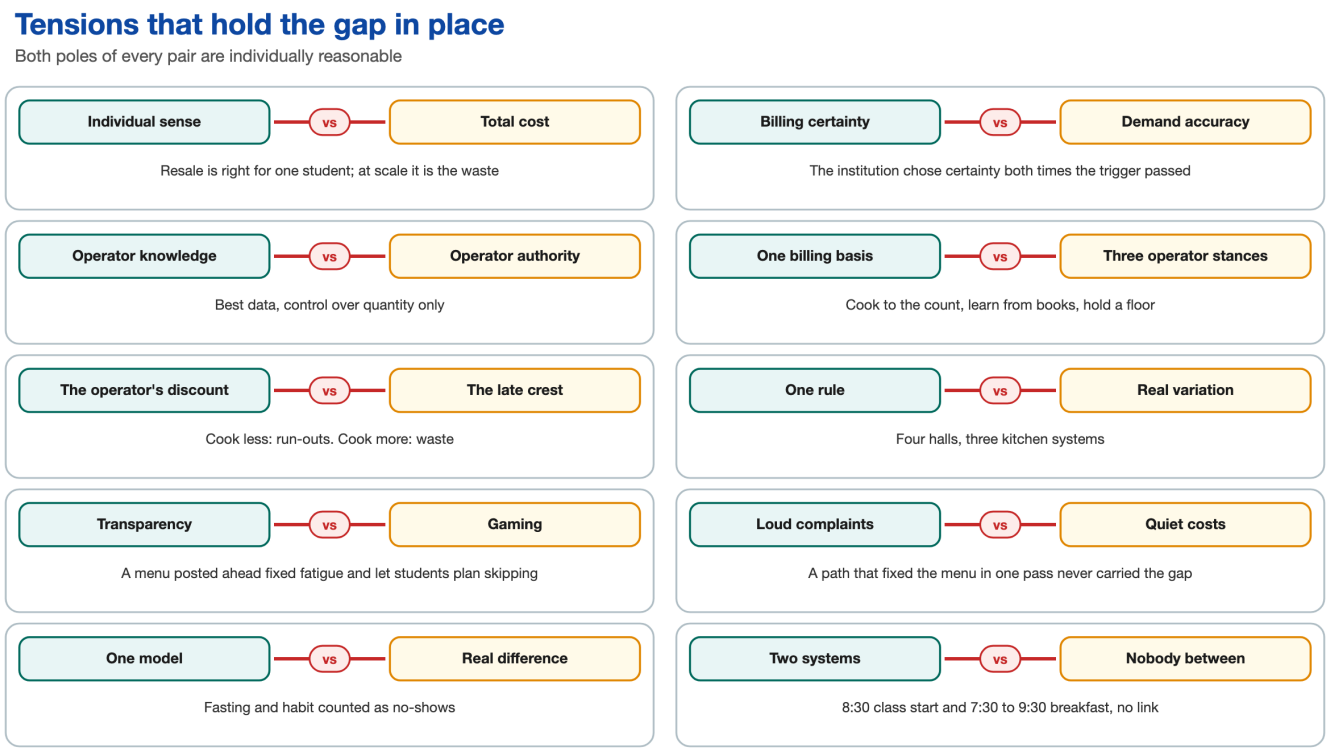


Figure 3. *The tensions: each pair is rational on both sides.*

Individual sense against total cost. Reselling what goes unused is the right move for one student. When everyone does it, the default produces the waste and crowding this project set out to explain.

Billing certainty against demand accuracy. The institution has run both sides. The standing model gives it a certain revenue base fixed four days out. The attendance-tracked model matches real attendance and gives up that certainty, and the institution chose certainty each time the trigger passed. The four-day lead does not protect the kitchen. Yuktāhār buys groceries fortnightly, vegetables two or three times a week and milk daily, and Kadamba buys dry goods a week ahead, some vegetables two or three days ahead and perishables the day before. Only dry goods are committed before the close, raw surplus can be carried forward, and cooked food lasts three to four hours. The binding constraint is the cooked first batch, which the pre-service safety sequence fixes before any turnout signal arrives (candidate).

Operator knowledge against operator authority. The actor with the most detailed demand data controls quantity only. Yuktāhār plans each breakfast in a daily meeting from portal bookings, the Wastage Book entry for the matching week of the repeating menu, a weekday register, the month's pattern and any events, and re-projects from the first half hour of service. Its store book must match what was cooked, it holds an impurity norm for every raw item, and it dropped commercial prediction software because the data already sits in its books and its staff are not digitally inclined (single-sourced). It cannot touch the registration close, the billing basis or the menu, and exams and events reach it only if someone mentions them in the meeting. Bakul's manager wants a say in the menu and objects to pairings, not dishes: kachori belongs with chole, and he would serve Raj Kachori with aloo curry. That is his view, untested with Bakul residents, and nobody asks them.

One billing basis against three operator stances. All three work under the same booking and a tendered plate payment whose basis is unconfirmed (section 7). Kadamba's operator cooks to the count because ”it is a responsibility and they are getting paid for it.” Yuktāhār names lower cost and higher profit as goals and built a book system that learns. Bakul's manager states that there should not be a shortage of food even if some wastage occurs, and keeps food for about 70% of registrations on mornings he calls very low (all single-sourced). The same rule yields one kitchen that measures and trims, one that cooks a fixed ratio, and one that adjusts but only down to a floor. Whether an operator produces the information that could correct the rules depends on its culture, not on the rules

(candidate).

The operator's discount against the late crest. This is the one place where shortage and waste share a morning. Kadamba's first batch of 70% of registrations is set against 37.1% turnout. On April's average day that is about 530 vegetarian portions against about 245 eaten, roughly 2.2 times, and about 220 non-vegetarian against about 153, roughly 1.4 times (candidate arithmetic), so a fixed ratio over-cooks most where turnout is lowest. The operator's 5% to 10% waste figure and a first batch near twice turnout can describe the same breakfast only if outside diners, reheating or small top-ups absorb most of the difference, or the figure uses another base. That is left open. Yet late students still meet run-outs, because the crowd crests at close: 26.5% of April scans fall between 9:15 and 9:30, about nine a minute by 9:29, and 9.2%, about 37 a day, come at or after 9:30 through a back door the mess cannot refuse (confirmed; back door single-sourced). Cooking less raises run-outs and cooking more raises waste. The crest is not universal: Bakul's manager puts his rush at 8:00 to 8:30 with 10 to 15 people in the last ten minutes, and holds food for about 50 at 9:20 because a top-up phoned to a kitchen 20 to 40 minutes away cannot land in time (single-sourced; the reason is a candidate).

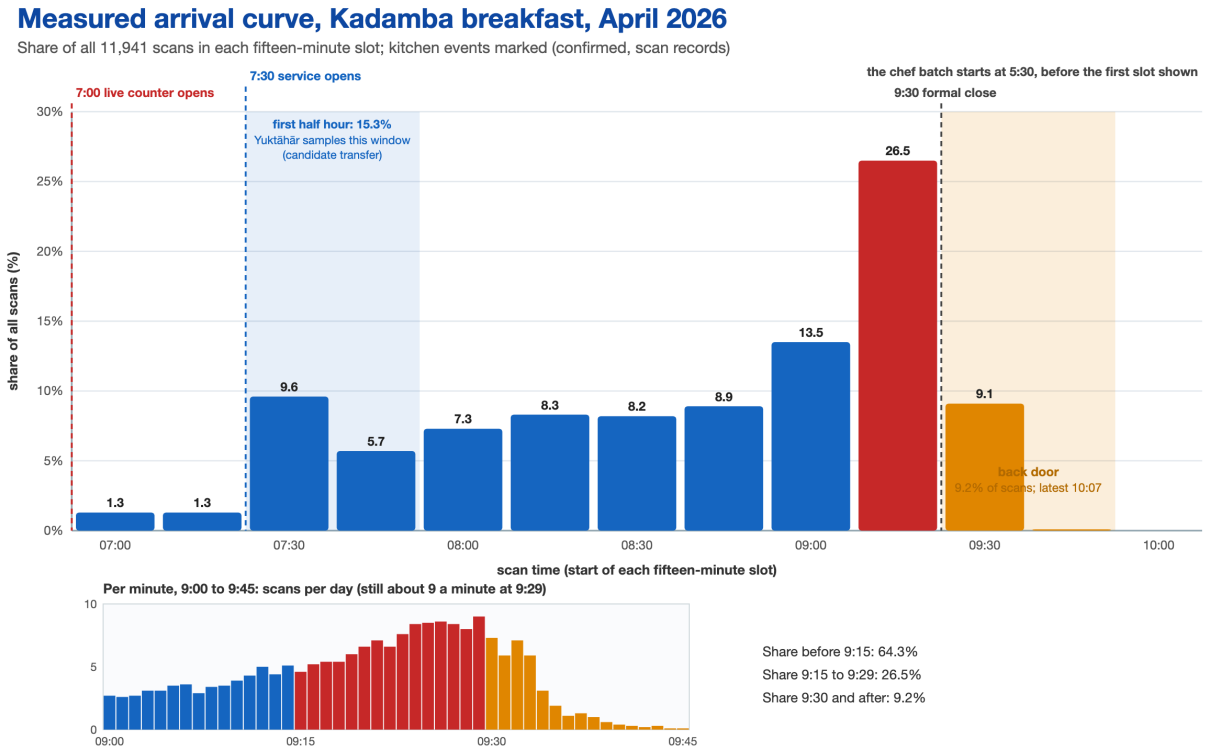


Figure 4. Kadamba's measured arrival curve: the crowd crests at the close, and about 9% arrive after it.

One rule against real variation. One registration rule and one billing rule cover four halls that run on three kitchen systems. Kadamba cooks 70% first and tops up in 10 to 15 minutes. Yuktāhār's first-half-hour projection samples the quietest part of service and so under-reads the crest (candidate). Bakul's top-up is a phone call, cooking and a drive of about 8 km, with a total lead time nobody has measured. Kadamba cooks biryani lunch at 100% of registrations, so its ratio follows the dish and not any record of turnout (candidate). Turnout varies by diet and weekday within one hall while the booking does not.

Transparency against gaming. Posting the menu two weeks ahead fixed menu fatigue, and by the same office's account it also let students plan their skipping. The same repeating cycle is what lets Yuktāhār look up the matching week.

Loud complaints against quiet costs. The path that fixed menu fatigue in one pass has never carried the larger, already measured registration gap. Complaints reach operators fast, from a director's email to the kitchen WhatsApp group, and nothing runs the other way. The facility team inspects safety and taste daily and never asks about quantity

(candidate). The same logic works inside the kitchen. At Bakul a run-out routes a complaint up to the managing director while a surplus is silent, so the operator's rule of no shortage at any cost of waste is the kitchen's form of a system that reacts only to what is loud (candidate).

One model against real difference, and two systems with nobody between them. Fasting, regional habits and religious observance fold into the same turnout number as ordinary no-shows. The 8:30 class start and the 7:30 to 9:30 breakfast window have no confirmed relationship, formal or informal, between their owners. The Kadamba record shows no surge before 8:30, and arrivals run at about 6% to 10% a quarter hour until 9:00. A first class ending near 9:30 could push students into the minutes the kitchen can least serve, which stays a candidate until checked against the timetable.

4. Reading It Through the Iceberg

The original framing stayed in the top two layers. The causes sit in the bottom two.

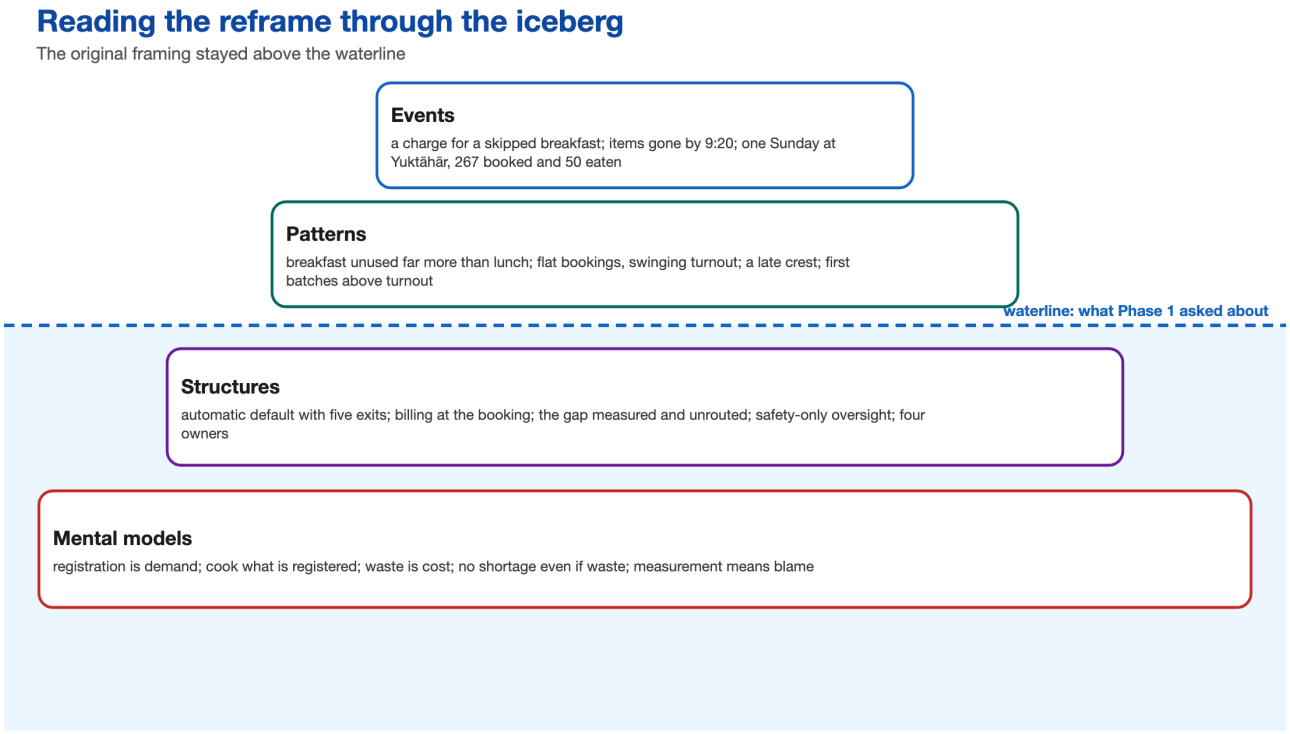


Figure 5. The paradox read at four depths.

Events. A student charged Rs. 48 for a Kadamba breakfast they were booked for and skipped. Idli, puri and bhatura gone by 9:20 with promised restocks that do not arrive. Sunday 6 September at Yuktāhār: 267 breakfasts booked, 50 eaten (confirmed). A Kadamba biryani day when 25 students arrived after close and the door was reopened (single-sourced). The wrapped display plate at Kadamba's entrance on the day of the visit (confirmed). Yuktāhār dropping prediction software, and its staff reporting in December that waste had fallen since weighing began in August (single-sourced). A frog in the chicken biryani in November 2024, the December 2024 tightening of cancellation rules over a student survey and two Parliament meetings (secondhand; candidate), the Felicity walk-in relaxation, and the vote that ended the fixed-semester menu.

Patterns. Breakfast bookings go unused at a far higher rate than lunch, bookings stay flat across the week while turnout swings, and arrivals crest in the last quarter hour and spill past close. The first batch sits above turnout at every kitchen. The same items waste at Kadamba while others sell out, and at Bakul demand follows the item. All six interviewed students are on a standing booking, none got value from Skip Meal, and four use Mess Cell. Complaints sit until an incident forces a response, with one loud exception.

Structures. Billing fires at the booking. Registration is automatic with five exits. Each kitchen converts bookings into a first batch by its own rule, and cooked quantity can rise but never fall. Bakul's food is cooked at Hafizpet, carried about 8 km in insulated boxes held at 60 to 72 °C, and topped up by phone across the same distance (single-sourced). Walk-in pricing runs 70% to 90% above the registered rate. Skip Meal pays the student nothing, and whether it lowers what Kadamba cooks is in conflict (section 8). The committee chooses the menu while the operator sees the waste. Complaints route down into the operator and nothing routes measurements up. Oversight inspects safety and taste, never quantity. Menu, kitchen, timetable and billing have four owners and none owns the outcome. Feedback is triaged by loudness, not cost.

Mental models. That a registration count is a good enough stand-in for demand, held by the rule setters. That the operator's job is to cook what is registered (Kadamba), or that waste is a cost to drive down (Yuktāhār), or that there should be no shortage even if food is wasted (Bakul). That prediction belongs close to the operator's own data, or that beyond a fixed ratio there is nothing to predict with (”we don't have that idea”, Kadamba). That measuring waste means assigning blame, the fear Yuktāhār's managers had to remove. That breakfast turnout is higher than it is. That no-shows are students away with a friend eating in their place. That a registration already made costs nothing extra to leave unused. That planning certainty always beats demand accuracy. That operational policy belongs to the Mess Office alone. That breakfast is a meal every student should be placed into by default.

Phase 1 asked about events and patterns. Everything that would move this system sits in the bottom two rows.

5. Why This System Holds Steady Instead of Breaking

A problem usually persists because nobody solved it, solving it costs too much, or somebody benefits. None of those is the story here. This gap holds steady because every consequence it creates is absorbed by something that works, before it reaches anyone who sets the rules.

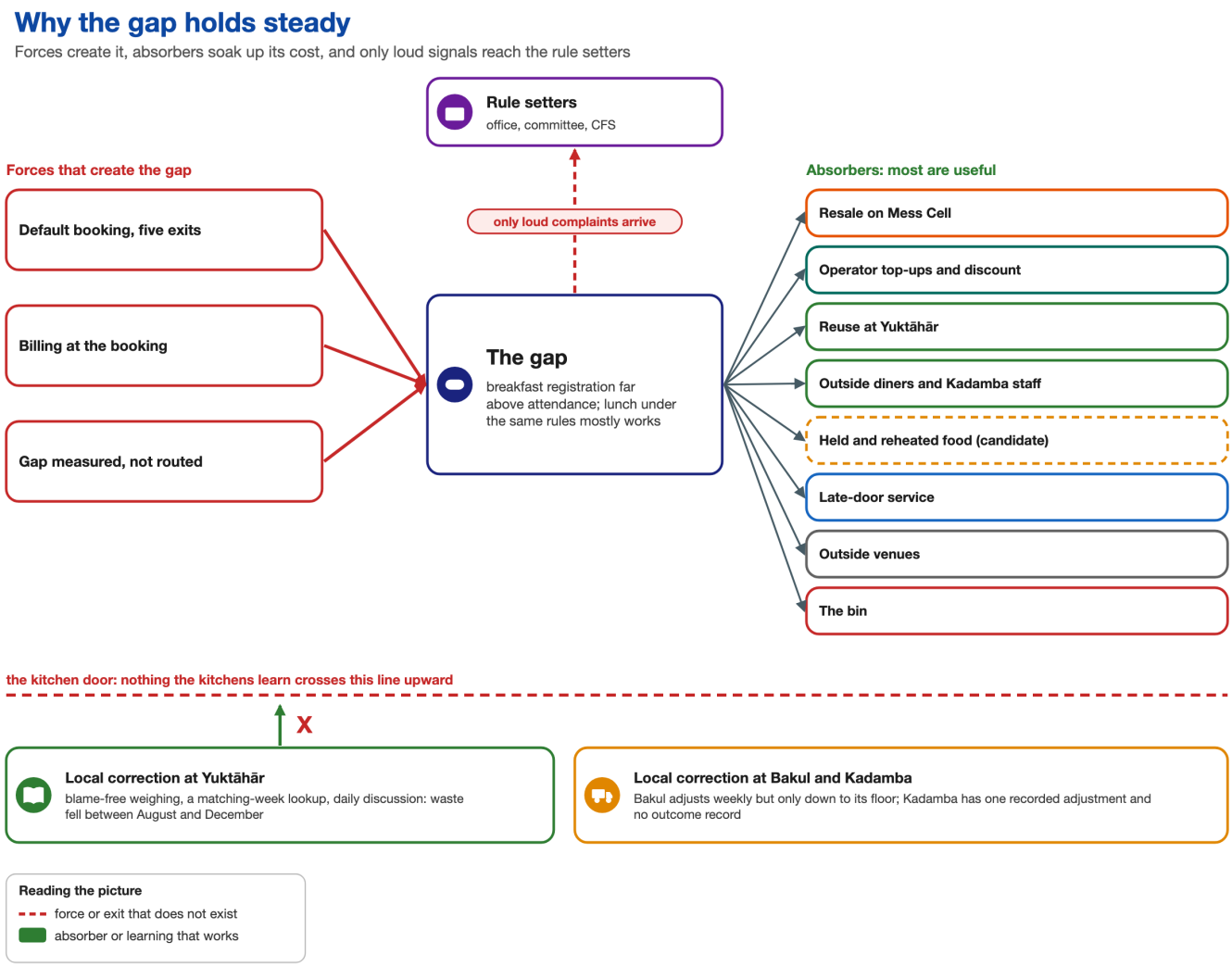


Figure 6. *Why the gap holds steady: three forces create it, eight absorbers soak up its cost, and only a loud signal reaches the rule setters.*

Resale absorbs the student's financial loss. The largest absorber is the operator's own daily adjustment: each operator converts the booking into its own cooked quantity, watches the counter and tops up or re-cooks, which turns the gap into a kitchen routine nobody upstream must answer for. Yuktāhār reuses leftovers: idli becomes idli upma or idli poha for lunch, dosa batter goes back to the fridge, surplus milk is set as curd (confirmed). At Kadamba about 30 portions are planned for 30 to 50 outside diners such as night-shift security, and about half of Kadamba's staff eat from the same food (single-sourced), an uncounted sink for first-batch surplus (candidate). The operator has not said whether those portions sit inside the first batch or on top of it, and whether they are the same allocation as the 30 to 40 extra portions held for walk-ins is unconfirmed. The door reopened for latecomers absorbs part of the crest. Holding and reheating converts surplus into lower quality: in the team's judgement Kadamba's food is cooked ahead, reheated and moved between vessels until it degrades (candidate). Whether that drives skipping is not evidenced, so it is an open chain. At Bakul, food held for about 50 at 9:20 carries part of the first batch into the last minutes, and some leftovers go back to Hafizpet and the rest are thrown (single-sourced). Hot holding keeps temperature while aloo paratha softens in its own steam, and the reserve for late diners is the food held longest (candidate). Outside venues absorb frustration that would otherwise become a complaint about hours, and informal workarounds absorb the dietary and cultural mismatch that never gets filed as a request. What none of these can use leaves in the bin. Kadamba has no composting: its waste goes to a dustbin collected by a garbage contractor (single-sourced), which the institution's account of an outside compost facility has not yet been reconciled with.

Most of this helps people. Students recover money, outside diners and staff are fed, and surplus is used where it can be. Any change to the first batch must plan their portions. Three absorbers protect no one: the over-cooked first batch, repeated reheating and the bin. Reducing those is a legitimate target in a way that removing resale or staff meals is not.

Together the absorbers mean no financial, volume or cultural signal reaches anyone positioned to act. At least one kitchen does weigh plate waste and kitchen waste item by item and records when each item ran out (confirmed), but that measurement never enters a rule-setting forum, and the system above the operator listens for complaint intensity, not records.

That changes the question from why this has not been fixed to what would make someone fix it. Something already does, locally. Yuktāhār cut its waste between August and December by weighing every meal, discussing it daily and making the weighing blame-free after staff feared blame, under managers who treat waste as cost against a profit goal (single-sourced, tracking confirmed). Kadamba has one recorded adjustment, cutting ragi idli in favour of plain idli, and no outcome record. Bakul adjusts weekly but only down to its floor. The local correction that goes furthest depends on an operator whose culture points at waste and on measurement nobody fears, and where the culture points at shortage it stops at the floor. Nothing carries these fixes up to the rules that keep regenerating the gap.

6. Developing Alternative Problem Framings: Other Ways to Frame This, and Why They Fall Short

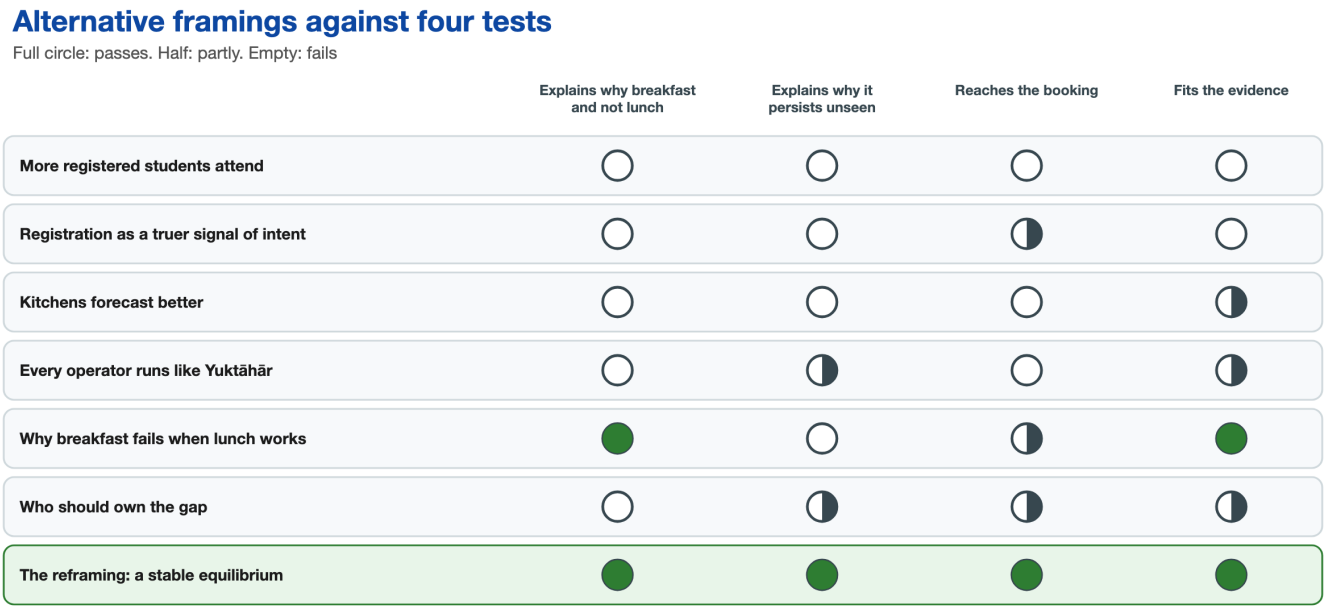


Figure 7. Six alternative framings against four tests; only the reframing passes all four.

"How do we get more registered students to attend?" This was the original framing. It treats attendance as the goal when a real share of non-attendance is legitimate, and it aims the fix at the people with the least power and the most defensible behaviour.

"How do we make registration a more accurate signal of intent?" Registration cannot express intent when it is assigned automatically, and the institution's own proven alternative needed no change in student behaviour. It removed the need for registration to be accurate.

"How do we help kitchens forecast better?" The operators already forecast and adapt, and the best-instrumented kitchen still believes up to 50% where its register says 28%, cannot un-cook a batch, and projects from a first half hour that misses the crest. Bakul already adjusts weekly and keeps food above its manager's own highest estimate

because its goal prefers waste to a run-out. A better forecast of a default booking cannot see a morning decision the booking never records.

”How do we get every operator to run like Yuktāhār?” Closer, because culture decides whether the gap is measured at all. It leaves automatic registration, billing at the booking and the missing upward route untouched, and Yuktāhār itself still overestimates breakfast. Culture can reduce the waste and not the gap.

”Why does breakfast fail when lunch works under the same rules?” Closer again. It isolates a default booking against a volatile morning decision, a late crest and legitimate skipping, but cannot say why so large and well recorded a difference never reaches a rule owner.

”Who should own the gap?” Fragmented ownership explains why a known fix was not picked up, and the operator owns the gap's consequences without owning any of its causes. It does not explain why the gap never got loud enough to demand an owner.

The framing at the top of this document is the one the evidence supports: a stable equilibrium specific to breakfast, held by a default booking, billing on that booking and a missing route from those who measure to those who set the rules, and protected by absorbers that are mostly useful.

7. Validating the Reframing with Evidence

The statement was tested against every assumption the project started with. Most were revised or overturned.

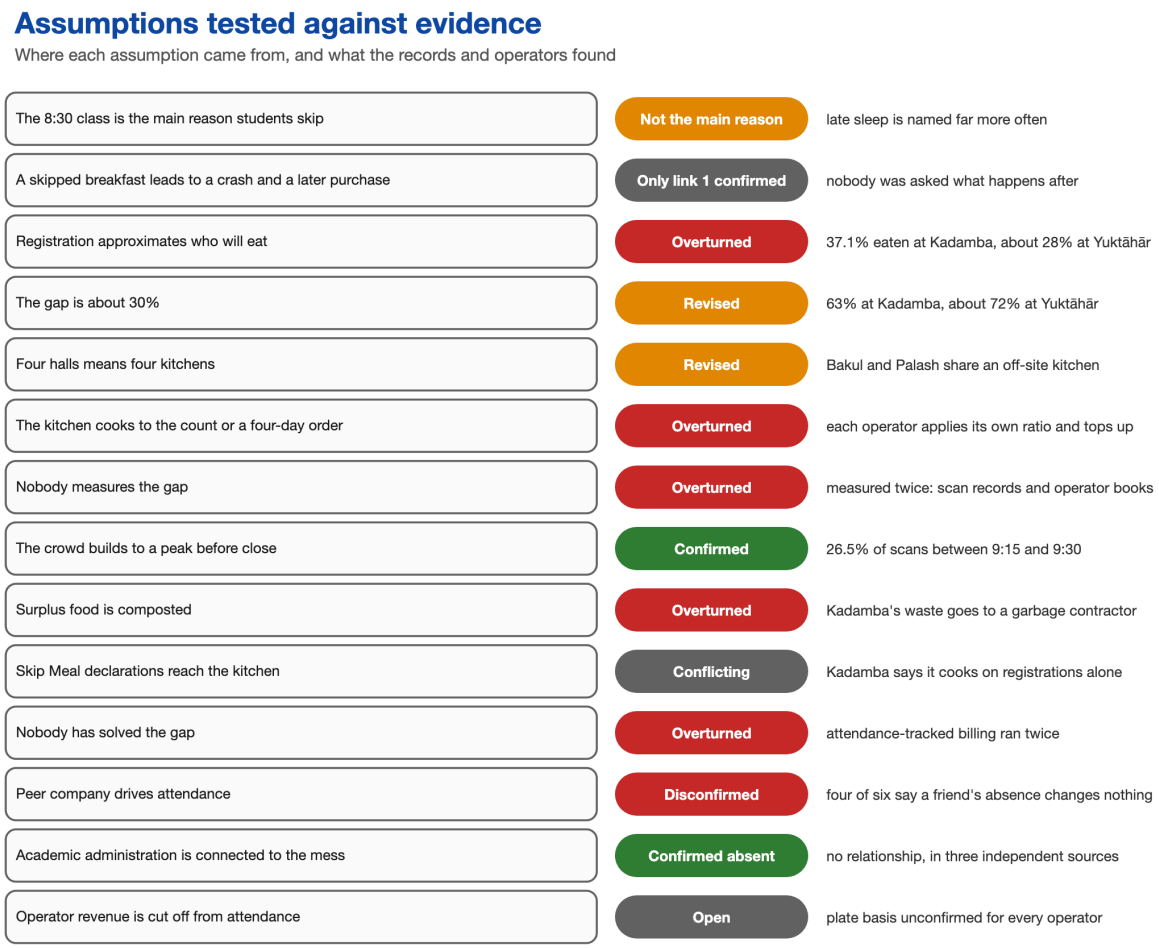


Figure 8. Assumptions tested against evidence: what held, what changed, what is open.

Seven pieces of evidence support the framing and one open question limits it.

- **A fix is possible.** The attendance-tracked model ran twice at this institution and closed the exact gap. The scan data it needs is captured daily under the standing model.
- **The booking is a default.** Confirmed three ways: the CDS poster's cancellation rules, a student's account of automatic registration with no way to nullify the charge, and Kadamba bookings flat across every weekday.
- **The gap is large and measured.** Two halls, from records, not recall (section 1).
- **This is a breakfast problem and billing alone does not explain it.** Yuktāhār's lunch runs at 56% to 89% under the same rules on the days breakfast runs at 18% to 40%.
- **The system adapts locally.** Three operators independently describe converting the booking to a lower cooked quantity and topping up, each under a different regime.
- **Fragmentation explains why the fix was not kept.** The CFS Chair has known the turnout figures for years without them entering a registration or billing conversation, and Yuktāhār's weekday and per-item records have reached no rule owner either. Its believed 50% against a recorded 28% shows nobody checks one number against another.
- **Attention follows loudness.** Menu fatigue travelled from Student Council to Committee and changed policy. The larger measured gap has never entered that process. A vendor that regards the count as what it is paid for has no stated reason to push the gap upward.

The pattern matches the clearest in the project. The system optimises what it can measure and bill, the registration count. Its own stated goal, printed on the CDS dining poster, is nutritious, hygienic, affordable meals with minimal food wastage through advance meal planning, and the CDS handbook names reduced wastage and better meal planning as the benefits of registering early. The easy metric and the stated goal have diverged, and the institution already knows it.

One question stays open and has narrowed without closing: what a plate counts for in each operator's monthly payment. Bakul's manager says plates served and Kadamba's remark points to booked plates (both single-sourced). Contracts may differ by operator, ”served” may be a normalised word for billed plates, or Kadamba's remark may describe duty and not the payment formula. No contract has been seen. The answer decides who bears the cost of a no-show, whether operator revenue depends on the gap, and whether a standing attendance-tracked model shifts risk onto the operator. If the terms rule it out the framing still holds, and the fix moves from adoption to renegotiation. If Bakul is paid on served plates it already bears the cost of what it over-cooks and accepts it (candidate).

8. What's Still Missing

Operations are now covered: three operators were interviewed, Yuktāhār walked through every book, a first-hand account fixed the cancellation rule and the Kadamba command chain, and April's scans measure the crowd curve. What remains open is stated and not merged.

- **Plate basis.** Whether operators are paid on booked or served plates, and the deductions, contract by contract. Nobody has asked directly. It separates a considered trade-off from an unexamined default.
- **Skip Meal.** Kadamba says it cooks on registrations only, while an earlier finding says declarations reach the kitchen. Skip Meal may adjust the count the operator sees while the operator treats any count as the obligation. One direct question settles it.
- **CDS head.** The operator names one person and earlier research another, and it is open whether these are two roles or one wrong name. Whether the operators' ”CDS team” is the Mess Office is also open, and the CDS head has never been interviewed.
- **Registration cadence.** Operators describe weekly blocks and students describe monthly, and the actual rule is unconfirmed.
- **Kadamba waste.** The base of the 5% to 10% figure, whether any outcome record exists beyond gram standards, whether outside-diner portions and the walk-in buffer are one allocation, and whether outside diners are registered anywhere (”some sort of registration done for outside”).
- **Holding and reheating.** The chain rests on the team's judgement and has not been observed at service level, and whether the roughly 37 scans a day at or after 9:30 are the back-door arrivals needs one check.
- **Skip Meal's low use.** Awareness, trust and portal friction would each point to a different fix, and this project cannot yet tell them apart.

- **Waste route.** Where the garbage contractor takes Kadamba's waste, whether any is composted, and Yuktāhār's disposal after reuse. Disposal has not been observed.
- **Bakul.** Whether Palash shares its kitchen, whether the weekly review is written down, how long a Hafizpet top-up takes, whether the 9:20 reserve rests on data or judgement, which menu decisions the manager can override, and why residents skip.
- **Records and timetable.** Lunch, dinner and other-hall scans have not been seen, and whether the 9:15 to 9:30 crest lines up with a first class ending needs one check against the timetable.

Design Opportunity Statement

IIIT Hyderabad Breakfast Mess System

Contents

Overview 1

1. Where the Leverage Is, and Isn't 1

2. The Design Opportunity Statement 3

3. Validate the Reframing with Evidence: What the Opportunity Rests On 3

Compare Stakeholder Perspectives: the Operator's View 4

4. The Four Intervention Points 4

4.1 On process: how a registration becomes a cooked quantity 5

4.2 On the rule: what a registration is and what triggers a bill 5

4.3 On decision rights: who owns the outcome 6

4.4 On information flow: what gets noticed 7

5. What Any Intervention Must Respect 7

Identify Contradictions and Tensions: What the Opportunity Must Hold 9

6. How the Four Interact 9

7. The Questions That Gate All of This 10

8. What Phase 3 Inherits 11

Overview

This is the second deliverable of Activity 8. The Reframed Systemic Problem Statement diagnoses the problem. This document turns the diagnosis into a design opportunity: where a change would do something, what any intervention has to respect, and what should not be attempted. It stops before designing interventions, which is Activity 9's job. What it hands over is a shortlist already filtered against the evidence, with the reasons obvious-looking directions are ruled out, so Phase 3 does not rediscover them.

The evidence is operator interviews at three messes, Yuktāhār's photographed books and registers, the institution's scan record of every Kadamba breakfast in April 2026, and a team member's first-hand account of two visits. Bakul (Vijayalakshmi Caterers) is known only through its manager's account, and no Bakul record of any kind exists, so every Bakul figure here is the operator's own statement. The Bakul account added three secondary opportunity areas: the quality of food as eaten after hot holding, demand by item, and the menu's pairings.

1. Where the Leverage Is, and Isn't

The reframe put the problem at the structure and mental-model layers of the iceberg, which says where effort pays.

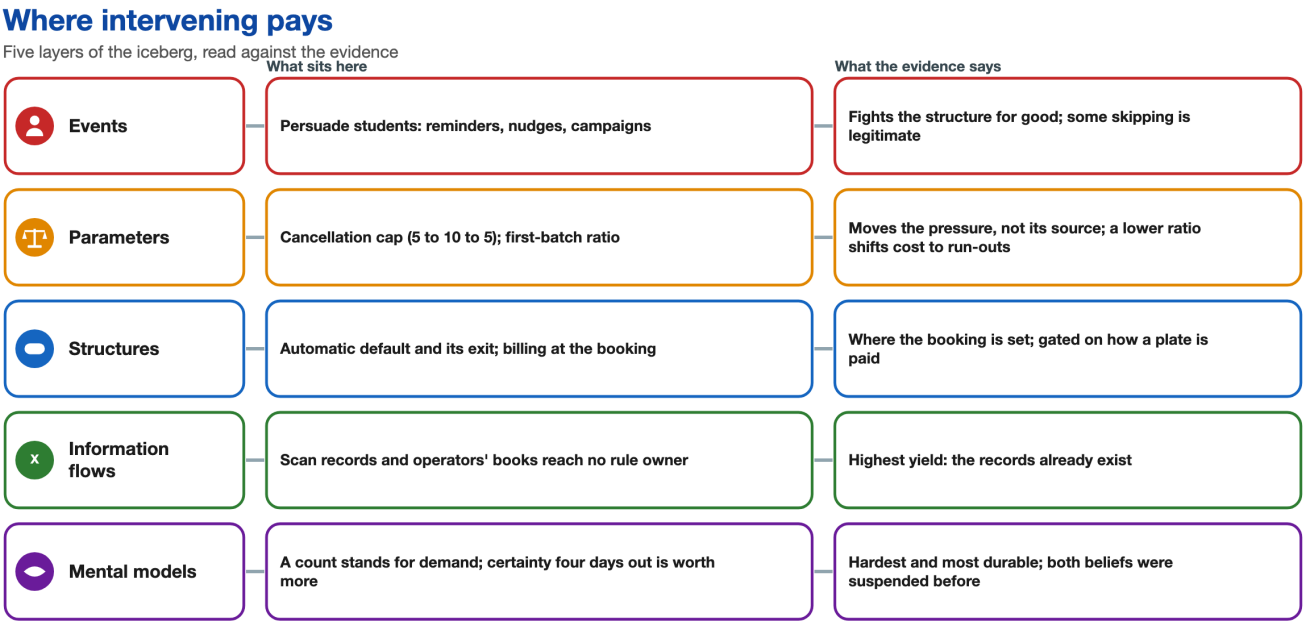


Figure 1. *Where intervening pays: five layers of the iceberg, read against what the evidence shows.*

Not at the event layer. Persuading students to attend with reminders, nudges or campaigns targets the most visible layer and the least movable one. Some non-attendance is legitimate and the rest is rational given the incentives.

Not at the parameter layer, though that is where the institution acts. The cancellation cap has moved from five to ten and back (confirmed), which changes where pressure escapes and leaves the default alone. The kitchen side has the same trap. The first-batch ratio does the most physical damage, but lowering it by decree moves the cost from surplus into run-outs at close. Kadamba cooks biryani lunch at 100% of registrations (single-sourced), so the ratio is a per-meal choice, and what makes it wrong at breakfast is the missing feedback that would set it. At Bakul some feedback already reaches the ratio weekly and a floor still holds it up, because the operator puts avoiding a shortage ahead of avoiding waste (single-sourced).

At the structural layer: what a registration is and what triggers a bill. Every student is registered automatically, on campus or not, and the only exit is five cancellations per meal type per month, each at least four days ahead, after which the charge stands (confirmed). Billing fires at the booking. Kadamba's registrations held at about 1,070 a day on every day of the week in April while turnout ran from 42% on Tuesdays to 27% on Sundays (confirmed). A default fixed that far ahead cannot carry the morning's demand. Yuktāhār's breakfast registrations are not flat (228 to 404 a day in September, confirmed), so the flat pattern is established for Kadamba only.

At the information-flow layer: a gap that is measured and goes nowhere. The scan system records every breakfast eaten against its registration, and Yuktāhār records turnout by weekday and cooked, eaten and wasted quantities per item. Kadamba keeps a gram standard per dish but no outcome record, and Bakul's manager reviews past attendance weekly, with no known written record. None of this reaches whoever owns the registration, billing or menu rules. Routing what already exists is the highest-yield change available.

At the mental-model layer. The institution holds that a registration count stands in for demand, and that certainty days ahead is worth more than a same-week signal. That second belief is not about procurement: the four-day lock applies to registration only, while Kadamba buys dry goods a week ahead, most perishables the day before and some vegetables two to three days ahead (single-sourced), and Yuktāhār takes vegetables two or three times a week and milk daily (confirmed). The binding physical constraint is the cooked first batch, since raw surplus is largely recoverable and cooked surplus is not (candidate). Each operator also carries a private picture of turnout that nothing corrects: Yuktāhār's site head says at most 50% against a register average of about 28% (confirmed), and Bakul's manager says at most about 60% (single-sourced). These layers are the hardest to change and the most durable, but the institution has already suspended both of its beliefs during a gas shortage and in fest periods, so

neither is as fixed as it looks.

2. The Design Opportunity Statement

The opportunity is to give this system a way to notice its own cost.

The Breakfast Paradox persists not because the fix is unknown, since it has been run twice, and not because the gap goes unmeasured. The gap is measured daily, in scan records and in the operators' own books, and none of those measurements reaches anyone with the authority to change the rules that produce it. Every consequence is absorbed before it becomes visible. The resale market takes the financial loss. The operators' discount, top-ups and reuse take the day-to-day mismatch. Outside diners, Kadamba's staff, the walk-in buffer, late service, reheated food and finally the bin take the physical surplus. Outside venues take the dissatisfaction and informal workarounds take the cultural mismatch. The institution's attention triggers on complaint intensity, and this problem is quiet.

The opportunity is therefore to build the missing routes: from the records that exist to the kitchen decision that fixes how much is cooked, and from the same records upward to a decision about registration and billing. Four intervention points follow. One sits in the kitchen, where a registration becomes a cooked quantity. Three sit in governance, in decreasing depth: the registration and billing rules, the ownership gap, and the attention mechanism. The last is the cheapest and the most overlooked, and it would have surfaced the CFS Chair's own figures years ago.

3. Validate the Reframing with Evidence: What the Opportunity Rests On

The opportunity rests on three claims, each checked against records.

The gap is large, specific to breakfast and not a fault of the platform. Kadamba's April 2026 breakfast turnout was 37.1%, a no-show rate of 62.9% across 32,162 registrations. Yuktāhār's September breakfast averaged about 28% while lunch on the same days ran at 56 to 89% (both confirmed from records). One platform and one operator give very different turnout at the two meals.

A private conversion step turns an inflated count into physical surplus. Each operator converts the portal's count into a cooked quantity by its own rule. Kadamba cooks about 70% of registrations first, by line, to a gram standard per dish, and its manager says of forecasting, ”We don't have that idea... it was wasted many times” (single-sourced). Against recorded turnout that batch is roughly 1.9 times what is eaten, about 750 portions against about 400 (candidate). Bakul's manager sets 70 to 80% by item each week and never goes below about 70%. On his own figures his batch and floor sit above the highest turnout he expects, so it exceeds turnout by design (candidate). Yuktāhār plans from its books and projects the rest of each meal from the first half hour (books confirmed, projection single-sourced). Once cooked, a batch can be topped up and never shrunk, and cooked food lasts three to four hours (single-sourced).

Kadamba's operator puts its waste at 5 to 10% a day with no stated base (single-sourced). Against a first batch near twice turnout, that fits only if outside diners and staff eat most of the surplus, or it is held and reheated, or top-ups are smaller than the ratio implies, or the percentage uses a different base. Which is true is unknown. To the extent diners and reheating absorb the surplus, its cost is paid in food quality and in other people's meals, not in bin weight.

The measurement exists and is stranded. The scan export holds one row per registration with a timestamp (confirmed). Yuktāhār's books record per item what was issued, cooked, left and wasted, with run-out time and re-cook quantity, and the issued book must match the cooked book or the mismatch goes to the daily meeting (one item cross-checked across two books held, confirmed). Operators take only the registration count from the portal, and exams and events reach Yuktāhār's kitchen only if someone raises them: ”Until we sit in the meeting, until we discuss that, we will never come to know” (confirmed). So four holders have the figures, none has a route to another, and none has a route to a rule owner.

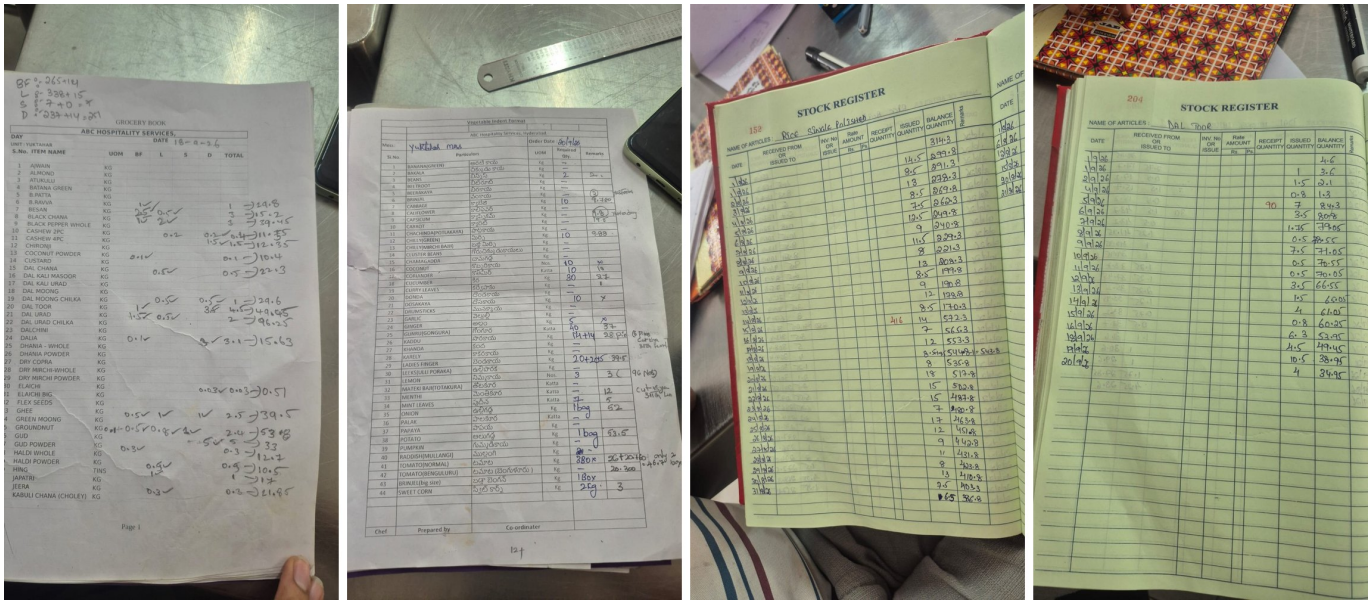


Figure 2. *Yuktāhār's grocery and stock books: the records exist, on paper, inside the kitchen.*

Compare Stakeholder Perspectives: the Operator's View

Operators feel the gap every morning, as food cooked for students who do not come, as a crush at closing that forces re-cooking, and as guesswork about the first batch. They hold the most outcome data and the least rule authority: they decide how much to cook but not how registration works, how it is billed or what is on the menu. The items they see wasted repeatedly, uttapam, set dosa and upma at Kadamba, have no route back to the menu.

The three read the gap differently under the same rules. Yuktāhār states quality, less waste, lower cost and higher profit as goals, so the gap is a cost it already acts on. Kadamba says it cooks on registrations because that is its responsibility and it is paid for it, so the gap is the institution's concern. Bakul holds that a shortage is the failure to avoid and waste its accepted price (all single-sourced). That the difference follows management culture and incentive is a candidate. For Yuktāhār the opportunity reads as a voice and a better input. For Kadamba a better input alone may not be wanted. For Bakul it is a check on a judgement it already makes, acceptable only if it does not raise the chance of a run-out.

4. The Four Intervention Points

These are stated as opportunities. Activity 9 works out what each would look like. The first acts closest to the physical waste and the other three act on the governance that sets its conditions. Three secondary areas rank below the four without changing their order.

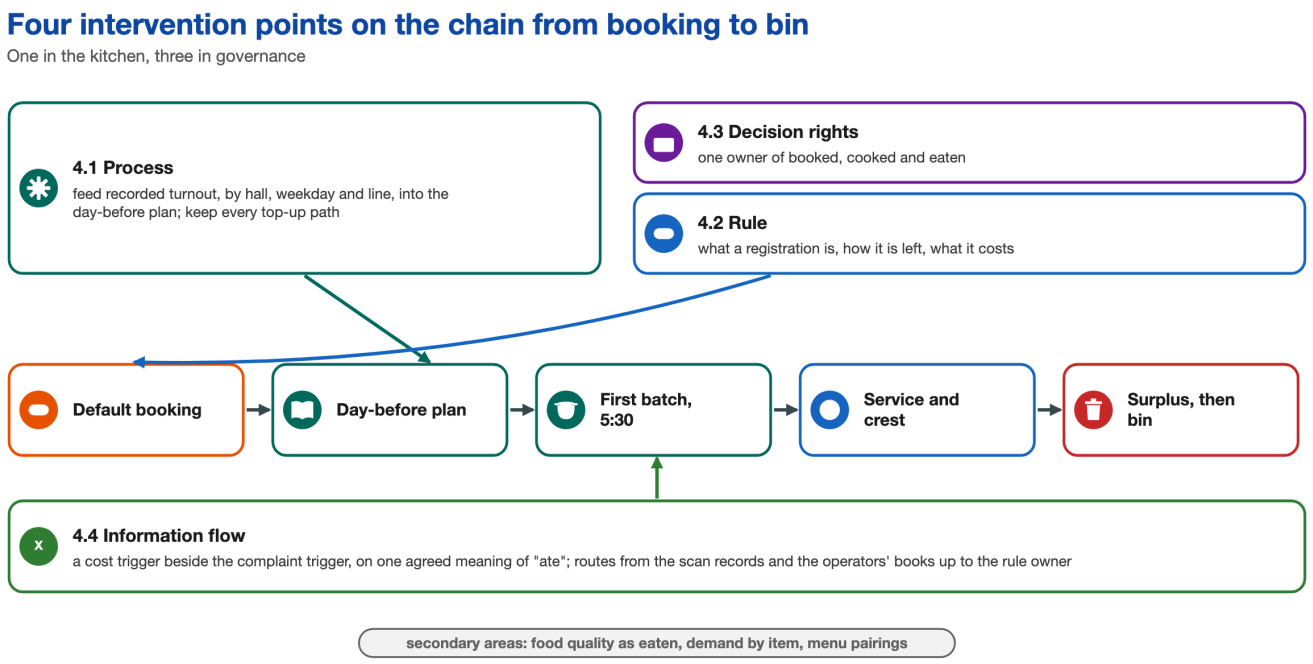


Figure 3. The four intervention points on the chain from booking to bin.

4.1 On process: how a registration becomes a cooked quantity

The first batch is the point of no return. Kadamba sets it by a fixed ratio not calibrated to turnout, Bakul by weekly judgement above a floor, and Yuktāhār from its books, anchored on a belief well above its records. The data that could calibrate it already exists, and Yuktāhār shows the loop can run in-house under the current rules. Waste tracking started there in August of last year, staff first resisted (”they are catching us”), the managers made the weighing blame-free and discussed it daily, and by December staff themselves reported that waste had fallen (single-sourced).

The opportunity is to feed recorded turnout by hall, weekday and line into the decision that sets the first batch, the evening before or at the planning meeting, and to keep each kitchen's top-up path. At Kadamba this can build on the gram standard, multiplied by expected turnout per line instead of 70% of registrations, with an allowance for items diners take more of. April turnout differs by line, 32.4% vegetarian against 48.5% non-vegetarian (confirmed), so one factor would not serve both. A projection from the first half hour is useful but will understate a breakfast that crests at close (candidate). At Bakul the opportunity is narrower: item-level records could let the weekly adjustment be checked against outcomes, provided the chance of a shortage does not rise.

Two secondary opportunities sit here. **Food quality as eaten:** Bakul's labour-heavy items such as aloo paratha start around 3 AM for a 7 AM arrival and are held at about 60 to 72 °C in insulated boxes, where paratha softens in its own steam (single-sourced). The reserve kept for late diners is the food held longest, so they likely get the softest (candidate). Holding time and texture should count alongside temperature and taste, and any change that holds more food for longer trades against quality as well as waste. Whether softer food changes what students eat has no student evidence. **Demand by item:** every kitchen already plans by item in some form (gram standards at Kadamba, per-item pages at Yuktāhār, a ratio that moves by item at Bakul, where pav bhaji draws students and muesli and Chocos go easily, single-sourced). Item-level consumption should reach the first-batch and menu decisions in a form someone can check. Whether it forecasts better than one ratio is a hypothesis.

4.2 On the rule: what a registration is and what triggers a bill

The institution has run attendance-tracked billing during the gas shortage and in holiday and fest periods such as Felicity. Registration is otherwise automatic, including over ordinary holidays (confirmed), so which holiday periods ran without compulsory registration is open and the precedent may cover only shocks and fests. Either way the

opportunity is a standing, partial version of a model already run. The institution already records consumption at the level such a rule needs, one scan per registration with a hand-written ID fallback when the scanner fails (single-sourced). Whether scans are retained per student is a candidate, since the export carries no student ID.

The rule has three parts. What a registration is: a default applied to every student, not a choice. How a student leaves it: the capped cancellations and a Skip Meal toggle that tells the kitchen not to prepare and removes no charge (confirmed). What it costs: a charge fixed at the booking, not at eating. Changing the default or widening its exit makes registration carry more of the morning's demand. Changing the billing trigger alone changes who bears a no-show and does nothing for the count.

Breakfast is the natural scope because it has the worst recorded turnout of any meal while lunch shows the same platform working. The case is not that little is at stake, since Kadamba alone registers about 1,070 breakfasts a day of which about 670 go uneaten (confirmed). It is that the problem and any trial can be confined to the meal where it lives. This opportunity is gated on how each operator's contract counts a plate (Section 7).

4.3 On decision rights: who owns the outcome

Decision rights over the outcome are split at least five ways. The Mess Office sets registration and cancellation rules (operators say ”CDS team”, and whether that is the same body is unconfirmed). The institution sets billing. Students and the Mess Committee choose the menu. Each operator decides privately how much to cook. The academic administration sets a calendar that reaches the kitchens by word of mouth. No actor holds both the knowledge and the authority, and the operator is the sharpest case, which is the confirmed reason a proven fix was never picked up again.

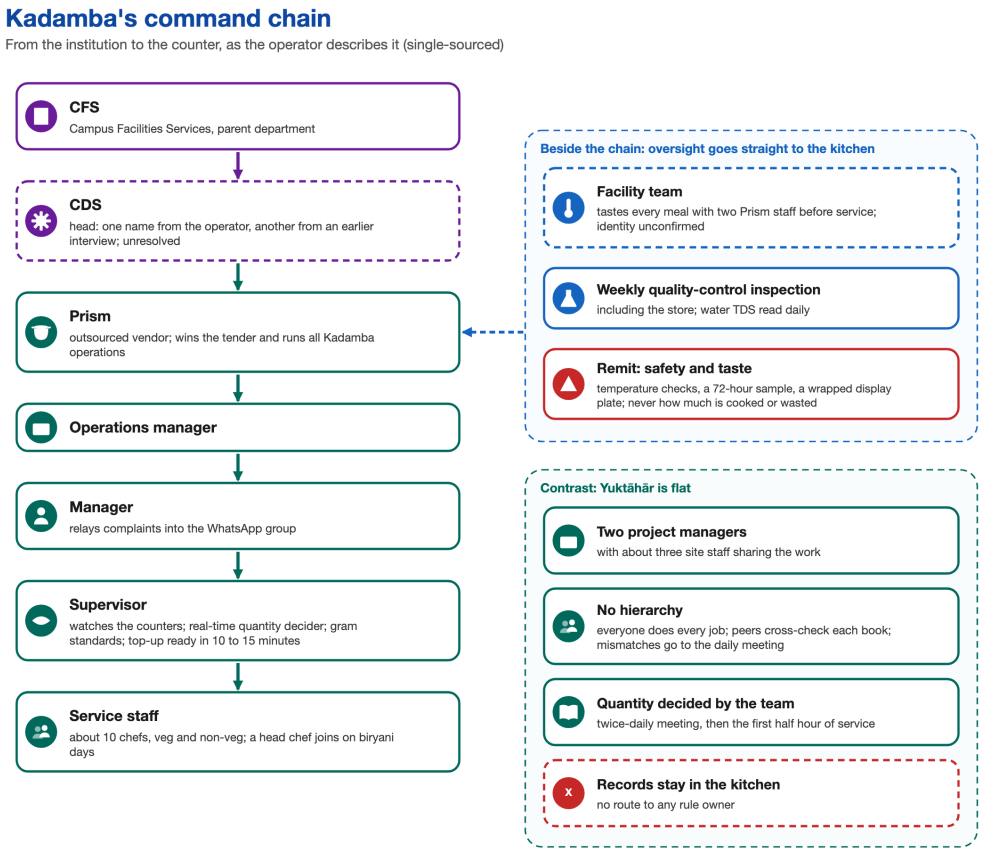


Figure 4. Kadamba's command chain, with the facility team's side channel and Yuktāhār's flat contrast.

At Kadamba the chain runs from CFS to CDS to Prism and down to a supervisor who decides within a meal how much more to cook (single-sourced). Who heads CDS is unresolved: the project has two names for the role, and

whether these are two roles or one wrong name is unknown. The only standing oversight in the kitchen, the facility team's tasting, weekly inspection and daily water testing, covers safety and taste and has no remit over quantity. Yuktāhār has no such hierarchy, only two project managers, staff who share every job and peer cross-checking.

The opportunity is one named owner accountable for the booked, cooked and eaten outcome end to end, with a standing route by which the operators' data reaches that owner. It is not a new committee, since the system already has escalation structures that work when something reaches them. What is missing is an actor for whom this gap is their problem. A third secondary area belongs here: **the menu's pairings.** Bakul's manager says the breakfast menu's problem is pairings, not dishes. Kachori belongs with chole, and Raj Kachori with aloo curry rather than chana. He sees students choosing for nutrition and preference and himself for traditional combinations, taste and what is eaten, from about 20 years with North Indian food (single-sourced). That the combinations reduce consumption is his view, untested with residents. What operators see eaten and left, item by item, should reach the forum that sets the menu and be checked there against what residents want. It is a voice, not authority over students' choices.

4.4 On information flow: what gets noticed

High-intensity feedback goes to a committee and low-intensity feedback to the CFS Head. Both respond to how loudly a problem is felt and neither to what it costs. The opportunity is an escalation trigger that fires on measured cost alongside the existing complaint trigger. It is the cheapest of the four by a wide margin and has the broadest reach, since it would also catch the next quiet but large problem.

The unlocking step is routing and a shared definition, not new measurement. Yuktāhār's two registers disagree for the breakfast of 3 September 2026: 77 eaten in the Wastage Book against 110 in the attendance register (confirmed), so a cost trigger needs one agreed meaning of ”ate”. Yuktāhār already reconciles issued against cooked, so an agreed definition of eaten extends a practice it already runs. Records exist for two halls only.

5. What Any Intervention Must Respect

These are constraints the evidence imposes.

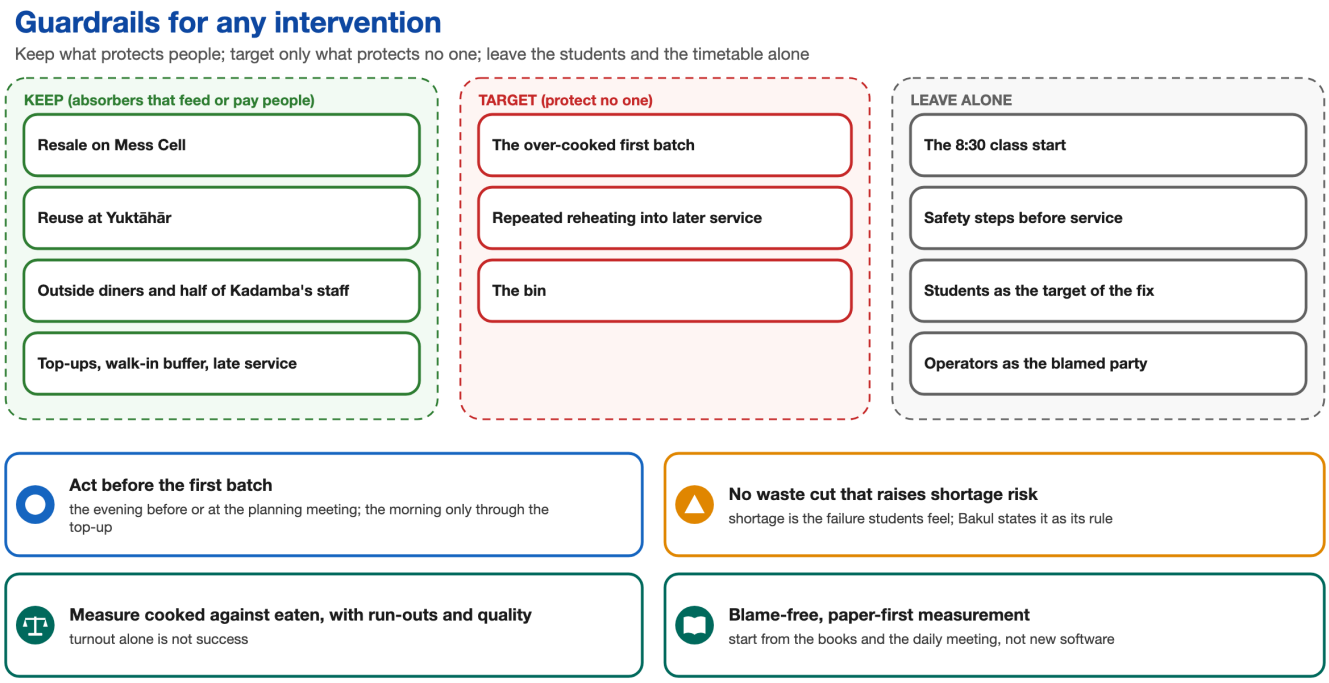


Figure 5. Guardrails: what to keep, what to target, and what to leave alone.

Do not remove an absorption mechanism to manufacture a signal. Making resale harder or ending anyone's

access to leftovers makes the problem visible by making real people worse off. The evidenced routes are Yuktāhār's reuse of leftovers (idli into idli upma or poha, batter refrigerated, surplus milk set as curd, confirmed); the 30 to 50 outside diners, such as night security who do not want to leave their post, and about half of Kadamba's staff, who eat the same food; top-ups; walk-in and paying portions, which may be the same allocation as the outside diners' (candidate); service to late arrivals; and the bin. Kadamba says its surplus goes to a dustbin collected by a garbage contractor, with no composting (single-sourced), which conflicts with the institution's account and is not reconciled. An intervention that shrinks the first batch takes food from those diners unless it provides for them by name. Two routes are not worth protecting. The bin benefits no one, and holding and reheating surplus, which the team judged to be Kadamba's main quality problem (candidate), absorbs surplus at the cost of what students eat. Any proposal that raises someone's pain to create visibility is the wrong kind of fix, because the signal can be read from records already held.

Act before the first batch is cooked. Cooking starts at 5:30 at Kadamba and Yuktāhār, and Bakul's labour-heavy items can start around 3 AM at Hafizpet. At Kadamba the batch then passes a safety sequence before any diner arrives: cooking temperature, a 72-hour sample, a wrapped display plate at the entrance (confirmed, observed), serving temperature and a tasting. No same-morning signal can reach that batch, so any change that cuts surplus has to act the evening before or at the planning meeting, and must not trade a safety step for speed. It must also keep a top-up path, and the lag differs by hall: 10 to 15 minutes on site at Kadamba, and at Bakul a call, cooking and a 20 to 40 minute drive of unmeasured total. No single ratio fits every hall.

Reduce waste without raising the chance of a shortage. Students need food to be there when they come. The run-outs they describe, idli, puri and bhatura gone by 9:20 with promised restocks that never arrive, are the failure they feel directly. One operator states the rule outright: ”There should not be a shortage of food, even if some wastage occurs.” A cut that raises shortage risk fails students, and an operator holding that rule is unlikely to adopt it (candidate). The risk concentrates differently by hall. At Kadamba 26.5% of April scans fell between 9:15 and 9:30 and the rate peaked at about nine scans a minute at 9:29 (confirmed), the operator's rule of thumb is about 150 diners in the last ten minutes, and Yuktāhār's head describes the same closing rush. Bakul's manager puts his main rush at 8:00 to 8:30 and keeps food for about 50 at 9:20, a reserve that stands in for a top-up that cannot arrive late from 8 km away (candidate). After 9:30 latecomers enter Kadamba by the back door and cannot be refused. The crest is the one place where surplus and shortage occur on the same morning, and where a first-half-hour projection is weakest (candidate). Any cut to the first batch has to protect late service in the form each hall needs, or it moves the cost onto the students who arrive last.

Do not treat turnout as the success metric. Part of the gap is students eating the way they live: intermittent fasting, regional patterns, religious observance. The right measure is the gap between cooked and eaten, read with run-outs, which Yuktāhār already logs per item, and with the quality of food as eaten, and whether the cost becomes visible and answerable. Waste falling while run-outs rise is not success.

Do not treat the 8:30 class start as the lever. The timetable solves real problems of its own and is a fixed boundary the mess works around. Nearly 60% of Kadamba's April scans fall at or after 8:45, after classes have begun, so class timing is not the only driver of late arrival. The window's end and the timing of batches sit inside the mess system.

Do not aim the fix at students or at operators. Students hold the least power and the most defensible behaviour in the system, and ”students should register more accurately” asks them to solve a structure they did not build, through an exit capped at five a month. Kadamba's manager mirrors the blame from the other side when he says students register ”even if you are in the village” and let a friend eat in their place (single-sourced). Operators over-prepare, and they also run the system's only same-day balancing loop. Two reasons are plausible: a shortage at close is complained about while a surplus produces nothing, which Bakul's manager states as his own rule (what makes a shortage costly to it, a complaint, a deduction or a contract risk, is unknown), and Kadamba's framing that the count is what it is paid to cook for (a candidate cause). Treating operators as the source of waste would discard the actors who hold the data.

Keep measurement blame-free, and work with the records operators already keep. Yuktāhār's weighing worked only after it was made blame-free. Any route that carries waste figures to a rule owner or a cost trigger must not become grounds for deductions on the staff who record them, or the records will be defended instead of kept. Yuktāhār also tried a commercial enterprise system for groceries and rush prediction and dropped it, because the

data was already in its books, the cost was not justified and its staff are not digitally inclined (single-sourced). Any route has to start from paper books and the daily meeting, and at Bakul from the manager's own weekly review.

Preserve what already works. The Student Council to Committee pathway produced the menu rotation, and the two-week cycle it created (weeks one and three identical, weeks two and four identical) is what makes Yuktāhār's matching-week lookup possible (confirmed from records). Yuktāhār's book routine is a working in-house calibration loop. Each operator's top-up is the only same-day balancing loop, and Bakul's weekly adjustment is a second calibration, bounded by its floor. Whether Skip Meal declarations reach kitchen forecasting is unresolved: the CFS side says they do, and Kadamba's operator says it cooks on registrations whether or not people want to eat (single-sourced), while no operator named Skip Meal as a planning input. Both can hold if Skip Meal lowers the count the operator reads and the operator then treats that count as its obligation (candidate). None of this needs replacing. The escalation pathway needs a second trigger, Skip Meal needs to be worth a student's time, and the kitchen loops need better inputs.

Identify Contradictions and Tensions: What the Opportunity Must Hold

The intervention points have to be designed with several tensions in view. Kadamba's stated waste does not square with its first batch, and surplus that looks like waste is also the meal of outside diners and staff. The same billing rules produce an operator that cuts waste for profit, one that cooks to the count as an obligation and one that accepts waste to avoid a shortage. The kitchen that forecasts best projects from the first half hour, while Kadamba's heaviest demand arrives in the last fifteen minutes. A reserve for late diners protects them from a run-out and is also the food held longest, so one hedge trades shortage against waste and texture (candidate). The safety sequence that protects diners fixes the first batch before any turnout signal exists. A menu chosen for nutrition may not be the menu that gets eaten, in Bakul's manager's view, untested with residents. Two accounts of Skip Meal, and two of whether holidays were ever non-compulsory, are unreconciled. Section 7 carries each as open.

6. How the Four Interact

They are not alternatives, and the order matters.

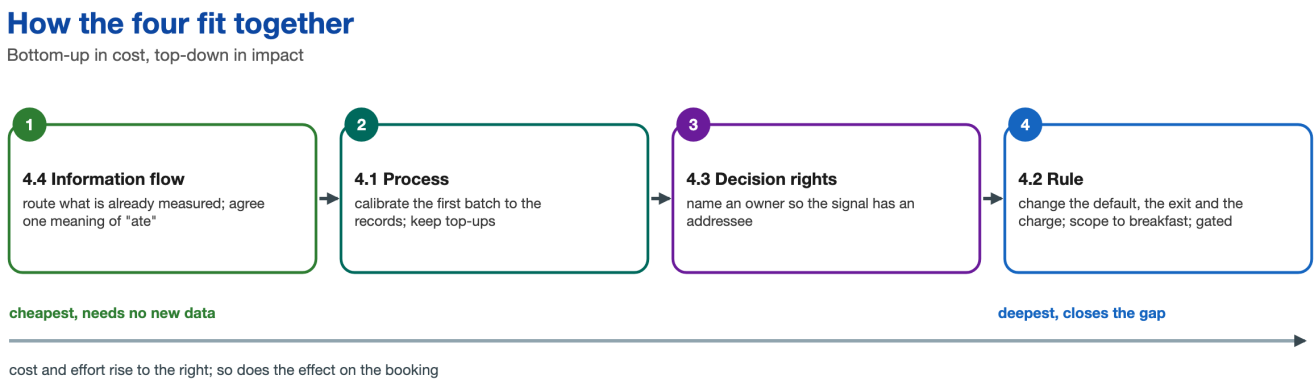


Figure 6. The sequence: cheap and enabling first, deep and gated last.

The information-flow change (4.4) enables the rest. Without a routed, agreed measure of cost there is nothing for an owner to act on and no evidence for a rule change, and it needs no new data. The process change (4.1) shows a physical effect fastest and can move without governance where the operator has a reason to, as Yuktāhār's loop already does. At Kadamba, whose operator treats the count as its paid obligation, the kitchen may not move alone, so 4.1 there is partly coupled to the plate-basis question. On its own it reduces surplus inside a registration system that still over-books. The ownership change (4.3) turns a visible number into a decision, since a signal with no addressee produces nothing, which is the current state. The rule change (4.2) is the deepest and the one that closes the gap, and it is gated on an external constraint, so it is best seen as what the first three make possible. The sequence runs

bottom-up in cost and top-down in impact: route the measurements already taken, calibrate the first batch to them, assign an owner, then change what a registration is, how a student leaves it and what it costs, scoped to breakfast and evaluated.

7. The Questions That Gate All of This

Two questions decide the shape of the deepest intervention point.

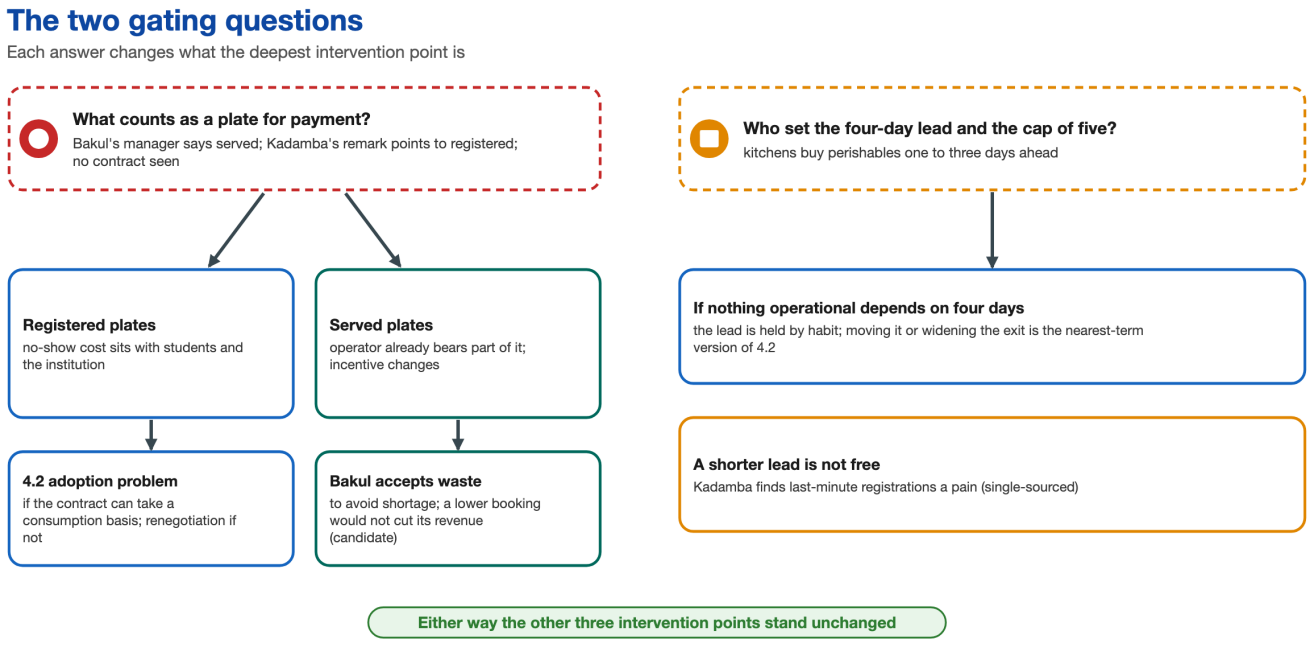


Figure 7. The two gating questions and what each answer would change.

What counts as a plate for payment. Bakul's operator won a tender and is paid monthly as a lump sum on plates, with deductions, and its manager says the plates counted are plates served (single-sourced). Kadamba's operator says it cooks to registrations because it is paid for them, which points to registered plates, but no contract term has been seen. The statements pull apart: contracts may differ, ”served” may be a normalised word for billed plates, or Kadamba's remark may be about duty and not the formula. One word from one operator does not change the gate. On registered plates, the no-show cost sits with students and the institution, and 4.2 is an adoption problem if the contract can absorb a consumption basis and a renegotiation if it cannot. On served plates, part of the no-show cost already sits with the operator and the incentive behind over-preparation changes. For Bakul that would mean it already bears the cost of what it over-cooks and accepts it to avoid a shortage, and a change to the breakfast default would not cut its revenue (both candidate).

Who set the four-day lead and the five-cancellation cap, and what depends on them. The kitchens buy perishables one to three days ahead, and Kadamba's team has challenged the four-day close. If nothing operational depends on four days, the lead is held by habit or billing convenience, and moving it or widening the exit is the nearest-term version of 4.2. A shorter lead is not free for operators, though, since Kadamba also finds last-minute registrations a pain (single-sourced). Either way the other three points stand, which is part of why they come first.

Other questions carried forward:

- Is the count Kadamba cooks against already net of Skip Meal declarations and cancellations? Who heads CDS, and does anyone with policy authority look at the scan records, retained per student or not?
- Which holiday periods ran without compulsory registration? This bears on June 2026 at Yuktāhār, where two Saturday breakfasts show 37 and 34 registered. The sheets write 89 and 91% eaten, but those totals include unregistered and paying diners, and registered students alone were 22 and 21, about 60% (confirmed from records).

If registration was non-compulsory then, a smaller booking closer to intent roughly doubles turnout, though still below lunch. Automatic registration over holidays makes that less likely, so it stays a candidate.

- What is the base of Kadamba's 5 to 10% waste figure, how much surplus is held and reheated, and does lost quality feed skipping? That last link is an open chain, not an evidenced loop.
- Are outside-diner portions the same allocation as walk-in portions, and is feeding them and Kadamba's staff a written entitlement or a norm? Which account of disposal is current, and where does Yuktāhār's residual waste go?
- Is over-preparation driven by an asymmetric penalty or by the count as a paid obligation? Which definition of ”ate” should a trigger use, and which body does the facility team belong to?
- How accurate is Yuktāhār's first-half-hour projection on breakfasts that crest at close? Which items other than aloo paratha lose quality in hot holding?
- What do turnout and waste look like at Bakul and Palash? At Bakul: is anything in the weekly adjustment written down, how long does a top-up take end to end, is the reserve of about 50 set from data or judgement, which menu decisions can the manager override, and why do its own residents skip breakfast, asked of them and not inferred from the manager?

8. What Phase 3 Inherits

Activity 9 asks for leverage points and design principles. This document hands it four intervention points, one in the kitchen and three in governance, ranked by depth and not by ease. It hands over constraints that rule out whole categories of intervention before they are designed: acting after the first batch is cooked, removing food from outside diners and staff, cutting waste in a way that raises shortage risk, penalising the people who record waste, depending on a new digital system, and aiming the fix at students or operators. It hands over three secondary areas (food quality as eaten, demand by item, the menu's pairings), a sequencing logic and two gating questions.

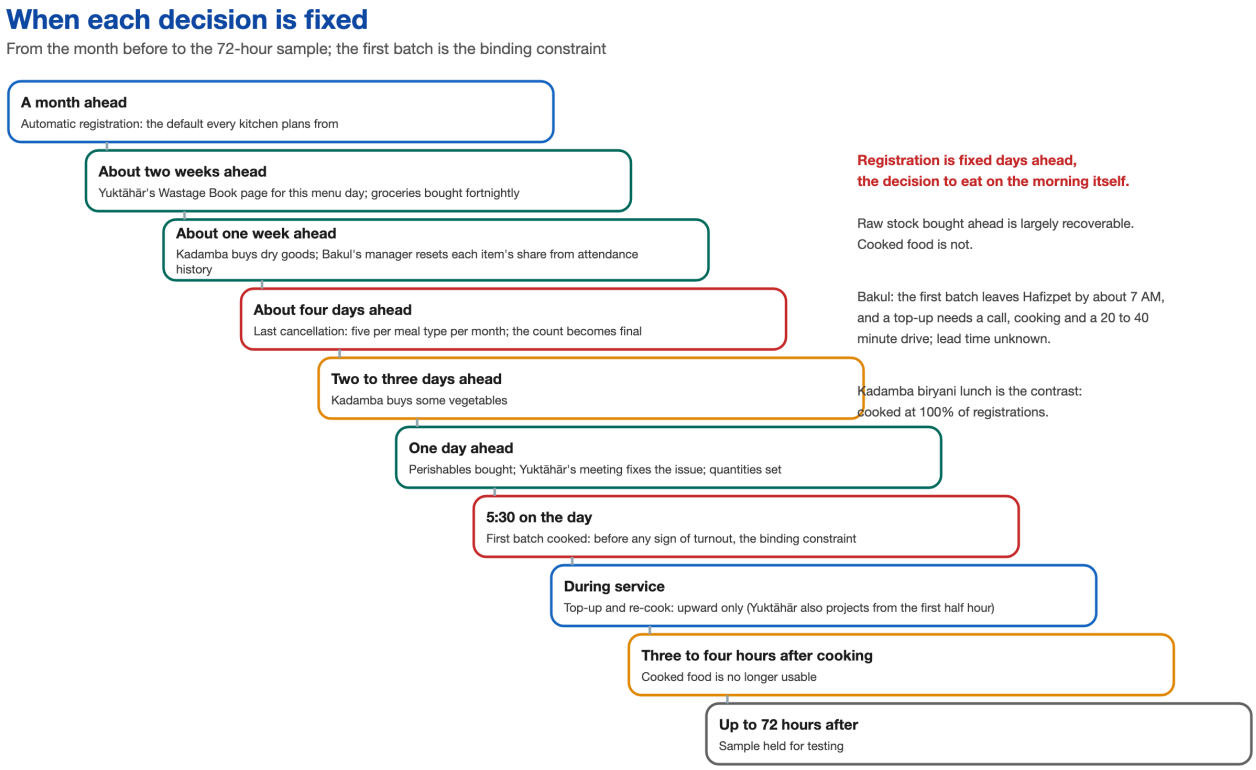


Figure 8. When each decision is fixed, from the month before to the 72-hour sample: the timeline any intervention works inside.

The evidence limits travel with it. Record-level turnout exists for Kadamba breakfast in April 2026, Yuktāhār in September 2026 and two Yuktāhār Wastage Book sheets from June 2026 only. The Kadamba ratio and gram standards are the operator's own statements, heard directly, and carry self-presentation bias. Bakul's rule, floor, timings and payment statement come from a team member's synthesis of its manager's account, with no Bakul records. Kadamba's only waste figure is a share with no stated base. None of these should become a headline number, and each hall was visited once.

The most useful thing carried forward is the negative result. The system has not fixed itself because it has no route for what it already knows, and nobody lacks capability, willingness or information. The gap is measured every morning and absorbed every morning, and nothing carries that known cost to the kitchen decision that sets the first batch or to the actors who set the rules. Designing those routes is the work.

Corrections Since the First Submission

Phase 2: IIIT Hyderabad Breakfast Mess System

Contents

What changed and why 1

What changed and why

The first Phase 2 documents were written before the operator visits and the Kadamba scan records. Each change below is stated once here and applied everywhere in the rebuilt documents.

First submission said	Now
Registration is a weekly or monthly standing booking	An automatic default for every student, with five cancellations per meal type per month, each at least four days ahead
The vendor forecast locks four days ahead; the four-day lock is the bottleneck	The four-day close applies to registration. Kitchens buy on their own cadences and set quantities the day before. The binding constraint is the cooked first batch
The operators are one actor that cooks to a four-day forecast	Three operators, each applying its own ratio to the closed count: a fixed 70% at Kadamba, 70 to 80% by item with a floor near 70% at Bakul, and a book-based lookup at Yuktāhār
Bakul's kitchen is near the airport and its ratio was raised from 70 to 80%	Hafizpet, about 8 km, 20 to 40 minutes by road. The raise is withdrawn: the share varies by item and day and is adjusted weekly
Bakul is a fixed-ratio kitchen like Kadamba	A third regime between Kadamba and Yuktāhār
Composting absorbs the surplus	Kadamba does not compost; its waste goes to a garbage contractor
20 to 30 staff eat leftovers at Kadamba	About 30 extra portions for 30 to 50 outside diners, plus about half of Kadamba's staff
Kadamba waste of 16 and 23 kg	Withdrawn. The operator reports 5 to 10% daily on an unstated base
Yuktāhār has a five-person project team	Two project managers, with about three site staff sharing the work
Prism is Kadamba's floor team	Prism is the vendor that holds the tender and runs all Kadamba operations
All three operators describe a closing rush, and Bakul peaks at 8:30 to 9:00	The crest is recorded at Kadamba and described at Yuktāhār. Bakul's manager places his rush at 8:00 to 8:30
Nothing in the kitchen reacts in real time; no one tracks waste	Supervisors, projections and phoned top-ups react all morning, upward only. Yuktāhār weighs waste per item and meal, and no route carries it to a rule owner
Kadamba 51 to 79% and Bakul and Palash 11 to 31% turnout by cuisine	Registrations as a share of capacity from one portal snapshot, not attendance
The loops were named R1 and B1 to B4, six of them closed	One scheme, L1 to L17 plus L3b. Three close with every link evidenced (L4, L11, L12), five are candidates, and L6 is a calendar switch, not a loop
The gap is about 30%	62.9% at Kadamba in April and about 72% at Yuktāhār in September, the figure of about 30% fits lunch